

Source files received / data types found

Survivors

1. [Survivors_SP_2025-01-26](#)

SP / TMNT survivors: Raphael, Leonardo, Donatello, Michelangelo, April, Splinter. Includes yellow-star upgrades, red awakening upgrades, passive skills, active skills, teamwork passive slots, crit rate, laceration, vulnerability, pet boosts, etc.

Tech Part Resonance

2. [Tech_Part_Resonance_Drone](#)

Drone resonance: Augment Wingbit + Attack Wingbit, damage scaling, cooldown, duration, what gets boosted.

3. [Tech_Part_Resonance_HI_Maintainer_RPG](#)

C4 / RPG / HI-Maintainer style resonance. Shows base damage, launch interval, explosion delay, max values.

Equipment

4. [Weapons_V2_2](#)

Weapons: Void Power, Lightchaser, Sword of Disorder, Kunai, Baseball Bat, Katana, Shotgun, Revolver.

5. [Armor_V2](#)

Chest armor: Eternal Suit, Voidwaker Windbreaker, Armor of Quietus, Full Metal Suit, Traveler's Jacket, Carapace, Army Uniform, Protective Suit.

6. [Belts_V2](#)

Belts: Twisting Belt, Eternal Belt, Stylish Belt, Leather Belt, Voidwaker Sash, Army Belt, Plated Sensor, Broad Waistguard.

7. [Boots_V2](#)

Boots: Voidwaker Treads, Eternal Boots, Light Runners, Shoes of Confusion, Army Boots, Prosthetic Legs, Layered Snowshoes, Highboots.

Forge / Cores

8. [Astral_Forge](#)

AF requirements, SS gear requirements, non-SS AF costs, salvage returns, SS breakpoints.

9. [Awakening_Core_Guide](#)

Awakening core priority paths for Yang, Metallia, King, Worm, Common, Catnips, TMNT survivors in Ender's Echo and Clan Expedition.

Debuffs / Damage Multipliers

10. [Boss_Debuffs](#)

Chill, Poison, Exposed, Weaken, Debilitate, Yin Yang, Laceration, Vulnerability. This is very important for DPS stacking logic.

11. [Crit_Rate_And_Damage](#)

Crit rate and crit damage sources from survivor levels, gear, collectibles, skills, pets, talents, combat harmony, survivor synergy, Don passive, Rex, etc.

Progression / Unlocks

12. [Chapter_Unlocks](#)

Unlock timing by chapter: shop, equipment, tech parts, pets, clans, Ender's Echo, Zone Ops, Special Ops, privilege card, etc.

Clan

13. [Clan_Expedition_Rewards](#)

Novice, Adept, Elite clan expedition rewards: chapter rewards, chest rewards, ranking rewards, clan tier limits.

14. [Clan_Level_Rewards](#)

Clan level perks, clan EXP, clan size, quick earnings, clan member ATK/HP boosts.

15. [Clan_Shop](#)

Clan shop items, costs, quantities, and how stock changes by clan level.

Collectibles

16. [Collectible_Chest_Odds](#)

Collectible chest rarity odds and shard odds.

17. [Collectible_Costs](#)

Collectible star costs, shard costs, exchange rates, collector's hearts, purchase costs.

18. [Collectibles_Edition_1](#)

1st Edition collectible chest item pool and drop rates.

19. [Collectibles_Edition_2](#)

2nd Edition collectible chest item pool and drop rates.

20. [Collectibles_Edition_3](#)

3rd Edition collectible chest item pool and drop rates.

Data type 1: SP Survivors / TMNT Survivors

Source: [Survivors_SP_2025-01-26](#)

This image is about **SP-grade survivor progression**, specifically the TMNT collab survivors:

Raphael, Leonardo, Donatello, Michelangelo, April, Splinter

This data type belongs in the optimizer as:

Survivor | SP Survivor | Star Upgrade | Awakening | Passive Skill |
Active Skill | Teamwork Passive | Debuff Source | Crit Source | Pet
Source

For your calculator, this should mostly affect **DPS**, not HP. The HP lines should still be stored as raw data, but scoring should mark them as **ignore / non-DPS**.

Shared SP survivor structure

Each SP survivor has:

Yellow stars

- 1-star: unlocks exclusive weapon / base survivor skill.
- 2-star: Survivor License, usually **Survivor ATK +400** and **Survivor HP +2000**.
- 3-star: improves the survivor's main weapon or attack mechanic.
- 4-star: Advanced Survivor License, usually **Survivor ATK +1000** and **Survivor HP +5000**.
- 5-star: **All Survivors: Crit Rate +8%**.
- 6-star: big personal DPS upgrade, usually main weapon damage + survivor ATK +20%.

Red awakening stars

- Adds active awakening skill improvements.
- Adds passive awakening skills.
- Some give **Teamwork Passive Slots**, which matter a lot because awakened passive skills can be shared/equipped.
- Costs appear to scale with awakening cores, pizza/slices, and blue materials.

For calculator logic, **5 yellow stars is extremely important** because it gives global crit rate. The HP parts are mostly trash for your DPS optimizer.

Raphael

Main association

Raphael is associated with:

Sai / Fury Sai / Bleed / Laceration / Lacerated-target damage / Rocket Assault

Yellow-star upgrades

1-star

- Exclusive weapon: **Sai**
- Raphael throws **2 sai** at enemies.
- Can evolve into **Fury Sai**.

2-star

- Survivor License:
 - Survivor ATK +400
 - Survivor HP +2000

3-star

- **Power Strike**
- For every 5 attacks, 1 more giant sai is thrown.
- Attacks inflict **Bleed**.
- Survivor HP +20%.

4-star

- Advanced Survivor License:
 - Survivor ATK +1000
 - Survivor HP +5000

5-star

- All Survivors: **Crit Rate +8%**

6-star

- **Hot-headed**
- Sai and Fury Sai damage +20%.
- Survivor ATK +20%.

Awakening upgrades

A1

- Active: **Rocket Assault**
- When Rocket Assault ends, fires a wave of **Assault Sai**.
- Passive: **Scattered Spikes**
- Throws piercing sai around every 20 seconds.
- Enemies hit get **+5 seconds of Laceration**.
- Damage to Lacerated enemies +10%.

A2

- Additional damage on Lacerated targets +10%.

A3

- More Assault Sai thrown during Rocket Assault.

A4

- Additional damage on Lacerated targets +10%.
- Unlocks 1 Teamwork Passive Slot.

A5

- Fires Assault Sai during Rocket Assault.

A6

- Additional damage on Lacerated targets +20%.
- Teamwork Passive Slot +1.

Optimizer notes

Raphael is a **Laceration damage amplifier**. He matters if the build uses Laceration stacking, boss debuffs, or teamwork passive sharing.

Database flags:

Debuff = Bleed, Laceration
DamageType = Lacerated target damage
GlobalValue = Crit Rate at 5-star
TeamworkSlot = A4 and A6
DPSRelevant = Yes

Leonardo

Main association

Leonardo is associated with:

Dual Katanas / Fullmoon Slash / Laceration / Whirlwind Slash /
Laceration duration

Yellow-star upgrades

1-star

- Exclusive weapon: **Swiftwind Dual Katanas**
- Leonardo performs long and short slashes with his katanas.
- Can evolve into **Fullmoon Slash**.

2-star

- Survivor License:
 - Survivor ATK +400
 - Survivor HP +2000

3-star

- **Voidbreaker Combo**
- Adds long and short slashes.
- Survivor HP +20%.

4-star

- Advanced Survivor License:
 - Survivor ATK +1000
 - Survivor HP +5000

5-star

- All Survivors: **Crit Rate +8%**

6-star

- **Secret Art Limit**
- Fullmoon Slash breakthrough art.
- Adds attack.
- Survivor ATK +20%.

Awakening upgrades

A1

- Active: **Whirlwind Slash**
- Whirlwind Slash damage +50%.
- Passive: **Undaunted Will**
- When hit by monster collisions, inflicts Laceration.
- Damage on Lacerated targets +2% for 5 seconds.

A2

- Additional damage on Lacerated targets +2%.
- Laceration duration +5 seconds.

A3

- Reduces Whirlwind Slash cooldown.

A4

- Additional damage on Lacerated targets +3%.
- Laceration duration +5 seconds.
- Unlocks 1 Teamwork Passive Slot.

A5

- Whirlwind Slash duration up.

A6

- Additional damage on Lacerated targets +3%.
- Laceration duration becomes permanent.
- Teamwork Passive Slot +1.

Optimizer notes

Leonardo is another **Laceration support/DPS source**, but his numbers look smaller than Raphael/Michelangelo. His value is more about Laceration uptime and duration.

Database flags:

Debuff = Laceration

DamageType = Lacerated target damage

Mechanic = Duration extension / permanent laceration

GlobalValue = Crit Rate at 5-star

TeamworkSlot = A4 and A6

DPSRelevant = Yes

Donatello

Main association

Donatello is associated with:

Bo / Savage Bo / Skill Damage / Crit Damage / Crit Rate / SP passive scaling / Brotherhood

This is one of the most important ones for the calculator because Donatello has **global SP passive scaling**.

Yellow-star upgrades

1-star

- Exclusive weapon: **Seeker Bo**
- Donatello spins his bo to attack enemies.
- Can be upgraded to **Savage Bo**.

2-star

- Survivor License:
 - Survivor ATK +400
 - Survivor HP +2000

3-star

- **Super Sweep**
- Seeker Bo and Savage Bo damage +30%.
- Survivor HP +20%.

4-star

- Advanced Survivor License:
 - Survivor ATK +1000
 - Survivor HP +5000

5-star

- All Survivors: **Crit Rate +8%**

6-star

- Staff Strike / Seeker Bo / Savage Bo:
 - Attack range up
 - Damage +30%
- Survivor ATK +20%

Awakening upgrades

A1

- Active: **Spinning Smash**
- Fires an energy wave when Spinning Smash ends.
- Passive: **Brotherhood**
- Exclusive global passive.
- Effective for all Survivors when unlocked.
- Cannot be equipped to Teamwork passive slots.
- For every SP Grade Survivor Awakening passive skill equipped:
 - Crit Rate +3%

A2

- Brotherhood adds:
 - Skill Damage +3%
 - Crit Damage +3%
 - Per SP-grade survivor awakening passive equipped.

A3

- Spinning Smash attack range up.

A4

- Brotherhood adds damage to special debuffed targets:
 - Damage to Lacerated targets +3%
 - Damage to Poisoned targets +3%

- Damage to Chilled targets +3%
- Unlocks 1 Teamwork Passive Slot.

A5

- Spinning Smash damage +50%.

A6

- Brotherhood adds:
 - Damage to Weakened targets +3%
 - Damage to Poisoned targets +3%
 - Damage to Chilled targets +3%
- Teamwork Passive Slot +1.

Optimizer notes

Donatello is a **global scaling engine**. He should not be treated like a normal survivor only. His Brotherhood passive creates value based on how many SP awakening passives are equipped.

Database flags:

```
GlobalPassive = Brotherhood
ScalingInput = Equipped SP Awakening Passive Count
CritRateSource = +3% per SP passive at A1
SkillDamageSource = +3% per SP passive at A2
CritDamageSource = +3% per SP passive at A2
DebuffDamage = Lacerated / Poisoned / Chilled / Weakened
TeamworkSlot = A4 and A6
DPSRelevant = Very High
```

This one is important for your optimizer because it affects **build synergy**, not just one survivor.

Michelangelo

Main association

Michelangelo is associated with:

Nunchucks / Windchaser Nunchucks / Whirlwind / Bleed / Laceration /
Lacerated-target damage / hit-count trigger

Yellow-star upgrades

1-star

- Exclusive weapon: **Arc Nunchucks**
- Swings nunchucks and launches whirlwinds.
- Can be upgraded to **Windchaser Nunchucks**.

2-star

- Survivor License:
 - Survivor ATK +400
 - Survivor HP +2000

3-star

- **Tubular Tornado**
- Arc Nunchucks / Windchaser Nunchucks whirlwind damage +50%.
- Whirlwind range up.
- Survivor HP +20%.

4-star

- Advanced Survivor License:
 - Survivor ATK +1000
 - Survivor HP +5000

5-star

- All Survivors: **Crit Rate +8%**

6-star

- **Apocalypse Tornado**
- Arc Nunchucks / Windchaser Nunchucks whirlwind damage +50%.
- Whirlwind inflicts **Bleed**.
- Survivor ATK +20%.

Awakening upgrades

A1

- Active: **Twin Throw**
- Nunchucks thrown by Twin Throw can be thrown again.
- Passive: **Fantastic Storm**
- Every 500 attacks on hit spawns 1 windstorm.
- Windstorm inflicts **Lacerate** for 5 seconds.
- Skill cooldown: 20 seconds.
- Damage to Lacerated targets +10%.

A2

- Fantastic Storm:
 - Required hits -100
 - Damage to Lacerated targets +10%

A3

- Twin Throw thrown nunchucks count up.

A4

- Fantastic Storm:
 - Required hits -100
 - Damage to Lacerated targets +10%
- Unlocks 1 Teamwork Passive Slot.

A5

- Twin Throw nunchucks attack range up.
- Attack damage +50%.

A6

- Fantastic Storm:
 - Required hits -200
 - Damage to Lacerated targets +20%
- Teamwork Passive Slot +1.

Optimizer notes

Michelangelo looks like a stronger Laceration DPS support than Leonardo because he has bigger Lacerated-target damage chunks and hit-count scaling.

Database flags:

Debuff = Bleed, Laceration
Mechanic = Hit count trigger
Trigger = Every 500 attacks, reduced by awakenings
DamageType = Lacerated target damage
GlobalValue = Crit Rate at 5-star
TeamworkSlot = A4 and A6
DPSRelevant = High

April

Main association

April is associated with:

Soundwave Balls / Mega Soundwave Balls / Vulnerability / Pet ATK / Pet HP / Reporter Snapshot

April matters because she connects **survivor awakening + pet damage + vulnerability**.

Yellow-star upgrades

1-star

- Exclusive weapon: **Microphone**
- April fires soundwave balls at enemies.
- Can evolve into **Mega Soundwave Balls**.

2-star

- Survivor License:
 - Survivor ATK +400
 - Survivor HP +2000

3-star

- **Increased Volume**
- Soundwave Ball damage +30%.
- Survivor HP +20%.

4-star

- Advanced Survivor License:
 - Survivor ATK +1000
 - Survivor HP +5000

5-star

- All Survivors: **Crit Rate +8%**

6-star

- **Sonic Boom**
- Mega Soundwave Balls explode in a small area when enemies are hit.
- Survivor ATK +20%.

Awakening upgrades

A1

- Active: **Reporter Snapshot**
- Reporter Snapshot damage +50%.
- Passive: **Pet Companion**
- When in battle, deployed pet's:
 - Attack +5%
 - HP +5%

A2

- Deployed pet additional:
 - Attack +5%
 - HP +5%

A3

- Reporter Snapshot damage +50%.
- Targets hit by Snapshot are inflicted with **5% Vulnerability**.

A4

- Deployed pet additional:
 - Attack +10%
 - HP +10%
- Unlocks 1 Teamwork Passive Slot.

A5

- Reporter Snapshot damage +50%.
- Targets hit by Snapshot get additional **5% Vulnerability**.

A6

- Deployed pet additional:
 - Attack +10%
 - HP +10%
- Teamwork Passive Slot +1.

Optimizer notes

April is not just personal damage. She is a **pet amplifier** and **vulnerability source**.

Database flags:

```

Debuff = Vulnerability
PetATKSource = Yes
PetHPSource = Ignore for DPS
Vulnerability = 5% at A3, additional 5% at A5
GlobalValue = Crit Rate at 5-star
TeamworkSlot = A4 and A6
DPSRelevant = High if pet build / vulnerability stacking
  
```

For your optimizer, April should become more valuable if the player uses a strong pet setup.

Splinter

Main association

Splinter is associated with:

```

Secret Art Blast / Secret Art Fusion / Slow / Enhanced Slow / Phantom
Assault / Pet Crit Rate / Pet Crit Damage / Pet Skill CD
  
```

Splinter is a pet-crit support survivor.

Yellow-star upgrades

1-star

- Exclusive weapon: **Secret Art Blast**
- Splinter brandishes his cane and fires a power wave at enemies.
- Can evolve into **Secret Art Fusion**.

2-star

- Survivor License:
 - Survivor ATK +400
 - Survivor HP +2000

3-star

- **Bonecrusher Blast**
- Secret Art Blast / Secret Art Fusion power wave damage +30%.
- Inflicts **Slow**.
- Survivor HP +20%.

4-star

- Advanced Survivor License:
 - Survivor ATK +1000
 - Survivor HP +5000

5-star

- All Survivors: **Crit Rate +8%**

6-star

- **Fatality Blast**
- Secret Art Blast / Secret Art Fusion power wave damage +30%.
- Inflicts **enhanced Slow**.
- Survivor ATK +20%.

Awakening upgrades

A1

- Active: **Phantom Assault**
- Adds 1 phantom.
- Phantom damage +20%.
- Passive: **Master Mentor**
- Equipped pet skill cooldown -5%.
- Pet Crit Rate +10%.

A2

- Master Mentor:
 - Equipped Pet Crit Rate +10%
 - Pet Crit Damage +20%

A3

- Phantom Assault:
 - +1 Phantom
 - Phantom damage +20%

A4

- Master Mentor:
 - Equipped Pet Skill cooldown -15%
- Unlocks 1 Teamwork Passive Slot.

A5

- Phantom Assault:
 - +1 Phantom
 - Phantom damage +20%

A6

- Master Mentor:
 - Equipped Pet Skill cooldown -10%
 - Pet Crit Rate +20%
 - Pet Crit Damage +20%
- Teamwork Passive Slot +1.

Optimizer notes

Splinter is heavily tied to **pet crit scaling** and pet skill cooldown. He should become stronger in the calculator if pets are a major DPS source.

Database flags:

PetCritRateSource = Yes

PetCritDamageSource = Yes

PetSkillCooldownSource = Yes

Debuff = Slow / Enhanced Slow

GlobalValue = Crit Rate at 5-star

TeamworkSlot = A4 and A6
DPSRelevant = High for pet builds

Important optimizer associations from this whole source

1. Global crit from SP survivors

Every SP survivor shown has:

5 yellow stars = All Survivors Crit Rate +8%

That means each 5-star SP survivor is a **global DPS upgrade**, even if you do not use that survivor.

For the optimizer:

CritRateGlobal += 8% per SP survivor at 5 yellow stars

This is very important for your damage-only calculator.

2. Laceration package

Laceration is tied to:

- Raphael
- Leonardo
- Michelangelo
- Donatello Brotherhood scaling
- Boss Debuffs source image also has a Laceration section

This means Laceration should be treated as a real debuff family.

Suggested field:

DebuffFamily = Laceration
DamageModifierType = Damage to Lacerated targets
StackingType = Needs testing / source-specific

3. Vulnerability package

April adds Vulnerability through Reporter Snapshot.

This connects directly to the **Boss_Debuffs** source because Vulnerability is a major damage amplifier.

Suggested field:

DebuffFamily = Vulnerability
Source = April Awakening A3/A5
DPSRelevant = Very High

4. Pet package

April and Splinter both connect to pets.

April:

- Pet ATK
- Pet HP

Splinter:

- Pet Crit Rate
- Pet Crit Damage
- Pet Skill cooldown

For your calculator:

PetATK = count for DPS
PetCritRate = count for DPS
PetCritDamage = count for DPS
PetSkillCDReduction = count for DPS if pet skill damage is modeled
PetHP = store but ignore

5. Teamwork Passive Slots

Most SP survivors unlock Teamwork Passive Slots at awakening levels:

- A4: Unlocks 1 Teamwork Passive Slot
- A6: Teamwork Passive Slot +1

This matters because survivor awakening is not just “upgrade survivor.” It changes how many passive skills can be equipped and shared.

Suggested fields:

```
TeamworkPassiveSlotUnlock_A4 = 1  
TeamworkPassiveSlotUnlock_A6 = +1
```

Clean database format for this data type

Use something like this:

SurvivorID
SurvivorName
Grade
CollabGroup
StarType
StarLevel
UpgradeName
UpgradeDescription
AffectedStat
StatAmount
TargetScope
AssociatedDebuff
AssociatedSkill
AssociatedPetStat
TeamworkPassiveSlotChange
DPSRelevant
IgnoreForDPS
Notes
SourceImage
SourceDate

Example row:

SP_RAPHAEL | Raphael | SP | TMNT | Yellow | 5 | All Survivors Crit Rate | Crit Rate +8% | CritRate | 8% | All Survivors | None | None | None | 0 | Yes | No | Global DPS upgrade | Survivors_SP_2025-01-26

Example row:

SP_APRIL | April | SP | TMNT | Awakening | A3 | Reporter Snapshot | Snapshot damage +50%; targets get 5% Vulnerability | Vulnerability | 5% | Enemy Hit By Snapshot | Vulnerability | Reporter Snapshot | None | 0 | Yes | No | Boss damage amplifier | Survivors_SP_2025-01-26

Next data type should be **Crit Rate/Crit Damage** or **Boss Debuffs**, because those explain how these survivor bonuses actually stack into DPS.

Data type 2: Crit Rate & Crit Damage

Source: [Crit_Rate_And_Damage_2025-03-09](#)

This image is a **stat-source registry**. It is not just showing one item. It shows every major place Crit Rate and Crit Damage can come from.

For the optimizer, this should become:

StatSource | CritRate | CritDamage | Global | Conditional | GearBased
| SurvivorBased | PetBased | SkillBased | CollectibleBased |
DPSRelevant

This is very important because crit is one of the biggest DPS multipliers in Survivor.io.

Main optimizer rule

Do **not** store crit as one single number only.

You need to split it into:

Global Crit Rate

Global Crit Damage

Active Survivor Crit Sources

Equipped Gear Crit Sources

In-battle Skill Crit Sources
Pet Crit Sources
Collectible Crit Sources
Conditional Crit Sources
Temporary / stance-based Crit Sources

Because some are always active, some require a specific survivor, some require a pet to be active, some require a gear condition, and some require in-battle setup.

Crit Rate section

1. Survivor level Crit Rate

This is associated with:

[Survivor Level Milestone](#) | [Global Stat](#) | [Account DPS](#)

The source shows **98% total global Crit Rate** possible from survivor levels / special survivor unlocks.

Normal survivor level sources shown

- King level 80: Crit Rate +5%
- Tsukuyomi level 80: Crit Rate +5%
- Worm level 80: Crit Rate +5%
- Wesson / other survivor icon level 120: Crit Rate +5%
- Master Yang level 120: Crit Rate +10%

SpongeBob event survivors

These are marked as special-event only.

- SpongeBob level 120: Crit Rate +5%
- Squidward level 120: Crit Rate +5%
- Patrick level 120: Crit Rate +5%
- Sandy level 120: Crit Rate +5%

TMNT SP survivors

Each TMNT survivor gives:

5 yellow stars = All Survivors Crit Rate +8%

Shown TMNT survivors:

- Raphael 5-star: Crit Rate +8%
- Leonardo 5-star: Crit Rate +8%
- April 5-star: Crit Rate +8%
- Michelangelo 5-star: Crit Rate +8%
- Donatello 5-star: Crit Rate +8%
- Splinter 5-star: Crit Rate +8%

Optimizer notes

This is a **global permanent DPS source**. It should have high value because it improves all builds, not just one survivor.

Database flag:

SourceCategory = SurvivorLevel

TargetScope = AllSurvivors

StatType = CritRate

DPSRelevant = Very High

ConditionType = PermanentGlobal

2. Gear Crit Rate

This is associated with:

Equipped Gear | Gloves / SS Gear / Necklace | Conditional DPS

The source shows multiple gear-based crit sources.

Important gear associations visible:

Eternal Gloves

From the gloves source, Eternal Gloves are heavily tied to crit.

Relevant effects:

- Excellent / purple: Attacks have 10% chance to crit.
- Legendary / red: Crit Rate +10%, Crit Damage +100%.

- AF3: Giantbane gives +3% Crit Rate when an elite or boss is defeated.

The crit source image also shows Eternal Gloves as:

0–40% Crit Rate, +10% with R5 collectible

This means Eternal Gloves crit value may depend on gear rarity, AF state, collectible support, or battle condition.

Leather Gloves

Leather Gloves are associated with:

- Crit Rate +5%
- Crit Damage +50%
- Crits inflict Bleed

This matters because it connects crit with the **Bleed / Laceration / debuff family**, not just raw crit.

SS / advanced gear sources

The image shows additional SS-style gear crit sources:

- 10% Crit Rate
- 20% Crit Rate
- AF3: +3% extra Crit Rate per boss / elite kill
- 30% Crit Rate
- V4: +20% Crit Rate during laser charge

Optimizer notes

Gear crit should **not** be treated as permanent global. It only counts if the item is equipped and the condition is active.

Database flags:

SourceCategory = Gear

Slot = Gloves / Necklace / SSGear

TargetScope = CurrentBuildOnly

ConditionType = Equipped / BattleConditional

DPSRelevant = High

Example row:

Eternal_Gloves | Gear | Gloves | Red | CritRate | +10% | CurrentBuild | Equipped | Yes | Also gives CritDamage +100%

Example row:

SS_Gear_Laser_Charge | Gear | UnknownSSSlot | V4 | CritRate | +20% | CurrentBuild | During laser charge | Yes | Temporary burst condition

3. Skill Crit Rate

This is associated with:

In-Battle Skills | Survivor Skill | Stance | Temporary DPS

The image shows skill-based crit rate sources:

- King-related skill source: Crit Rate +40%
- King-related evolved / upgraded source: Crit Rate +50%
- Yang in Palm stance: Crit Rate +20%

Optimizer notes

These should be stored as **in-run conditional sources**, not account stats.

Important: Yang's Palm stance crit should only count when the optimizer is modeling **Master Yang in Palm stance**. Do not apply it to every survivor.

Database flags:

SourceCategory = Skill

TargetScope = ActiveRunOnly

ConditionType = SkillSelected / StanceActive

DPSRelevant = High

Example row:

Yang_Palm_Stance | Skill | MasterYang | CritRate | +20% | ActiveSurvivorOnly | Palm stance active | Yes

4. Collectible Crit Rate

This is associated with:

Collectibles | Red-star thresholds | Global DPS

The source shows two collectible groups for Crit Rate:

Red collectible group

- 10 collectibles
- Each gives Crit Rate +10%
- Total shown: **100% global Crit Rate**

Yellow collectible group

- 11 collectibles
- Each gives Crit Rate +5%
- Total shown: **55% global Crit Rate**

Optimizer notes

This is extremely important later-game account scaling.

For now, because the image uses icons, the optimizer should store each collectible by icon/source until we map the exact collectible names from the collectible edition images.

Database flags:

SourceCategory = Collectible

TargetScope = Global

ConditionType = OwnedCollectibleStarThreshold

DPSRelevant = Very High

Example row:

Collectible_CR_Icon_01 | Collectible | UnknownNamePending | CritRate | +10% | Global |
Requires listed star threshold | Yes | Map exact name from Collectibles Edition source

5. Master Yang awakening Crit Rate

This is associated with:

Master Yang | Awakening Red Stars | Active Survivor DPS

The source shows Yang / panda crit rate from red awakening stars:

- 1 red star: Crit Rate +5%
- 4 red stars: additional +10%
- 6 red stars: additional +15%
- Total: **30% Crit Rate**

Optimizer notes

This should probably be treated as **Master Yang active-survivor value**, not global, unless another source confirms it is global.

Database flags:

SourceCategory = SurvivorAwakening
Survivor = MasterYang
StatType = CritRate
TargetScope = ActiveSurvivorLikely
ConditionType = AwakeningLevel
DPSRelevant = High

6. Talent Crit Rate

This is associated with:

Talents | Expose Weakness | Global Account Stat

The source shows:

- Talent: **Expose Weakness**
- Level 40
- Crit Rate +8%
- Marked as global

Optimizer notes

This is a clean global stat.

Database row:

Expose_Weakness_Lv40 | Talent | CritRate | +8% | Global | Permanent | Yes

7. Pet Crit Rate

This is associated with:

Pet Skill | Motivation | Pet / build-dependent DPS

The source shows:

- Pet skill: **Motivation**
- Crit Rate +3% / +4% / +5%

The value likely depends on the pet skill level / rarity.

Optimizer notes

This should only count when the pet has the skill equipped/active.

Database flags:

SourceCategory = PetSkill

PetSkill = Motivation

StatType = CritRate

Value = 3/4/5%

TargetScope = CurrentPetBuild

ConditionType = EquippedPetSkill

DPSRelevant = Medium/High

8. Donatello Crit Rate passive

This is associated with:

Donatello | Brotherhood | TMNT Passive Count | Global / Loadout Scaling

The source shows:

- Donatello 1 red star
- Crit Rate: 3% / 6% / 9%
- Text: **3% per equipped TMNT passive**

This connects directly to the previous SP survivor source.

Optimizer notes

This is one of the most important synergy stats because it scales based on how many TMNT/SP passives are equipped.

Database flags:

SourceCategory = SurvivorPassive
Survivor = Donatello
Passive = Brotherhood
StatType = CritRate
ScalingInput = EquippedTMNTPassiveCount
ValuePerPassive = +3%
MaxShown = +9%
ConditionType = DonatelloAwakeningUnlocked + TMNTPassivesEquipped
DPSRelevant = Very High

Example row:

Donatello_Brotherhood_CR | SurvivorPassive | Donatello | CritRate | +3% per equipped TMNT passive | LoadoutScaling | Requires Don A1 | Yes

9. Combat Harmony Crit Rate

This is associated with:

Combat Harmony | Star Level | Global Account Stat

The source shows:

- Combat Harmony level 1: Crit Rate +2%
- Level 3: additional +3%
- Level 6: additional +5%
- Level 9: additional +5%
- Level 12: additional +5%
- Level 15: additional +5%
- Level 18: additional +5%

Total shown:

30% global Crit Rate

Optimizer notes

This is global and should have high priority in the account stat model.

Database row:

Combat_Harmony_CR | CombatHarmony | CritRate | +30% total | Global | Based on harmony level | Yes

Crit Damage section

1. Default Crit Damage

The source shows:

Default Crit Damage = 100%

This should be stored as the baseline, not as an upgrade.

Important: do not accidentally double count this as a bonus source.

Database row:

Default_Crit_Damage | BaseStat | CritDamage | 100% | AllBuilds | Baseline | Yes | Do not treat as upgrade

2. Survivor level Crit Damage

This is associated with:

Survivor Level | Global Crit Damage | Account DPS

The source shows **60% total global Crit Damage** from survivor level bonuses.

Visible sources include:

- One survivor at level 40: Crit Damage +5%
- King level 120: Crit Damage +5%
- Tsukuyomi level 120: Crit Damage +5%
- Worm level 120: Crit Damage +5%

- Metallia / red-haired survivor level 120: Crit Damage +10%
- Catnips / cat-like survivor level 120: Crit Damage +10%
- SpongeBob level 40: Crit Damage +5%
- Squidward level 40: Crit Damage +5%
- Patrick level 40: Crit Damage +5%
- Sandy level 40: Crit Damage +5%

Optimizer notes

This is permanent account DPS scaling.

Database flags:

SourceCategory = SurvivorLevel
 StatType = CritDamage
 TargetScope = AllSurvivors
 ConditionType = PermanentGlobal
 DPSRelevant = Very High

3. Gear Crit Damage

This is associated with:

Equipped Gear | Gloves / SS gear / Necklace | Conditional DPS

The source shows several gear Crit Damage sources:

- V2 gear source: Crit Damage +36%
- V4 gear source: Crit Damage +60%
- Leather Gloves: Crit Damage +50%
- Eternal Gloves red: Crit Damage +100%
- Gear + collectible condition: Crit Damage +20%
- SS necklace / energy shield condition:
 - E5: 0–50%, listed as 10% per energy level
 - E1: if Crit Rate >100%, Crit Damage +30%
 - E5: if Crit Rate >150%, Crit Damage +100%

Optimizer notes

This is where crit math becomes conditional. The optimizer needs to know the player's **Crit Rate** before applying some Crit Damage bonuses.

Especially important:

If Crit Rate >100%, add +30% Crit Damage

If Crit Rate >150%, add +100% Crit Damage

That means Crit Rate can unlock more Crit Damage. This makes Crit Rate more valuable than it looks.

Database flags:

SourceCategory = Gear

StatType = CritDamage

ConditionType = Equipped / Threshold / EnergyLevel

DependsOn = CritRate / EnergyLevel / GearRarity

DPSRelevant = Very High

Example row:

SS_Necklace_CR_Threshold_E1 | Gear | Necklace | CritDamage | +30% | CurrentBuild |
Requires CritRate >100% | Yes

Example row:

SS_Necklace_CR_Threshold_E5 | Gear | Necklace | CritDamage | +100% | CurrentBuild |
Requires CritRate >150% | Yes

4. Skill Crit Damage

This is associated with:

Survivor Skill | Awakening Stars | Active Survivor DPS

The source shows:

Metallia / red-haired survivor skill

- 1 red star: Crit Damage +20%
- 3 red stars: additional +20%
- 5 red stars: additional +20%

Total shown:

+60% Crit Damage

King evolved skill

- King-related evolved skill: Crit Damage +25%

Optimizer notes

These are not global. They depend on active survivor and skill state.

Database flags:

SourceCategory = Skill

StatType = CritDamage

TargetScope = ActiveRunOnly

ConditionType = ActiveSurvivor / SkillUnlocked / AwakeningLevel

DPSRelevant = High

5. Collectible Crit Damage

This is associated with:

[Collectibles](#) | [Red-star thresholds](#) | [Global DPS](#)

The source shows two collectible groups:

Red collectible group

- 10 collectibles
- Each gives Crit Damage +10%
- Total: **100% global Crit Damage**

Yellow collectible group

- 13 collectibles
- Each gives Crit Damage +5%
- Total: **65% global Crit Damage**

Optimizer notes

This should be connected to the collectible database later. For now, store by icon ID and source image.

Database flags:

SourceCategory = Collectible

StatType = CritDamage

TargetScope = Global

ConditionType = OwnedCollectibleStarThreshold

DPSRelevant = Very High

6. Master Yang awakening Crit Damage

This is associated with:

Master Yang | Awakening Red Stars | Active Survivor DPS

The source shows:

- 1 red star: Crit Damage +5%
- 4 red stars: additional +10%
- 6 red stars: additional +25%

Total shown:

40% Crit Damage

Optimizer notes

Treat this as Yang-specific unless another source confirms it is global.

Database flags:

SourceCategory = SurvivorAwakening

Survivor = MasterYang

StatType = CritDamage

ValueTotal = +40%

TargetScope = ActiveSurvivorLikely

ConditionType = AwakeningLevel

DPSRelevant = High

7. Survivor Synergy Crit Damage

This is associated with:

Survivor Synergy | Global Account Stat | Crit Damage

The source shows:

- Synergy level 5: Crit Damage +5%
- Synergy level 10: additional +5%
- Synergy level 15: additional +10%
- Synergy level 20: additional +10%

Total:

30% global Crit Damage

Optimizer notes

This is clean global DPS.

Database row:

Survivor_Synergy_CD | SurvivorSynergy | CritDamage | +30% total | Global | Based on synergy level | Yes

8. Pet Crit Damage

This is associated with:

Pet | Rex | Inspiration | Active Pet DPS

The source shows two pet-related Crit Damage sources.

Rex

- Rex at 3 yellow stars: Crit Damage +5%
- Rex at red stars: Crit Damage +10%
- Note: **Rex must be active**

This means Rex is not just a passive account source. It only counts when Rex is deployed.

Inspiration

- Pet skill: **Inspiration**
- Crit Damage +6% / +10%

Optimizer notes

Pet crit damage should only be counted if the correct pet/skill is active.

Database flags:

SourceCategory = Pet

Pet = Rex

StatType = CritDamage

ConditionType = ActivePetRequired

DPSRelevant = Medium/High

Example row:

Rex_Active_CD | Pet | Rex | CritDamage | +5%/+10% | CurrentPetBuild | Rex must be active | Yes

Example row:

PetSkill_Inspiration | PetSkill | Inspiration | CritDamage | +6%/+10% | CurrentPetBuild | Skill equipped | Yes

9. Donatello Crit Damage passive

This is associated with:

[Donatello | Brotherhood | TMNT Passive Count | Loadout Scaling](#)

The source shows:

- Donatello 2 red stars
- Crit Damage: 3% / 6% / 9%
- Text: **3% per equipped TMNT passive**

This connects with Donatello's Brotherhood passive from the SP survivor source.

Database flags:

SourceCategory = SurvivorPassive

Survivor = Donatello

Passive = Brotherhood
StatType = CritDamage
ScalingInput = EquippedTMNTPassiveCount
ValuePerPassive = +3%
MaxShown = +9%
ConditionType = DonatelloAwakeningUnlocked + TMNTPassivesEquipped
DPSRelevant = Very High

Most important optimizer takeaways

1. Crit Rate is not just a damage stat — it unlocks Crit Damage bonuses

Because some gear gives extra Crit Damage only when Crit Rate is above a threshold:

Crit Rate >100% = extra Crit Damage

Crit Rate >150% = much bigger extra Crit Damage

So the calculator should give extra value to Crit Rate when it helps cross those breakpoints.

2. Global crit sources are the best long-term investments

Highest-value global sources in this image:

Survivor Levels
Collectibles
Combat Harmony
Talents
Survivor Synergy
TMNT 5-star bonuses

These help every build and every mode.

3. Yang, King, Metallia, Donatello, and pets should be conditional

These should not be blindly added to all builds.

Use conditions like:

ActiveSurvivor = MasterYang

ActiveSurvivor = King

ActiveSurvivor = Metallia

DonatelloPassiveUnlocked = True

EquippedTMNTPassiveCount = 0/1/2/3

ActivePet = Rex

PetSkill = Inspiration or Motivation

Clean database format for this data type

Use this table structure:

CritSourceID

StatType

SourceCategory

SourceName

AssociatedCharacter

AssociatedItem

AssociatedPet

AssociatedSkill

UnlockRequirement

Value

ValueType

TargetScope

ConditionType

IsGlobal

IsConditional

DPSRelevant

StackingGroup

Notes

SourceImage

SourceDate

Example rows:

TMNT_Raphael_5Star_CR | CritRate | SurvivorStar | Raphael | None | None | None | 5 yellow stars | 8% | FlatPercent | AllSurvivors | Permanent | Yes | No | VeryHigh | GlobalCritRate | Special event survivor | Crit_Rate_And_Damage_2025-03-09

Donatello_Brotherhood_CR | CritRate | SurvivorPassive | Donatello | None | None | Brotherhood
| Don A1 + equipped TMNT passives | 3% per passive | ScalingPercent | CurrentLoadout |
PassiveCount | Maybe | Yes | VeryHigh | LoadoutCritRate | Max shown 9% |
Crit_Rate_And_Damage_2025-03-09
SS_Necklace_CR_Threshold_CD | CritDamage | Gear | None | SS Necklace | None | None |
Crit Rate >150% | 100% | FlatPercent | CurrentBuild | Threshold | No | Yes | VeryHigh |
ConditionalCritDamage | Requires crit breakpoint | Crit_Rate_And_Damage_2025-03-09
Rex_Active_CD | CritDamage | Pet | None | None | Rex | None | Rex active | 5%/10% |
FlatPercent | CurrentPetBuild | ActivePet | No | Yes | MediumHigh | PetCritDamage | Rex must
be active | Crit_Rate_And_Damage_2025-03-09

Formula logic for the optimizer

Store the pieces separately first:

TotalCritRate =
GlobalCritRate
+ GearCritRate_Equipped
+ SkillCritRate_Active
+ PetCritRate_Active
+ SurvivorPassiveCritRate_Active
+ ConditionalCritRate_Active
TotalCritDamage =
DefaultCritDamage
+ GlobalCritDamage
+ GearCritDamage_Equipped
+ SkillCritDamage_Active
+ PetCritDamage_Active
+ SurvivorPassiveCritDamage_Active
+ ConditionalCritDamage_Active

Then calculate DPS after checking conditions:

If TotalCritRate > 100:
 apply CritDamage threshold bonuses that require CR > 100

If TotalCritRate > 150:
 apply CritDamage threshold bonuses that require CR > 150

Do **not** hardcode the final damage formula yet until we verify Survivor.io's exact crit math. But for ranking choices, this source clearly says Crit Rate and Crit Damage are major DPS multipliers and should be weighted very heavily.

Data type 3: Pets, Pet Skills, Pet Awakening

Sources used here:

Pets_2024-11-28
Pet_Skills_2024-09-09
Pet_Awakening_Guide_2024-10-25
Pet_Awakening_Costs_and_Toy_Bonuses_2024-12-04
Pet_Awakening_Guide_-_Murica_2024-10-10
Pet_Awakening_Guide_-_Croaky_2024-10-11
Pet_Awakening_Guide_-_Rex_2024-10-11

This data type belongs in the optimizer as:

Pet | Pet Skill | Pet Awakening | Assist Skill | Pet Toy | Pet Affection | Pet DPS | Owner DPS Buff
| Pet Crit | Pet Cooldown | Pet Inherited ATK | Pet Utility

For your calculator, pets matter in **two different ways**:

1. **Direct pet damage** — Murica/Eagle whirlwind, Crab bullets, Robot lasers, etc.
2. **Owner DPS buffs** — Crit Rate, Crit Damage, Skill Damage, cooldown, pet skill damage, inherited ATK.

HP, damage reduction, revival time, and invincibility should be stored, but mostly marked as **ignore for DPS** unless later you make a survival-mode version.

Main pet structure

Every pet has:

Primary Attack
Fixed pet skill pool
Awakening/star progression
Assist skill compatibility
Disposition
Toy/affection bonus

The optimizer should not treat all pets equally. A pet is valuable based on:

PetBaseAttack
PetAttackCooldown
PetSkillDamage
PetCooldownReduction
PetCritRate
PetCritDamage
PetInheritedATK
OwnerCritRateBuff
OwnerCritDamageBuff
OwnerSkillDamageBuff
AssistSkillSetup
AwakeningLevel

Pet list from **Pets_2024-11-28**

1. Frog / Croaky

Main association

Frog | Croaky | Corrosive Saliva | Poison/DOT style damage | Owner Skill Damage | Crit Rate | Crit Damage

Primary attack

Frog spits corrosive saliva that deals:

200% ATK damage every 1 second for 2 seconds
Cooldown: 1 second

This is a damage-over-time style attack.

Built-in skill themes

Frog/Croaky is associated with:

Owner Skill Damage
Pet inherited HP
Battle EXP gain
Owner Crit Rate
Owner Crit Damage

Visible pet effects include:

Owner Skill Damage +3%

Pet inherited HP +10%

Owner Battle EXP gain +7.5%

Owner Crit Rate +4%

Owner Crit Damage +10%

Optimizer notes

Croaky is not just a pet attack pet. It also has **owner DPS buff value** because it can connect to Skill Damage, Crit Rate, and Crit Damage.

The pet awakening guide says **Rex or Croaky should be taken to R5 first**, because both are close in total damage.

Database flags:

Pet = Croaky

PrimaryAttack = Corrosive Saliva

DamageType = DOT / Poison-like

OwnerSkillDamageSource = Yes

OwnerCritRateSource = Yes

OwnerCritDamageSource = Yes

PetDirectDamage = Medium/High

DPSRelevant = High

2. Fish

Main association

Fish | Explosive Bubbles | Explosion Damage | Battle EXP | Time Stop

Primary attack

Fish spits explosive bubbles:

Bubble hit damage: 200% ATK

Explosion damage: 200% ATK

Cooldown: 1 second

Built-in skill themes

Fish is associated with:

Battle EXP gain
Collision damage reduction
Shot damage reduction
Pet size/range
Time Stop

The red skill shown is:

When owner's HP is below 30%, stops time and freezes all enemies for 5 seconds.
During this, player and pet are invincible.
Cooldown: 60 seconds.

Optimizer notes

Fish has some damage from explosion, but a lot of its value is **utility/survival**, not pure DPS.

Database flags:

Pet = Fish
PrimaryAttack = Explosive Bubbles
DamageType = Explosion / AoE
Utility = Time Stop
DPSRelevant = Medium
IgnoreForDPS = Time Stop, damage reduction, survival effects

3. Dog / Rex

Main association

Dog | Rex | Yell Attack | Owner Crit Rate | Owner Crit Damage | Damage Transfer | Pet HP Absorb

Primary attack

Dog yells at enemies with loud noises:

250% ATK damage

Cooldown: 1 second

Built-in skill themes

Dog/Rex is associated with:

Owner Crit Rate
Owner Crit Damage
Pet movement speed
Pet HP absorb
Damage transfer

Visible pet effects include:

Owner Crit Rate +3%
Owner Crit Damage +6%
Pet movement speed +75%
Pet HP absorb from attacks +2%
Takes 30% damage in place of owner

Rex awakening guide

The Rex guide recommends assist skills:

Raid*
Strong
Anything

Disposition:

Hot-blooded
Battler

Rex shares the same key owner-buff skills as Croaky:

Battle Lust
Diligence
Encouragement
Inspiration
Motivation
Vampirism

Optimizer notes

Rex matters because it can support **owner Crit Rate and Crit Damage**, and the awakening cost guide specifically says Rex or Croaky should be pushed to R5 first.

Database flags:

Pet = Rex
PrimaryAttack = Yell
OwnerCritRateSource = Yes
OwnerCritDamageSource = Yes
SurvivalUtility = Damage transfer
DPSRelevant = High

4. Turtle

Main association

Turtle | Shockwave | Damage Reduction | Revival | Tank Pet

Primary attack

Turtle leaps up and smashes the ground:

Shockwave damage: 300% ATK
Cooldown: 1 second

Built-in skill themes

Turtle is associated with:

Shot damage reduction
Collision damage reduction
Pet revival time reduction
Pet inherited HP
Pet damage taken reduction

Visible effects include:

Owner Shot Damage Reduction +10%
Owner Collision Damage Reduction +10%
Pet revival time -30%
Pet inherited HP +15%

Pet damage taken -60%

Optimizer notes

Turtle is mostly a **survival/tank pet**. For your damage-only calculator, most of this gets stored but not scored.

Database flags:

Pet = Turtle

PrimaryAttack = Shockwave

DamageType = AoE

TankUtility = High

DPSRelevant = Low/Medium

IgnoreForDPS = Most defensive effects

5. Eagle / Murica

Main association

Eagle | Murica | Whirlwind | Penetrating Damage | Cooldown Reduction | Crush | Pet Skill Damage

Primary attack

Eagle flaps wings to create a whirlwind:

350% ATK damage

Whirlwind penetrates all enemies

Cooldown: 1 second

This is why Murica is so strong early/mid game. Penetrating attacks are very valuable when they hit multiple enemies or bosses consistently.

Built-in skill themes

Murica is associated with:

Pet size/range

Pet ATK damage

Pet cooldown reduction
Pet movement speed
Crush / 10x damage chance

Visible effects include:

Pet becomes bigger
Pet ATK and skill range +15%
Pet ATK damage +15%
Pet ATK and skill cooldown -20%
Pet movement speed +75%
Attacks have 5% chance to deal 10x damage to enemies below 50% HP

Murica awakening guide

The guide recommends assist skills:

Raid*
Powerful
Boost
Penetration

Disposition:

Shrewd
Hot-blooded
Battler

Important note from the source:

Replace Motivation with Inspiration if high Crit Rate %
After Y5, look at switching to Rex or Croaky.
Otherwise, use the Pet Optimizer.

Optimizer notes

Murica should be modeled as a **direct pet DPS pet**.

Its strongest associations are:

Whirlwind
Penetration
Cooldown reduction
Skill damage
Crush

Pet ATK damage

Database flags:

Pet = Murica

PrimaryAttack = Whirlwind

DamageType = Penetrating

PetCooldownReduction = Yes

PetSkillDamage = Yes

Crush = Yes

DirectPetDPS = Very High

DPSRelevant = Very High

6. Cat

Main association

Cat | Claw Attack | Pet Crit Rate | Pet Cooldown | Nine Lives

Primary attack

Cat scratches enemies with sharp claws:

200% ATK damage

Cooldown: 0.5 seconds

This means it attacks faster than most pets.

Built-in skill themes

Cat is associated with:

Pet ATK and skill Crit Rate

Pet inherited HP

Pet cooldown reduction

Pet ATK damage

Fatal-damage protection

Visible effects include:

Pet ATK and skill Crit Rate +15%

Pet inherited HP +10%
Pet ATK and skill cooldown -20%
Pet ATK damage +20%
When player and pet take fatal damage, nullify that attack and heal allies for 50% HP
Cooldown: 100 seconds

Optimizer notes

Cat has real DPS value because of **fast attack speed, pet crit, pet cooldown, and pet ATK damage**. The fatal-damage protection is survival, so store it but ignore for DPS.

Database flags:

Pet = Cat
PrimaryAttack = Claws
AttackCooldown = 0.5s
PetCritRateSource = Yes
PetCooldownReduction = Yes
PetATKDamageSource = Yes
DPSRelevant = High

7. Crab

Main association

Crab | Water Bullets | Pet ATK Damage | Pet Crit Rate | Pet Inherited ATK | Pet Cooldown

Primary attack

Crab fires powerful water bullets:

500% ATK damage
Cooldown: 1 second

Built-in skill themes

Crab is associated with:

Pet HP absorb
Pet ATK damage

Pet ATK and skill Crit Rate
Pet inherited ATK
Pet cooldown reduction

Visible effects include:

Pet HP absorb from attacks +1%
Pet ATK damage +15%
Pet ATK and skill Crit Rate +20%
Pet inherited ATK +15%
Pet ATK and skill cooldown -30%

Optimizer notes

Crab is a clean **pet DPS scaling pet**. Unlike Turtle/Fish, most of Crab's good stats are directly damage-related.

Database flags:

Pet = Crab
PrimaryAttack = Water Bullets
PrimaryDamage = 500% ATK
PetCritRateSource = Yes
PetCooldownReduction = Yes
PetInheritedATKSource = Yes
PetATKDamageSource = Yes
DPSRelevant = High/Very High

Since you have cared about Crab before, this source confirms Crab is not random trash. It has a real DPS identity, especially through cooldown, crit rate, inherited ATK, and 500% primary attack.

8. Robot

Main association

Robot | Burning Lasers | Satellite Cannon | Pet Cooldown | Inherited ATK

Primary attack

Robot fires 3 burning lasers:

3 lasers

Each laser deals 400% ATK damage

Cooldown: 1 second

If all lasers connect, this can be very high theoretical damage.

Built-in skill themes

Robot is associated with:

Pet cooldown reduction

Pet revival time

Pet damage reduction

Pet inherited ATK

Satellite Cannon

Visible effects include:

Pet ATK and skill cooldown -15%

Pet revival time -20%

Pet damage taken -40%

Pet inherited ATK +15%

Satellite blast deals 1000% ATK damage

Cooldown: 5 seconds

+250% damage when owner levels up

Optimizer notes

Robot has high theoretical DPS because of lasers and Satellite Cannon, but the calculator needs to eventually model **hit consistency**. If the lasers miss or split badly, the real value drops.

Database flags:

Pet = Robot

PrimaryAttack = Burning Lasers

PrimaryDamage = 3 x 400% ATK

SpecialSkill = Satellite Cannon

SpecialDamage = 1000% ATK

Cooldown = 5s

LevelUpScaling = +250% damage when owner levels up

DPSRelevant = High but hit-dependent

Pet skill registry

Source: [Pet_Skills_2024-09-09](#) and [Pet_Awakening_Guide_2024-10-25](#)

This is the skill dictionary. These skills should be stored separately from pets because different pets unlock different skills at different rarity/star levels.

Owner / Survivor buff skills

These affect the player, not only the pet.

Barrier

Increases player's Range Damage Reduction by 10/15%

DPSRelevant = No

Diligence

Increases player's Battle EXP gain by 5/7.5/10%

DPSRelevant = Indirect only

Encouragement

Increases player's Skill Damage by 3/5%

DPSRelevant = Yes

Guard

Increases player's Collision Damage Reduction by 10%

DPSRelevant = No

Inspiration

Increases player's Crit Damage by 6/10%

DPSRelevant = Yes

Motivation

Increases player's Crit Rate by 3/4/5%

DPSRelevant = Yes

Optimizer notes

For your damage-only calculator, the big owner-buff pet skills are:

Encouragement = Skill Damage

Inspiration = Crit Damage

Motivation = Crit Rate

Diligence should be stored because faster leveling can matter in chapters, but it is not a direct boss DPS stat.

Pet DPS skills

These affect the pet's own damage.

Crush

Attacks have 5% chance to deal 10x damage

DPSRelevant = Very High

Battle Lust

Reduces pet cooldowns by 15/20/30%

DPSRelevant = Very High

Frenzy

Increases pet Crit Rate by 15/20/60%

DPSRelevant = Very High

Gargantua

Pet becomes bigger and range increases by 15/20%

DPSRelevant = Medium/High because it improves hit coverage

High Windchaser

Pet movement speed +75%

DPSRelevant = Low/Medium, mostly hit consistency

Painstrike

Increases pet Crit Damage by 60%

DPSRelevant = High

Sharp Claws

Increases pet Skill Damage by 15/20/60%

DPSRelevant = Very High

Tough Skin

Reduces pet damage taken by 40/60%

DPSRelevant = No

United Effort

Increases pet inherited HP by 10/15/35%

DPSRelevant = No

United Stand

Increases pet inherited ATK by 15/35%

DPSRelevant = High

Unyielding Will

Reduces pet revival time by 20/30/40%

DPSRelevant = Mostly no

Utmost Loyalty

Pet takes 30% damage in place of player
DPSRelevant = No
Vampirism
Increases pet HP absorb from attacks by 1/2%
DPSRelevant = No, unless survival model is added

Misc pet skills

Nine Lives
When player and pet take fatal damage, nullifies that attack and heals allies for 50% HP.
Cooldown: 100 seconds.
DPSRelevant = No
Satellite Cannon
Guides satellite to blast a shot at a certain area.
Damage: 1000% ATK.
Cooldown: 5 seconds.
+250% damage when owner levels up.
DPSRelevant = Yes
Time Stop
When player's HP is below 30%, stops time and freezes enemies for 5 seconds.
Player and pet are invincible during it.
Cooldown: 60 seconds.
DPSRelevant = No, but useful utility

Assist skills

Assist skills are separate from the deployed pet's normal skill set. These should be modeled as:

AssistPetSkill | AssistPet | DeployedPetBuff | InBattleScaling

The assist skills shown:

Boost
For every 5 owner levels, pet Range increases by 5/6/7.5/10%.
DPSRelevant = Medium/High
Durable
For every 5 owner levels, pet shot Duration increases by 8/10/12/15%.
DPSRelevant = Medium/High if pet projectile duration matters
Penetration
For every 10/8/6/5 owner levels, penetrated units of pet shot +1.

DPSRelevant = High for projectile/whirlwind pets

Powerful

For every 5 owner levels, pet Skill Damage increases by 5/6/7.5/10%.

DPSRelevant = High

Strong

For every 5 owner levels, pet Max HP increases by 10/12/15/20%.

DPSRelevant = No

Raid

For every 5 owner levels, pet Skill Damage increases by 6/7.5/10%.

Bonus trait: deployed pet cooldowns reduced by 10/15/20%.

DPSRelevant = Very High

Important note:

Raid is marked as Gary exclusive.

That means the optimizer needs an availability check:

If GaryAvailable = false:

RaidAssistSkill = unavailable

Recommended pet awakening setups

Murica

Source: [Pet_Awakening_Guide_-_Murica_2024-10-10](#)

Murica recommended assist skills:

Raid*

Powerful

Boost

Penetration

Disposition:

Shrewd

Hot-blooded

Battler

Important source note:

Replace Motivation with Inspiration if high Crit Rate %
After Y5, look at making the switch to Rex or Croaky.
Otherwise, use the Pet Optimizer.

Optimizer interpretation

Murica is great when the build needs:

Direct pet DPS
Whirlwind damage
Cooldown reduction
Skill damage scaling
Penetration/range scaling

But once your account has high crit and stronger pet awakenings, the guide says to consider Rex or Croaky after Y5.

Croaky

Source: [Pet_Awakening_Guide_-_Croaky_2024-10-11](#)

Croaky recommended assist skills:

Boost
Raid*
Strong

Disposition:

Shrewd
Hot-blooded
Battler

Important note:

Use a lower tier of Battle Lust for every tier of Raid used.

Optimizer interpretation

Croaky is strong because it mixes:

Owner Skill Damage

Owner Crit Rate
Owner Crit Damage
Pet damage scaling
Assist scaling

Croaky should be scored as a strong long-term pet, not just as an early pet.

Rex

Source: [Pet_Awakening_Guide_-_Rex_2024-10-11](#)

Rex recommended assist skills:

Raid*
Strong
Anything

Disposition:

Hot-blooded
Battler

Optimizer interpretation

Rex is valuable because it has:

Owner Crit Rate
Owner Crit Damage
Damage transfer utility
Pet HP absorb

The pet cost guide also says Rex or Croaky should be taken to R5 first because they are close in total damage.

Pet awakening costs and affection bonuses

Source: [Pet_Awakening_Costs_and_Toy_Bonuses_2024-12-04](#)

This is a cost/progression table. It matters because the optimizer needs to decide where awakening crystals should go.

Total awakening crystals

Total crystals to full pet awakening path shown: 350

Awakening crystal costs by stage

Y1: +5 crystals
Y2: +5 crystals
Y3: +10 crystals
Y4: +20 crystals
Y5: +30 crystals
R1: +40 crystals
R2: +60 crystals
R3: +60 crystals
R4: +60 crystals
R5: +60 crystals
Total: 350 crystals

Deploy count shown

Y1: 3 deployed pets
Y2: 3 deployed pets
Y3: 5 deployed pets
Y4: 5 deployed pets
Y5: 7 deployed pets
R1: 7 deployed pets
R2: 9 deployed pets
R3: 9 deployed pets
R4: 9 deployed pets
R5: 15 deployed pets

Affection level caps

Base: Level 10
Y1: Level 20
Y2: Level 30
Y3: Level 40
Y4: Level 50
Y5: Level 60
R1: Level 70
R2: Level 80

R3: Level 90
R4: Level 105
R5: Level 130

Player ATK from affection

The table gives player ATK per affection level and max ATK.

Important max ATK values:

Base: 300 max ATK
Y1: 900 max ATK
Y2: 1800 max ATK
Y3: 2700 max ATK
Y4: 3600 max ATK
Y5: 4800 max ATK
R1: 6000 max ATK
R2: 7500 max ATK
R3: 9000 max ATK
R4: 11.3k max ATK
R5: 15k max ATK

Optimizer notes

This is very important because pet awakening is not only pet damage. It gives **player ATK** through affection.

For the calculator:

PetAffectionATK = Global/player ATK source
PetAwakeningCrystalCost = investment cost
PetAwakeningStage = unlock requirement

HP values should be stored but not scored for your DPS optimizer.

Preferred pet toys

The source groups toys by pet pair.

Dog + Cat
Robot + Frog

Fish + Eagle
Turtle + Crab

Toy stat values shown:

Blue toy: 3 ATK, 12 HP
Purple toy: 15 ATK, 60 HP
Yellow toy: 45 ATK, 180 HP

Important note:

You must level up affection before gaining the benefits.

Optimizer notes

Toy stats are small individually but permanent. Since your optimizer cares about damage:

ToyATK = count
ToyHP = store but ignore

Pet awakening investment advice from the source

The source gives a clear priority rule:

1. Take Rex or Croaky to R5 first. Both are very close in total damage.
2. After that, take each pet to Y5 with a balanced approach to get the most ATK per crystal.
Example: all pets to Y2, then all to Y3, etc.
3. Finally, take one pet from each gift type to R5.

It also says:

Since Y5 to R5 takes 280 crystals, the average for all 5 levels is about 36.5 crystals.
It is not worth rushing R1-R3 at worse rates when cheaper Y-level sources still exist.

Optimizer interpretation

This means the optimizer should not blindly push one pet from Y5 to R1/R2/R3 if the player has cheaper account-wide ATK available from other pets.

The calculator needs a value function like:

ATK gained per awakening crystal
DPS gained per awakening crystal
Unlock value from new skill slot
Direct pet damage gained
Owner DPS buff gained

Biggest DPS-relevant pet stats

For the optimizer, the pet system should prioritize:

Owner Crit Rate
Owner Crit Damage
Owner Skill Damage
Pet Skill Damage
Pet ATK Damage
Pet Crit Rate
Pet Crit Damage
Pet Cooldown Reduction
Pet Inherited ATK
Pet Range / Penetration
Pet Primary Attack Damage
Pet Affection ATK

Ignore or down-rank:

Pet HP
Owner HP
Damage reduction
Revival time
Invincibility
Damage transfer
Time stop

Store them anyway, but mark:

DPSRelevant = No

Clean database format for this data type

Use these tables.

Pet base table

PetID
PetName
BaseAnimal
PrimaryAttackName
PrimaryAttackDescription
PrimaryDamagePercent
PrimaryCooldown
DamageType
HitType
DPSRelevant
Notes
SourceImage
SourceDate

Example:

PET_MURICA | Murica | Eagle | Whirlwind | Creates penetrating whirlwind | 350% ATK | 1s |
Penetrating | Multi-hit/line | Yes | Strong direct pet DPS | Pets_2024-11-28

Example:

PET_CRAB | Crab | Crab | Water Bullets | Fires powerful water bullets | 500% ATK | 1s |
Projectile | Bullet | Yes | Strong pet DPS scaling from crit/CD/ATK | Pets_2024-11-28

Pet skill table

PetSkillID
SkillName
SkillCategory
AffectedTarget
AffectedStat
ValuePurple
ValueYellow
ValueRed
Condition
DPSRelevant
Notes
SourceImage
SourceDate

Example:

PETSKILL_MOTIVATION | Motivation | SurvivorBuff | Owner | CritRate | 3% | 4% | 5% | Pet skill unlocked/equipped | Yes | Owner DPS crit source | Pet_Skills_2024-09-09

Example:

PETSKILL_BATTLE_LUST | Battle Lust | PetBuff | Pet | CooldownReduction | 15% | 20% | 30% | Pet skill unlocked/equipped | Yes | Very high pet DPS scaling | Pet_Skills_2024-09-09

Example:

PETSKILL_CRUSH | Crush | PetBuff | Pet | DamageProc | None | None | 5% chance 10x damage | Pet attacks | Yes | Strong burst pet DPS | Pet_Skills_2024-09-09

Pet awakening cost table

AwakeningStage
CrystalCost
TotalCrystalCost
DeployCountRequired
SacrificeRequirement
AffectionLevelCap
PlayerATKPerLevel
MaxPlayerATK
HPPerLevel
MaxHP
DPSRelevant
SourceImage
SourceDate

Example:

Y5 | 30 | 70 total to Y5 | 7 | Red pet sacrifice | 60 | +120 | 4800 | +480 | 28.8k | Yes for ATK, no for HP | Pet_Awakening_Costs_and_Toy_Bonuses_2024-12-04

Example:

R5 | 60 | 350 total | 15 | Red pet sacrifice | 130 | +150 | 15k | +600 | 78k | Yes for ATK, no for HP | Pet_Awakening_Costs_and_Toy_Bonuses_2024-12-04

Pet recommendation table

PetID
RecommendedStage

RecommendedAssist1
RecommendedAssist2
RecommendedAssist3
RecommendedAssist4
PreferredDisposition1
PreferredDisposition2
PreferredDisposition3
SpecialNote
DPSPriority
SourceImage
SourceDate

Example:

PET_MURICA | Y5+ | Raid | Powerful | Boost | Penetration | Shrewd | Hot-blooded | Battler |
Replace Motivation with Inspiration if high Crit Rate; after Y5 consider Rex/Croaky | Very High
early/mid | Pet_Awakening_Guide_-_Murica_2024-10-10

Example:

PET_CROAKY | R5 target | Boost | Raid | Strong | None | Shrewd | Hot-blooded | Battler | Use
lower tier Battle Lust for each Raid tier used | Very High long-term |
Pet_Awakening_Guide_-_Croaky_2024-10-11

Example:

PET_REX | R5 target | Raid | Strong | Anything | None | Hot-blooded | Battler | None | Rex or
Croaky to R5 first; close total damage | Very High long-term |
Pet_Awakening_Guide_-_Rex_2024-10-11

Formula logic for pet DPS

Keep pet DPS separate first:

PetDPS =
PetBaseAttackDamage
* PetAttackFrequency
* PetATKMultiplier
* PetSkillDamageMultiplier
* PetCritMultiplier
* PetCooldownMultiplier
* HitConsistencyMultiplier

Then owner DPS buffs from pets:

OwnerDPSFromPetBuffs =
OwnerCritRateFromPet
+ OwnerCritDamageFromPet
+ OwnerSkillDamageFromPet
+ PlayerATKFromPetAffection

Do not mix these too early. A pet can be good because it does direct damage, or because it buffs the owner.

Important conclusion for the optimizer

The pet system is not “just pick Murica forever.”

The source data says:

Murica = strong direct pet DPS, especially through whirlwind, cooldown, Crush, and assist scaling.

Croaky = strong long-term owner/pet DPS mix.

Rex = strong long-term owner crit support and total damage contender.

Crab = real pet DPS scaling through 500% attack, crit rate, cooldown, and inherited ATK.

Robot = high theoretical laser/satellite damage but hit consistency matters.

Turtle/Fish = more utility/survival than damage.

For your damage-only optimizer, the highest-value pet stats are:

Battle Lust
Raid
Powerful
Sharp Claws
Frenzy
Crush
Motivation
Inspiration
Encouragement
United Stand
Pet Affection ATK

Data type 4: SS Gear / Relic Core Gear Pathing

Source: [SS_Gear_2025-02-06](#)

Game version shown: **3.6**

Updated: **2025-02-06**

This image is not just “equipment.” It is a full **SS gear upgrade map** showing:

- SS Gear Slot
- Base rarity effects
- Eternal forge path
- Void forge path
- Chaos forge path
- Chaos Fusion Power
- Collectible set bonuses tied to SS gear
- Relic/core/fodder cost rows
- Damage/debuff/crit/shield associations

For the optimizer, this should become:

SSGear | Slot | ForgePath | ForgeLevel | EffectName | StatAffected | Condition | DPSRelevant | Cost | AssociatedDebuff | AssociatedCollectibleSet

Important note: some official item names are not written clearly in the table, so I’m labeling them by **slot + effect name** when needed. That is safer than inventing a wrong item name.

SS gear slots shown

The image has 6 SS gear columns:

- SS Weapon
- SS Necklace
- SS Gloves
- SS Armor
- SS Belt
- SS Boots

DPS value by slot

For your damage-only optimizer:

SS Weapon = Very High DPS

SS Gloves = Very High DPS

SS Belt = Very High DPS

SS Armor = High DPS

SS Boots = High but condition-based DPS

SS Necklace = Medium/High, mostly skill/shield conditional

HP effects should be stored but mostly ignored for DPS.

1. SS Weapon

Main association

SS Weapon | Starforged Havoc | Havoc Nova | Eternity Ray | Void Fissure | Void Power | Energy Overload | Chaos Meteor | Chilled/Weakened damage

This is one of the most important DPS systems because it combines:

Skill damage

Attack speed/interval changes

Void Fissure damage taken debuff

Energy Overload stacking

Chaos path max-level increases

Chilled target damage

Weakened target damage

Boss lock-on behavior

Base / rarity effects

ATK +15%

Starforge:

Triggers Energy Overload 1 time for every 2 Eternity Ray attacks.

Increases own skill damage by 30% for 10 seconds.

Max 2 stacks.

Starforged Havoc:

Higher thrust damage with lower HP.

ATK +25%

Havoc Nova:

Releases Void power during a thrust.

Targets hit by Void Fissure take an additional 10% damage.

Havoc Nova:

Outermost rim of Eternity Ray has suppression force.

Eternal forge path

E1:

Energy Overload count also recorded when Eternity Ray retracts.

Starforge and Havoc start at Lv.3.

E2:

ATK +10%

E3:

Super Star:

Each Energy Overload stack's damage increased to 35%.

E4:

Final ATK +4000

E5:

Jade Cluster:

Energy Overload stacks up to 3 times.

Void forge path

V1:

ATK +10%

V2:

Disordered Death:

Void power damage +80%.

Targets hit by Void Fissure take an additional 20% damage.

V3:

Final ATK +4000

V4:

Blink:

Thrusts performed at low HP reduce base attack intervals by up to 1 second.

V5:

Final ATK +8000

Chaos forge path

This source shows Chaos path only under the weapon.

C1:

Max Level +5

C2:

Meteorite:

When Void Power is fired, additional homing Chaos Meteor fires are launched.

Inflicts 10% Vulnerability on hit for 10 seconds.

Does not stack.

C3:

Max Level +5

C4:

Damage to Weakened targets +20%

C5:

Max Level +5

C6:

Shatterstar:

When Chaos Meteorite hits enemies, Chaos Shatterstar is triggered.

C7:

Max Level +5

C8:

Damage to Chilled targets +30%

C9:

Max Level +10

C10:

Warp:

After Void Power hits a boss, it locks onto the boss aura and homes back in after flying out to deal damage again.

Void Power base damage -30%.

Chaos Fusion Power

3x:

Eternally Immortal:

Thrusts no longer depend on HP.

All effects trigger at full HP.

Targets hit by Void Fissure take +10% additional damage.

6x:

Infinity Void:

Void Power additionally gains 30% of Overload's damage bonus.

Targets hit by Void Fissure take +20% additional damage.

9x:

Chaos Wind:

Removes Eternity Ray and fuses its energy into Void Power, transforming it into Chaos Wind.

Post-fusion Chaos Wind triggers Energy Overload.

Each Overload stack's damage boost effect +25%.

Optimizer notes

This is a **core DPS item**. The calculator should track:

EnergyOverloadStacks
SkillDamageBonus
VoidFissureDamageTakenBonus
VoidPowerDamage
ChaosMeteorVulnerability
DamageToWeakened
DamageToChilled
BossLockOnBehavior
AttackIntervalReduction
MaxSkillLevelIncrease

Database flags:

Slot = Weapon

DPSRelevant = Very High

Debuffs = Vulnerability, Void Fissure damage taken, Weakened synergy, Chilled synergy

ModeValue = Very High for boss modes and general DPS

2. SS Necklace

Main association

SS Necklace | Skill Choice | Energy Shield | Clarity | Skill Damage | Reverse Scale | Shield HP

This is a mixed item. A lot of it is shield/HP, but the useful DPS parts are:

Extra skill choice

Skill damage

Shield-based skill damage

Clarity upgrade

Reverse Scale

Base / rarity effects

Skill choice chances when match starts +1

HP +15%

Adversity:

Gain an energy shield with 30% of Max HP when dying.

Once per battle.

Resets on revival.

HP +25%

Clarity:

Skill damage +20%.
Effect fades on death.
Preservation:
HP recovery through supplies also restores 10% of energy shield.
Shield recovery cannot exceed max HP.
Max 100% HP.

Eternal forge path

E1:
Halberd Shield:
Shield damage +10%.
Damage up when shield is active.
Taking damage lasts 10 seconds.
Energy shield slowly restores 2.5% HP every second.
Cannot exceed max shield HP.
E2:
HP +10%
E3:
Shield damage +10%.
Violent Backlash:
When own HP is higher than 10%, own HP is deducted first before shield HP when taking damage.
E4:
Final HP +16000
E5:
Omnicombat:
The higher the energy shield's HP, the higher the damage.
For every 25% shield HP, shield damage +5%.

Void forge path

V1:
HP +10%
V2:
Reverse Scale:
When shield HP is higher than own HP, skill damage +10%.
V3:
Final HP +16000
V4:
Clarity increases skill damage by +15%.
V5:
Final HP +32000

Optimizer notes

The necklace is not a simple raw DPS piece. It needs condition checks.

Track:

ExtraSkillChoice
SkillDamage
EnergyShieldPercent
ShieldHigherThanHP
DeathReset
ClarityActive
ShieldDamageBoost

Database flags:

Slot = Necklace
DPSRelevant = Medium/High
ConditionType = ShieldBased / DeathSensitive / SkillChoice
IgnoreForDPS = HP-only lines

3. SS Gloves

Main association

SS Gloves | Weaken | Damage to Weakened | Red Star passive upgrades | Berserk | Crit
Damage from passive levels

This is a very important DPS slot because it connects to **Weakened target damage** and **passive skill scaling**.

Base / rarity effects

Hatred:

On hit, inflicts Weaken on attacker, reducing its Attack by 10%.

ATK +15%

Defensive Stance:

Restores 15% of Energy Shield when skills level up.

ATK +25%

Potential Unleashed:

All passive skills learned in battle can be upgraded to Red Star, including character-exclusive passives.

Amplify:
+25% damage to Weakened targets.

Eternal forge path

E1:

Berserk:

+20 seconds Berserk when skill levels up.

+50% skill damage during Berserk.

Defeat monsters to extend Berserk duration.

Max 200 seconds.

E2:

ATK +10%

E3:

Wondermight:

The higher the shield value, the more damage dealt to Weakened targets.

For every 20% of Energy Shield value, +2% damage to Weakened targets.

E4:

Final HP +16000

E5:

Limit Break:

+40% Max Energy Shield.

Void forge path

V1:

ATK +10%

V2:

Stealth Attack:

+2% Crit Damage for every 2 levels of passive skills learned.

V3:

Final ATK +4000

V4:

Divine Might:

Initial passive skill level +1.

For each Red Star passive upgrade, Crit Damage +10%.

V5:

Final ATK +8000

Optimizer notes

This item should be scored very high when the build uses:

Weaken

High passive level count
Red Star passive upgrades
Crit Damage stacking
Skill Damage
Energy Shield

Database flags:

Slot = Gloves
DPSRelevant = Very High
Debuff = Weaken
ScalingInputs = PassiveSkillLevels, RedStarPassiveCount, EnergyShieldValue
StatTypes = SkillDamage, CritDamage, DamageToWeakened

4. SS Armor

Main association

SS Armor | Revival | Toughened Armor | Energy Orbs | Energy Level | Damage Multiplier | Skill Damage | Crit Damage

This armor has survival, but also real DPS through **energy level damage multipliers**.

Base / rarity effects

Revival Effect +35%
HP +15%
Toughened Armor:
Triggers Toughened Armor for 5 seconds when attacked.
60-second cooldown.
Immune to all damage during this period.
Collisions instakill minions.
HP +25%
Eternal Recharge:
Energy orbs drop every 50th kill.
Collect orbs to raise energy level, up to Lv.5.
+5% damage multiplier for each energy level.
Energy drains over time.
Super Enhance:
When total energy is above 90%, +30% additional skill damage.

Eternal forge path

E1:

Peerless:

-100 required hits for energy orb drops.

+5% additional damage multiplier for each energy level.

E2:

HP +10%

E3:

Suppression:

Able to trigger Super Enhance at Lv.4.

+5% additional damage multiplier for each energy level.

E4:

Final HP +16000

E5:

Ultimate Eternity:

-100 required hits for energy orb drops.

+10% additional Crit Damage for each energy level.

Void forge path

V1:

HP +10%

V2:

Cabalistic:

Energy level is maxed at the start.

Restore additional 2% of energy shield when looting energy orbs.

Shield damage boost +10%.

V3:

Final HP +16000

V4:

Destructive Restraint:

Energy drain speed halved.

Shield damage boost +20%.

V5:

Final HP +32000

Optimizer notes

This armor is not just tank gear. It creates a full **energy-level DPS system**.

Track:

EnergyLevel

EnergyOrbDropRequirement
DamageMultiplierPerEnergyLevel
SkillDamageWhenEnergyAbove90
CritDamagePerEnergyLevel
EnergyDrainRate
StartingEnergyLevel

Database flags:

Slot = Armor
DPSRelevant = High
ScalingInput = EnergyLevel
StatTypes = DamageMultiplier, SkillDamage, CritDamage
IgnoreForDPS = Revival/HP unless survival mode is added

5. SS Belt

Main association

SS Belt | Crit Rate ramp | Energy Shield restore | Eternal Laser | Crit Rate threshold | Crit Damage threshold | Skill Damage threshold

This is one of the most important optimizer items because it has **crit breakpoints**.

Base / rarity effects

Crit Rate +5% for every 500 effective attacks in battle.

Max 30%.

ATK +15%

0.5% of energy shield restored when a critical hit defeats a monster.

Max 5 times every second.

ATK +25%

Shield damage +10%

Light Judgment:

Fires 1 Eternal Laser every 10 seconds.

Marks 1 boss 10 seconds before firing and enters Charging Attack state.

During this time, 100% of total damage dealt to boss is converted into Charging Damage.

Charging Damage fires as a single laser when charge phase ends.

Charging damage is not affected by attack multiplier bonuses.

Eternal forge path

E1:

Mega Crit I:

If Crit Rate exceeds 100%, Crit Damage +30%.

E2:

ATK +10%

E3:

Mega Crit II:

If Crit Rate exceeds 130%, Skill Damage +30%.

E4:

Final ATK +4000

E5:

Mega Crit III:

If Crit Rate exceeds 150%, Crit Damage +100%.

Void forge path

V1:

ATK +10%

V2:

Revolution:

Shield damage boost +20%.

Restore 10% energy shield after laser is fired.

V3:

Final ATK +4000

V4:

Final Evolution:

During Eternal Laser charging period, Crit Rate +20%.

Stored Damage +30%.

V5:

Final ATK +8000

Optimizer notes

This belt should be a major DPS engine if the player can hit Crit Rate breakpoints.

Track:

EffectiveAttackCount

CritRateFromBelt

TotalCritRate

CritRateAbove100

CritRateAbove130

CritRateAbove150

EternalLaserChargeState

StoredDamage
BossMarked
ShieldRestoreOnCritKill

Database flags:

Slot = Belt
DPSRelevant = Very High
Breakpoints = 100% CR, 130% CR, 150% CR
StatTypes = CritRate, CritDamage, SkillDamage, StoredDamage
ModeValue = Very High for boss modes

This connects directly to the earlier **Crit Rate & Crit Damage** data type.

6. SS Boots

Main association

SS Boots | Movement Speed | Chilled enemies | Crystal Trail | Ice Blocks | Glacial Bloodline | Vulnerability | Damage stacks

The boots are very conditional. They are strong if the build can maintain chill/glacial stacks.

Base / rarity effects

Base Movement Speed +3
HP +15%
Crystal Trail:
Moving leaves a trail of ice blocks lasting 10 seconds.
Ignites key explosions when they expire.
Enemies touching blocks or hit by explosions are Chilled for 10 seconds.
Unstackable.
HP +25%
Damage to Chilled enemies +15%
Glacial Bloodline:
Move to start charging.
The faster the speed, the faster the charge.
Charge weakens when movement stops.
At full charge, gain 1 Glacial Bloodline stack.
Each stack gives +12% damage, lasting 15 seconds.
Max 5 stacks.

Repeat to extend duration.

Eternal forge path

E1:

Ice Armor:

2% Energy Shield.

+4% shield damage boost for each Glacial Bloodline stack.

E2:

HP +10%

E3:

Enhanced Ice Armor:

Shield damage boost +21%.

For every 25% shield HP gained, Movement Speed +10%.

E4:

Final HP +16000

E5:

Rime Blast:

Charge stacks +1.

Void forge path

V1:

HP +10%

V2:

Frost Blast:

Damage on Chilled enemies +20%.

V3:

Final HP +16000

V4:

Ice Arrow:

Damage on Chilled enemies +20%.

When crystal trail reaches 30 blocks in length, the trail immediately explodes, firing homing ice arrows.

Ice arrows inflict Chill and 10% Vulnerability for 10 seconds.

Unstackable.

V5:

Final HP +32000

Optimizer notes

Boots are good, but they are not always active. The calculator should score them based on whether the mode allows movement and whether chill uptime is realistic.

Track:

MovementSpeed
ChillUptime
DamageToChilled
GlacialBloodlineStacks
ChargeState
VulnerabilityFromIceArrow
ShieldDamageBoost
MovementRequired = True

Database flags:

Slot = Boots
DPSRelevant = High but conditional
Debuffs = Chill, Vulnerability
ConditionType = MovementBased / ChillBased / StackBased

Forge cost structure

The SS gear source also shows costs for each path.

The pattern is:

ForgePath
ForgeLevel
Relic/Core requirement
Faction core requirement
Fodder gear requirement

The image shows separate paths:

E1-E5 = Eternal path
V1-V5 = Void path
C1-C10 = Chaos path, shown for weapon

For the calculator, do not only store “effect.” Store **cost per upgrade** too, because relic cores are a major bottleneck.

Suggested fields:

ForgeCostID
GearSlot

ForgePath
ForgeLevel
RelicCoreCost
FactionCoreType
FactionCoreCost
FodderRarity
FodderCount
SourceImage

Important optimizer rule:

Relic cores should be spent where the next forge level gives the most DPS per relic core.

That means the recommender should compare:

DPSGain / RelicCoreCost
DPSGain / FactionCoreCost
DPSGain / FodderGearCost

Collectible sets tied to SS gear

At the bottom, the source shows SS gear collectible-set bonuses.

Erudite Heirloom

Associated mostly with **SS Weapon / Starforged Havoc**.

Each 3 yellow stars:

Starforged Havoc higher thrust damage with lower HP.

Max damage +20%.

Each 3 red stars:

Super Star E3:

Each Energy Overload stack's damage +10%.

25 stars total:

Disordered Death V2:

Targets take additional +15% damage.

Otherworld Treasure

Associated with **shield / Clarity / Reverse Scale**.

Each 3 yellow stars:
Omnicombat E5:
-5% required shield value to trigger each damage buff.
Each 3 red stars:
Clarity increases skill damage by +10%.
25 stars total:
Reverse Scale V2:
Triggered skill damage +20%.

Meaning of Life

Associated with **Fury / Weakened targets / Stealth Attack crit damage**.

Each 3 yellow stars:
When Fury is active, +10% additional skill damage.
Each 3 red stars:
+10% additional damage dealt to Weakened targets.
25 stars total:
Stealth Attack V2:
+1% additional Crit Damage for every 2 levels of passive skills learned.

Interdimension Movement

Associated with **SS Armor energy/shield system**.

Each 3 yellow stars:
Cabalistic shield damage boost +10%.
Each 3 red stars:
Suppression:
Super Enhance deals additional +5% Skill Damage every level.
25 stars total:
Eternal Recharge:
Each Energy level additionally adds +1% damage multiplier.

Optimizer notes

Collectibles should not be treated as separate random bonuses. They modify specific SS gear effects.

Database flags:

CollectibleSetBonus
AssociatedGearSlot

AssociatedForgeEffect
RequiredStars
BonusType
DPSRelevant

Most important DPS connections from this source

1. Crit Rate breakpoint system

The SS Belt creates major breakpoints:

Crit Rate >100% = Crit Damage +30%
Crit Rate >130% = Skill Damage +30%
Crit Rate >150% = Crit Damage +100%

This means Crit Rate is not linear. If an upgrade pushes the player over 100/130/150%, it becomes more valuable.

2. Debuff families involved

This SS gear source connects to multiple debuffs:

Weaken
Chill
Vulnerability
Void Fissure damage taken

That means this source must connect to your [Boss_Debuffs](#) table later.

3. Energy Shield / Energy Level are separate systems

Do not mix these up.

Energy Shield = shield HP system
Energy Level = armor/orb damage multiplier system

Energy Overload = weapon stack system

They sound similar, but they are different optimizer variables.

4. SS weapon, belt, gloves are probably the highest DPS modeling priority

For your calculator, the most urgent SS gear models are:

SS Weapon:

Energy Overload, Void Fissure, Chaos path, Void Power

SS Belt:

Crit Rate ramp, Crit thresholds, Eternal Laser

SS Gloves:

Weakened damage, Red Star passive upgrades, Crit Damage scaling

Armor and boots matter too, but they are more conditional.

Clean database format for this data type

SSGearEffectID

GearSlot

GearNameOrEffectFamily

ForgePath

ForgeLevel

EffectName

EffectDescription

AffectedStat

StatAmount

TargetScope

ConditionType

AssociatedDebuff

AssociatedMechanic

DPSRelevant

IgnoreForDPS

CostRelicCore

CostFactionCoreType

CostFactionCoreAmount

CostFodderType

CostFodderAmount

SourceImage

SourceDate

Example rows:

SS_WEAPON_E3_SUPER_STAR | Weapon | Starforged Havoc | Eternal | E3 | Super Star | Each Energy Overload stack damage increased to 35% | SkillDamage | Stack-based | CurrentBuild | EnergyOverloadActive | None | EnergyOverload | Yes | No | TBD | EternalCore | TBD | GearFodder | TBD | SS_Gear_2025-02-06

SS_BELT_E5_MEGA_CRIT_III | Belt | SS Belt | Eternal | E5 | Mega Crit III | If Crit Rate exceeds 150%, Crit Damage +100% | CritDamage | +100% | CurrentBuild | CritRateAbove150 | None | CritBreakpoint | VeryHigh | No | TBD | EternalCore | TBD | GearFodder | TBD | SS_Gear_2025-02-06

SS_GLOVES_V4_DIVINE_MIGHT | Gloves | SS Gloves | Void | V4 | Divine Might | Initial passive skill level +1; each Red Star passive upgrade gives Crit Damage +10% | CritDamage | +10% per Red Star passive | CurrentBuild | RedStarPassiveCount | None | PassiveSkillScaling | VeryHigh | No | TBD | VoidCore | TBD | GearFodder | TBD | SS_Gear_2025-02-06

SS_BOOTS_V4_ICE_ARROW | Boots | SS Boots | Void | V4 | Ice Arrow | Crystal trail explosion fires homing ice arrows that inflict Chill and 10% Vulnerability for 10s | Vulnerability | 10% | EnemiesHit | TrailLength30Blocks | Chill/Vulnerability | MovementTrail | High | No | TBD | VoidCore | TBD | GearFodder | TBD | SS_Gear_2025-02-06

Optimizer scoring rule

For SS gear recommendations, don't rank upgrades only by rarity. Rank by:

1. Does it increase direct DPS?
2. Does it unlock a crit breakpoint?
3. Does it add a debuff damage multiplier?
4. Does it improve boss damage specifically?
5. Does it scale with stats the player already has?
6. What is the relic core cost?
7. Is the condition realistic for the player's mode?

For your current damage-only style, the calculator should heavily favor:

Crit Rate breakpoint upgrades

Void Fissure / Vulnerability upgrades

Energy Overload stack upgrades

Damage to Weakened / Chilled targets
Skill Damage and Crit Damage scaling
Passive Red Star / passive level scaling
Boss laser / stored damage mechanics

1. Main chapter unlocks

From [Chapter_Unlocks-2024-09-17](#).

Chapter	Feature unlocked	What it does
1, after losing	Talent Tree	Permanent account upgrades using gold/oil-style progression. DPS value depends on ATK/crit nodes.
1, after losing	Shop	Basic item purchasing.
1, after losing	Equipment	Unlocks gear system: weapon, necklace, gloves, armor, belt, boots.
1, after winning	7 Days Challenge	Starter reward ladder.
1, after winning	Rookie Login Gifts	Starter login rewards.
1, after winning	Monthly + Super Monthly Cards	Paid/passive reward cards.
2, after losing	Growth Fund	Gem reward progression.
2, after losing	Daily Shop	Daily purchasable items.
2, after losing	Friends	Social/friend rewards.
2, after winning	Mission Control	Dailies and weeklies. Important for gem/pass/pass EXP income.
2, after winning	Main Challenges	Extra challenge content tied to chapters.

2, after winning	Quick Earnings	Idle rewards. Important for designs, gold, EXP.
2, after winning	Survivor Costumes	Cosmetic system; some may have stats depending on source.
2, after winning	Events	Limited-time events and shops. Big source of keys, shards, cores, collectibles.
3, after losing	Daily Rewards	Login/daily collection.
3, after winning	Clans	Unlocks clan system, clan shop, clan level rewards, clan expedition.
3, after winning	Daily Discount	Discount shop section.
4, after winning	Durian skill	In-battle skill unlock.
4, after winning	Laser Launcher skill	In-battle skill unlock.
5, after winning	Daily Event: Gold Mine	Gold farming mode.
6, after losing	Regular Challenge	Recurring challenge mode; also tied to Survivor Pass EXP.
6, after losing	Tech Parts after Regular Challenge	Unlocks tech part system. Important for Drone, Molotov, RPG, resonance.
7, after winning	Skill Visuals	Visual/cosmetic skill display feature.
8, after winning	Ender's Echo	Boss damage mode. Very important for DPS optimizer.
8, after winning	Survivor Pass / Super Survivor Pass	Paid/free pass reward systems.
8, after winning	Special Ops	Mission system for gems, gear, rewards.
9, after winning	Pets	Unlocks pet system, pet skills, pet awakening, pet toys.

9, after winning	Survivor level up	Unlocks leveling survivors for global stats.
10, after losing	Modular Mine	In-battle skill unlock.
10, after losing	Thunderbolt Bomb	In-battle skill unlock.
10, after losing	Inferno Bomb	In-battle skill unlock.
11, after losing	Permanent Privilege Card	Permanent card purchase. Important for long-term F2P/low-spend gem planning.
30, after winning	Zone Ops & Extreme Ops	Late challenge system. Useful for rewards and progression checks.

Big rule for the calculator: **chapter unlocks control what systems the optimizer is allowed to recommend**. Example: before Chapter 9, don't recommend pet awakening. Before Chapter 6, don't recommend tech parts.

2. Equipment system

Sources:

[Weapons_V2_2_2024-11-28](#)
[Necklaces_V2_2024-11-28](#)
[Eternal_Necklace_2024-11-28](#)
[Gloves_V2_2024-11-28](#)
[Armor_V2_2024-11-28](#)
[Belts_V2_2024-11-28](#)
[Boots_V2_2024-11-28](#)
[Merging_-_Equip_2024-11-28](#)
[Equip_Design_Costs_2024-11-14](#)
[Astral_Forge_2024-10-06](#)
[SS_Gear_2025-02-06](#)

What it does

Equipment is the main build system. It gives:

- Base ATK / HP
- Gear-specific effects
- Skill damage
- Crit rate
- Crit damage
- Boss damage
- Shield effects
- Revive effects
- Movement bonuses
- Debuff effects like Chill, Weaken, Vulnerability, Bleed, Laceration
- SS gear forge paths

Unlock requirement

- Equipment unlocks at **Chapter 1 after losing**.
- Merging requires duplicate gear and rarity progression.
- Astral Forge requires high-rarity gear and cores/materials.
- SS gear requires SS item creation/unlock plus relic cores/faction cores for upgrades.

Optimizer rule

For your calculator, equipment should be split into:

Normal Gear

S Gear

SS Gear

Astral Forge Upgrades

Collectible Gear Bonuses

Merge Cost / Upgrade Cost

HP-only gear effects should be stored but marked **DPSRelevant = No**.

3. SS gear system

Source: **SS_Gear_2025-02-06**

What it does

SS gear is the late-game gear layer. Each slot has special systems:

Slot	Main function
SS Weapon	Starforged Havoc, Void Power, Energy Overload, Void Fissure, Chaos Meteor, boss damage loops.
SS Necklace	Extra skill choices, skill damage, energy shield effects.
SS Gloves	Weaken, damage to Weakened targets, Red Star passive scaling, crit damage scaling.
SS Armor	Energy levels, energy orbs, damage multiplier, skill damage, crit damage by energy level.
SS Belt	Crit rate ramping, Mega Crit breakpoints, Eternal Laser stored damage.
SS Boots	Movement speed, Chill, Glacial Bloodline stacks, damage to Chilled enemies, Vulnerability from ice arrows.

Unlock / upgrade requirement

SS gear upgrades use:

Relic Cores

Eternal Cores

Void Cores

Chaos Cores

SS gear/fodder requirements

Forge path level: E1-E5, V1-V5, Chaos levels where available

Most important DPS breakpoints

From SS Belt:

Crit Rate >100% = Crit Damage +30%

Crit Rate >130% = Skill Damage +30%

Crit Rate >150% = Crit Damage +100%

This means the optimizer must value Crit Rate extra hard when it crosses one of those breakpoints.

4. Relic core usage

Sources:

[Relic_Core_Usage_-_Clan_Expedition_2024-11-29](#)
[Relic_Core_Usage_-_Clan_Expedition_mini_2024-11-29](#)
[Relic_Core_Usage_-_Enders_Echo_2024-11-29](#)
[Relic_Core_Usage_-_Enders_Echo_mini_2024-11-29](#)

What it does

These files show the recommended SS forge spread for different numbers of relic cores.

There are separate tables for:

Clan Expedition
Ender's Echo

because the best SS gear spread changes depending on mode.

Unlock / usage requirement

You need:

SS gear unlocked
Relic cores
Eternal/Void/Chaos cores
Enough faction cores/fodder

Optimizer rule

This becomes a relic-core recommender:

Input:

- Player relic core count
- Mode = Ender's Echo or Clan Expedition
- Current SS gear levels
- Crit rate
- Equipped gear

Output:

- Best next relic core spend

Important notes from the source:

- SS Gloves E3 requires a minimum of **101% Crit Rate from other sources, not including gloves**.
- For E5, you need about **121% Crit Rate** to have 100% uptime.
- The Clan Expedition table has special notes around **10, 29, 50 relic cores** and when to switch priorities.

So yes, the relic core data is already in your source pack.

5. Tech parts

Sources:

Tech_Parts_2024-10-09

Merging_-_Tech_Parts_2024-11-28

Tech_Part_Resonance_Cost_2024-11-28

Resonance_2024-10-30

Tech_Part_Resonance_Drone_2024-11-28

Tech_Part_Resonance_HI_Maintainer_RPG_2024-11-28

Twinborn_Part_2025-02-23

Unlock requirement

Tech parts unlock at:

Chapter 6 after losing Regular Challenge

What tech parts do

Tech parts modify in-battle skills. They turn normal skills into much stronger versions.

Major offensive tech parts include:

Tech part	Associated skill	What it does
Drone tech	Drones / Destroyer	One of the most important DPS techs. Unlocks Drone improvement and later Drone Resonance.

Lightning tech	Lightning Emitter / Supercell	Boosts lightning/electric skill damage and behavior.
Soccer tech	Soccer Ball / Quantum Ball	Improves soccer ball damage, duration, and behavior.
RPG tech	RPG / Sharkmaw Gun / C4 resonance	Improves rockets/explosions and becomes important in resonance.
Laser tech	Laser Launcher	Improves lasers and laser behavior.
Molotov tech	Molotov / Fuel Barrel	Improves fire pools, burn/fire effects, and Molotov resonance.

Major defensive/utility tech parts include:

Tech part	Associated skill	What it does
Brick tech	Brick / Dumbbell	Improves brick/dumbbell behavior and damage.
Guardian tech	Guardian	Improves knockback, guardian behavior, and damage.
Forcefield tech	Forcefield	Improves area field damage and utility.
Boomerang tech	Boomerang / Magnetic Dart	Improves boomerang behavior.
Durian tech	Durian	Improves durian movement/damage.
Rocket/defensive tech shown	Defensive skill modifiers	Some are more survival/utility than DPS.

Optimizer rule

For damage-only optimizer:

Drone tech = very high priority

RPG/C4 resonance = high priority if using that resonance path

Molotov resonance = high if fire/SF/poison/deep wound synergies are used

Defensive tech = only count if it gives real damage

HP/defense-only tech = store but ignore

6. Tech resonance

Sources:

[Tech_Part_Resonance_Cost_2024-11-28](#)

[Resonance_2024-10-30](#)

[Tech_Part_Resonance_Drone_2024-11-28](#)

[Tech_Part_Resonance_HI_Maintainer_RPG_2024-11-28](#)

What it does

Tech resonance lets a main tech part gain bonus effects from support tech parts. It adds:

Resonance Energy

ATK/HP from resonance

Special resonance skills

Multiplier upgrades

Drone wingbits

RPG/C4 explosion bonuses

Molotov/SF fire and poison bonuses

Resonance energy values

From [Tech_Part_Resonance_Cost](#):

Yellow tech part = 50

Yellow +1 = 100

Yellow +2 = 150

Yellow +3 = 200

Red = 300

Red +1 = 400

Red +2 = 550

Red +3 = 700

Red +4 = 850

Rainbow / resonance-tier part = 1000

Multiplier upgrade cost

1.0 → 1.2 = 1 resonance chip

1.2 → 1.4 = 1

1.4 → 1.6 = 2

1.6 → 1.8 = 3

1.8 → 2.0 = 5
2.0 → 2.2 = 5
2.2 → 2.4 = 7
2.4 → 2.6 = 9
2.6 → 2.8 = 12
2.8 → 3.0 = 15
Total = 60 resonance chips

Drone resonance

Drone resonance gives:

Attack Wingbit
Augment Wingbit
Attack Wingbit damage
Augment Wingbit damage
Wingbit interval reduction
Extra shots
Precision Device ATK bonuses
ATK/HP from resonance milestones

The special Drone source also says:

Augment Wingbit increases Destroyer/Drone damage.
Attack Wingbit attacks enemies with bullets.
Attack Wingbit base damage = x8 base attack.
Attack duration = 2s.
Base cooldown = 4s, can go down to 1s.
Augment Wingbit buff duration = 3s.
Base buff cooldown = 3s, can go down to 1s.

RPG / C4 resonance

RPG resonance is tied to:

RPG
Sharkmaw Gun
C4
Hi-Maintainer
Explosion radius
Launch interval
Explosion delay
C4 explosion damage

The C4 source says:

Base damage = x8 base attack

Maximum damage increase from resonance skills can be +240%

Base launch interval = 1.5s, max 0.6s

Throws new C4 every 1.5s

Base explosion delay = 4s, max 1s

C4 explodes after 4s from launch time

Molotov resonance

Molotov resonance is tied to:

Molotov

Fuel Barrel

SF / fire field effects

Deep Wound

Poison effect damage

Fire damage

SF generation interval

Optimizer rule

Tech resonance must be its own system. Do not just score it as “tech part rarity.”

Use:

MainTechPart

SupportTechParts

ResonanceEnergy

ResonanceMultiplier

ResonanceChipCost

SpecialMilestoneEffects

DPSGain

7. Survivor system

Sources:

Survivor_2024-09-07

Survivors_2025-01-26

Survivors_S_2025-01-26
Survivors_SP_2025-01-26
Survivor_Stat_Boosts_2024-12-12
Survivor_Upgrade_Costs_2024-12-19
Master_Yang_Skills_2024-11-28
Awakening_Core_Guide_2024-10-04

Unlock requirement

Survivor level-up unlocks at:

Chapter 9 after winning

What it does

Survivors give:

Active survivor skills
Passive skills
Global ATK / Crit / Crit Damage boosts
Star upgrades
Awakening skills
Teamwork passive slots
Mode-specific damage value

Survivor stat milestones

Important global milestone types:

Level 40
Level 80
Level 120
5 yellow stars
Red awakening stars

High-value DPS stats

ATK %
Crit Rate
Crit Damage
Skill Damage
Vulnerability

Laceration
Poisoned target damage
Chilled target damage
Weakened target damage
Teamwork passive slots

SP/TMNT survivors

TMNT survivors are SP-grade and event-limited.

Each 5-star TMNT survivor gives:

All Survivors Crit Rate +8%

That is huge for the optimizer because it feeds SS Belt crit breakpoints.

Awakening cores

Awakening cores are used for survivor awakening. The guide gives different priority paths for:

Ender's Echo
Clan Expedition
Yang
Metallia
King
Worm
Common
Catnips
TMNT survivors

Optimizer rule:

Awakening cores should be spent by DPS gain per core, not just favorite survivor.

8. Combat Harmony / Survivor Synergy

Source:

[Survivor_Combat_Harmony_Skills_and_Costs_2024-12-20](#)

What it does

Combat Harmony gives extra skill/passive effects and global bonuses.

It includes:

- Combat Harmony skill upgrades
- Survivor synergy levels
- Crit Rate bonuses
- Crit Damage bonuses
- Level cost requirements

Important DPS value

From the crit source and harmony source:

- Combat Harmony can give up to 30% global Crit Rate.
- Survivor Synergy can give up to 30% global Crit Damage.

Optimizer rule

This is a global account system. It should be treated like:

- GlobalCritRateSource
- GlobalCritDamageSource
- SurvivorInvestmentSystem

Very high long-term DPS value.

9. Crit Rate / Crit Damage system

Source:

[Crit_Rate_And_Damage_2025-03-09](#)

What it does

This tracks every major source of:

- Crit Rate
- Crit Damage
- Global crit

Conditional crit
Pet crit
Gear crit
Skill crit
Collectible crit
Survivor crit

Why it matters

Crit Rate does not only increase average damage. It also unlocks SS Belt thresholds:

>100% Crit Rate
>130% Crit Rate
>150% Crit Rate

Optimizer rule

Crit must be stored by source:

GlobalCritRate
GearCritRate
SkillCritRate
PetCritRate
CollectibleCritRate
ConditionalCritRate
GlobalCritDamage
GearCritDamage
SkillCritDamage
PetCritDamage
CollectibleCritDamage

Do not just make one flat crit number.

10. Debuff / multiplier system

Sources:

[Boss_Debuffs_2024-12-27](#)

[Shield_and_Skill_Damage_Multipliers_2024-12-27](#)

What it does

Tracks damage multipliers and enemy debuffs, including:

Chill
Poison
Exposed
Weaken
Debilitate
Yin Yang
Laceration
Vulnerability
Shield damage multipliers
Skill damage multipliers

Optimizer rule

Debuffs should be treated as their own damage layer.

Important reason: multiple systems say “damage to Chilled,” “damage to Weakened,” “damage to Lacerated,” or “Vulnerability.” Those only matter if the debuff exists and has uptime.

Use:

DebuffSource
DebuffDuration
DebuffStackLimit
DebuffUptime
DamageToDebuffedTarget
ModeReliability

11. Pet system

Sources:

[Pets_2024-11-28](#)
[Pet_Skills_2024-09-09](#)
[Pet_Awakening_Costs_and_Toy_Bonuses_2024-12-04](#)
[Pet_Awakening_Guide_2024-10-25](#)
[Pet_Awakening_Guide_-_Murica_2024-10-10](#)
[Pet_Awakening_Guide_-_Croaky_2024-10-11](#)

Unlock requirement

Pets unlock at:

Chapter 9 after winning

What pets do

Pets give:

Direct pet damage
Owner Crit Rate
Owner Crit Damage
Owner Skill Damage
Pet Skill Damage
Pet Crit Rate
Pet Crit Damage
Pet cooldown reduction
Pet inherited ATK
Pet affection ATK
Assist skill bonuses
Toy bonuses

Important pets

Pet	Main use
Murica / Eagle	Direct pet DPS, whirlwind, penetration, cooldown, Crush.
Croaky / Frog	Owner skill damage, crit, long-term DPS support.
Rex / Dog	Owner crit rate/crit damage and long-term DPS support.
Crab	Strong pet DPS scaling: high attack, crit, cooldown, inherited ATK.
Robot	High theoretical damage with lasers/satellite, hit consistency matters.
Cat	Fast attacks, pet crit, cooldown, survival passive.
Fish	Utility/time stop, medium damage.

Turtle

Mostly defensive/tank.

Pet awakening requirement

Pet awakening uses awakening crystals and pet sacrifices.

Important cost path:

Y1 = 5 crystals

Y2 = 5

Y3 = 10

Y4 = 20

Y5 = 30

R1 = 40

R2 = 60

R3 = 60

R4 = 60

R5 = 60

Total = 350 crystals

Optimizer rule

The source says:

Take Rex or Croaky to R5 first.

After that, raise other pets in a balanced way to Y5 for better ATK per crystal.

For your damage-only calculator, pet HP and survival should be stored but ignored.

12. Collectibles

Sources:

[Collectible_Chest_Odds_2024-11-28](#)

[Collectible_Costs_2024-10-11](#)

[Collectibles_Edition_1_2024-11-28](#)

[Collectibles_Edition_2_2024-11-28](#)

[Collectibles_Edition_3_2024-11-28](#)

What they do

Collectibles give:

- Global ATK
- Crit Rate
- Crit Damage
- Gear-specific bonuses
- SS gear set bonuses
- Skill damage bonuses
- Target-specific damage bonuses

Unlock requirement

The source set has collectible chest odds and costs, but the exact chapter unlock is not shown in the chapter unlock chart. So in the optimizer, mark unlock as:

UnlockRequirement = Collectible system available / event or chest access

Do not invent a chapter until confirmed.

Optimizer rule

Collectibles must connect to exact gear/skill systems.

Example:

- Some collectibles boost SS Weapon Energy Overload.
- Some boost SS Belt/Armor/Gloves effects.
- Some give global Crit Rate or Crit Damage.

Collectibles are later-game but very important for final DPS.

13. Clan system

Sources:

[Clan_Shop_2025-02-17](#)

[Clan_Level_Rewards_2024-11-28](#)

[Clan_Expedition_Rewards_2024-11-28](#)

Unlock requirement

Clans unlock at:

Chapter 3 after winning

What clan does

Clan gives:

- Clan shop
- Clan expedition
- Clan level perks
- Clan member stat boosts
- Quick earning boosts
- Clan EXP rewards
- Clan currency spending

Clan shop

The clan shop source includes:

- S selector/chests
- S survivor shards
- Survivor shards
- Collectible keys
- Tech keys
- Pet/equipment related chests
- Relic/core-style items
- Other upgrade currencies

The chart shows stock changing by clan level:

- Level 1-11
- Level 12
- Level 13
- Level 14
- Level 15
- Level 16

Important: some clan shop items are icon-only in the image, so I'll store them by icon ID first if the exact item name is not readable.

Example:

ClanShopItem_01 | Cost 40000 | Qty 1 | Stock scales by clan level

Then later we map exact names.

Optimizer rule

Clan shop should be valued by:

Gem-equivalent value
DPS relevance
Rarity bottleneck
Player progression stage
Clan currency cost
Monthly/weekly stock limit

High-priority shop items are usually rare bottleneck items like relic cores, S selectors, shards, tech keys, collectible keys, and resonance/awakening materials depending on stage.

14. Survivor Pass

Sources:

[Survivor_Pass_2024-12-19](#)

[Survivor_Pass_Experience_2024-09-16](#)

Unlock requirement

Survivor Pass unlocks at:

Chapter 8 after winning

What it does

Survivor Pass gives reward tracks for:

Survivor resources
Tech parts
Equipment resources
Joey shards
Essence

Survivor shards
Reset vial key
Awakening core
Tech keys
S keys
Relic core
Collectible key

Pass EXP sources

From the pass EXP chart:

Dailies
Defeat monsters
Clear main chapters/challenges
Regular Challenge
Path of Trials
Ender's Echo
Clan Expedition
Special Ops

Season max shown:

15,590 to 15,740 EXP range

Optimizer rule

Pass value should be treated as:

RewardValue / Cost
DPSMaterialValue
RelicCoreValue
ShardValue
KeyValue

15. Gems, oil, essence, designs, item values

Sources:

Sources_Of_Gems_2024-10-07
Sources_of_Gems_2024-12-18
Sources_Of_Oil_2024-11-03
Sources_of_Oil_2024-12-18
Energy_Essence_Cost_2024-11-28
Energy_Essence_Cost_Full_2024-11-28
Equip_Design_Costs_2024-11-14
Item_Values_2024-10-02

What they do

These are economy/value sources.

They track:

Where gems come from
Where oil comes from
Survivor level essence costs
Equipment design costs
Item gem-equivalent values
Upgrade costs

Optimizer rule

This is needed for event/shop decisions.

Use it to answer:

Is this worth buying?
Should I spend gems?
Should I save for Black Friday?
What item has better gem value?
What is my bottleneck?

For your account, this matters because you are trying to avoid wasting gems and maximize DPS.

16. Ender's Echo rewards

Source:

EE_Rewards_2024-09-15

Unlock requirement

Ender's Echo unlocks at:

Chapter 8 after winning

What it does

Ender's Echo is a boss damage/ranking mode. It gives rewards based on performance/ranking.

Optimizer rule

For Ender's Echo, favor:

Boss DPS
Crit Rate breakpoints
Relic core spread from EE guide
Debuff uptime
Single-target damage
Pet direct DPS
Yang/Metallia/Joey mode-specific paths

Do not overvalue mob-clearing effects for EE unless they directly boost boss damage.

17. Skills

Source:

Skills_2024-11-28

What it does

This contains in-battle skills and likely their evo/passive pairings.

Skills are needed because tech parts, drones, resonance, survivor skills, and SS gear modify them.

Unlock requirement

Some skills are chapter-unlocked:

Durian = Chapter 4 after winning

Laser Launcher = Chapter 4 after winning

Modular Mine = Chapter 10 after losing

Thunderbolt Bomb = Chapter 10 after losing

Inferno Bomb = Chapter 10 after losing

Optimizer rule

Skills should be stored as:

SkillName

EvoName

PassiveRequired

BaseFunction

TechPartAssociation

ResonanceAssociation

DPSRole

MapClearValue

BossValue

18. Wishlist probabilities

Source:

[Wishlist_Probabilities_2024-11-28](#)

What it does

Shows odds/probabilities for wishlist pulls.

Unlock requirement

Requires the relevant chest/wishlist system to be available.

Optimizer rule

Use this for:

Expected value

Chest opening strategy

How many keys/gems needed

Chance to get target gear

19. Timed Muster medal conversion

Source:

[Timed_Muster_Medal_Conversion_2024-12-12](#)

What it does

Shows conversion/value for timed muster medals.

Optimizer rule

Use for event value and whether to convert/spend medals now or save/use differently.

20. Master source index for the calculator

This is the source pack I'll use going forward:

Chapter_Unlocks

Skills

Boss_Debuffs

Shield_and_Skill_Damage_Multipliers

Crit_Rate_And_Damage

Survivor

Survivors

Survivors_S

Survivors_SP

Survivor_Stat_Boosts
Survivor_Upgrade_Costs
Survivor_Combat_Harmony_Skills_and_Costs
Master_Yang_Skills
Awakening_Core_Guide

Weapons
Necklaces
Eternal_Necklace
Gloves
Armor
Belts
Boots
SS_Gear
Astral_Forge
Relic_Core_Usage_Enders_Echo
Relic_Core_Usage_Clan_Expedition

Tech_Parts
Tech_Part_Resonance_Cost
Resonance
Tech_Part_Resonance_Drone
Tech_Part_Resonance_HI_Maintainer_RPG
Twinborn_Part
Merging_Tech_Parts

Pets
Pet_Skills
Pet_Awakening_Costs_and_Toy_Bonuses
Pet_Awakening_Guide
Pet_Awakening_Guide_Murica
Pet_Awakening_Guide_Croaky
Pet_Awakening_Guide_Rex
Merging_Pets

Collectible_Chest_Odds
Collectible_Costs
Collectibles_Edition_1
Collectibles_Edition_2
Collectibles_Edition_3

Clan_Shop
Clan_Level_Rewards
Clan_Expedition_Rewards

Survivor_Pass
Survivor_Pass_Experience
EE_Rewards

Sources_Of_Gems
Sources_Of_Oil
Energy_Essence_Cost
Energy_Essence_Cost_Full
Equip_Design_Costs
Item_Values
Merging_Equip
Wishlist_Probabilities
Timed_Muster_Medal_Conversion

Batch 2 — Twinborn Lightning, Boomerang, Drill, Rocket/RPG Resonance

Sources extracted from the images you gave:

Twinborn Lightning Resonance
Twinborn Boomerang Resonance
Twinborn Drill Resonance
Twinborn Rocket / RPG Resonance
Game version 4.5
Updated 2026-02-23

Shared structure for these charts:

ResonanceAmount
TotalATK
TotalHP
DamageStatColumns
BuffText
DPSRelevant
AssociatedDebuff
AssociatedSkill

HP is stored but ignored for your DPS optimizer.

1. Twinborn Lightning Resonance

Main association

Twinborn Lightning
Twinborn Maelstorm
Thunderburst Field
Falling Thunder
Phase Driver
Crit Damage
Skill Damage
Vulnerability

Stat columns

Crit Damage
Skill Damage
Vulnerability

Optimizer identity

Twinborn Lightning is a **crit + skill damage + vulnerability tech**. It becomes much more important later because it gives Vulnerability and large crit/skill damage chunks.

DPSRelevant = Very High later

DebuffFamily = Vulnerability

ScalingType = Thunderburst / Falling Thunder / Crit Damage / Skill Damage

Extracted resonance rows

50:

+200 ATK, +800 HP

100:

Twinborn Maelstorm Damage +10%

150:

+400 ATK, +1600 HP

200:

Crit Damage +40
Buff: Crit Damage +40%

250:
+600 ATK, +2400 HP

300:
Falling thunder deals +14% damage to enemies damaged by swirling Thunderburst field

350:
+800 ATK, +3200 HP

400:
Phase Driver ATK +5%, HP +5%

450:
Skill Damage +40
Buff: Skill Damage +40%

500:
+1000 ATK, +4000 HP

550:
Phase Driver ATK +5%, HP +5%

600:
After 8 falling thunder strikes, the next 4 strikes deal +36% damage

700:
+1200 ATK, +4800 HP

800:
Phase Driver ATK +5%, HP +5%

900:
Vulnerability +50
Enemies damaged by swirling Thunderburst field are inflicted with 50% Vulnerability

1000:
+1400 ATK, +5600 HP

1100:
Phase Driver ATK +5%, HP +5%

1200:
Twinborn Maelstorm Damage +6%

1350:
+1600 ATK, +6400 HP

1500:
Phase Driver ATK +5%, HP +5%

1650:
Skill Damage +70
Buff: Skill Damage +30%

1800:
+1800 ATK, +7200 HP

1950:
Phase Driver ATK +5%, HP +5%

2100:
Falling thunder deals +10% damage to enemies damaged by swirling Thunderburst field

2250:
+2000 ATK, +8000 HP

2400:
Phase Driver ATK +5%, HP +5%

2550:
Crit Damage +70
Buff: Crit Damage +30%

2700:
+2500 ATK, +10000 HP

2850:
Phase Driver ATK +5%, HP +5%

3000:
Twinborn Maelstorm Damage +5%

3500:
+3000 ATK, +12000 HP

4000:

Phase Driver ATK +5%, HP +5%

4500:

Vulnerability +80

Enemies damaged by swirling Thunderburst field are inflicted with 30% Vulnerability

5000:

+3500 ATK, +14000 HP

5500:

Phase Driver ATK +5%, HP +5%

6000:

After 8 falling thunder strikes, the next 4 strikes deal +25% damage

6500:

+4000 ATK, +16000 HP

7000:

Phase Driver ATK +5%, HP +5%

7500:

Crit Damage +90

Vulnerability +100

Enemies damaged by swirling Thunderburst field are inflicted with 20% Vulnerability

Crit Damage +20%

8000:

+6000 ATK, +24000 HP

8500:

Phase Driver ATK +5%, HP +5%

9000:

Crit Damage +100

Skill Damage +100

Skill Damage +30%

Crit Damage +10%

9500:

+7500 ATK, +30000 HP

10000:

Phase Driver ATK +5%, HP +5%

10500:

Crit Damage +120

Twinborn Maelstorm Damage +20%

Crit Damage +20%

11000:

+7500 ATK, +30000 HP

11500:

Phase Driver ATK +5%, HP +5%

12000:

Vulnerability +175

Enemies damaged by swirling Thunderburst field are inflicted with 75% Vulnerability

12500:

+7500 ATK, +30000 HP

13000:

Phase Driver ATK +5%, HP +5%

13500:

Skill Damage +120

Falling thunder deals +30% damage to enemies damaged by swirling Thunderburst field

Skill Damage +20%

14000:

+7500 ATK, +30000 HP

14500:

Phase Driver ATK +5%, HP +5%

15000:

Crit Damage +160

Skill Damage +160

Skill Damage +40%

Crit Damage +40%

Database flags

TwinbornID = TB_LIGHTNING

Mode = Lightning / Maelstorm

PrimaryStats = CritDamage, SkillDamage, Vulnerability
Debuff = Vulnerability
DPSRelevant = VeryHigh
BestUse = Late-game DPS / vulnerability scaling
IgnoreForDPS = HP

2. Twinborn Boomerang Resonance

Main association

Twinborn Boomerang
Twinborn Magvolt Fang
Lightning Zone
Vulnerability stacks
Skill Damage
Weakened target damage
Phase Driver

Stat columns

Vulnerability
Skill Damage
Weakened Damage

Optimizer identity

Twinborn Boomerang is not like old trash Boomerang. The Twinborn version becomes a **Vulnerability stack + Weakened damage + Skill Damage** system.

DPSRelevant = High later
DebuffFamily = Vulnerability / Weaken
ScalingType = Vulnerability stacks + Weakened target damage

Extracted resonance rows

50:
+200 ATK, +800 HP

100:
Max lightning zone strikes +3
Twinborn Boomerang does not require passive skill to upgrade

150:
+400 ATK, +1600 HP

200:
Vulnerability +30
Lightning zone strikes inflict 1 Vulnerability stack on hit
Increases target's damage taken by 5% per stack
Max 6 stacks

250:
+600 ATK, +2400 HP

300:
Twinborn Magvolt Fang Damage +25%

350:
+800 ATK, +3200 HP

400:
Phase Driver ATK +5%, HP +5%

450:
Skill Damage +40
Buff: Skill Damage +40%

500:
+1000 ATK, +4000 HP

550:
Phase Driver ATK +5%, HP +5%

600:
Weakened Damage +40
Damage to Weakened targets +40%

700:
+1200 ATK, +4800 HP

800:
Phase Driver ATK +5%, HP +5%

900:
Vulnerability +50

Skill Damage +50
Weakened Damage +50
Lightning zone strike Vulnerability max stacks +4
Skill Damage +10%
Damage to Weakened targets +10%

1000:
+1400 ATK, +5600 HP

1100:
Phase Driver ATK +5%, HP +5%

1200:
Skill Damage +70
Skill Damage +20%

1350:
+1600 ATK, +6400 HP

1500:
Phase Driver ATK +5%, HP +5%

1650:
Twinborn Magvolt Fang Damage +20%

1800:
+1800 ATK, +7200 HP

1950:
Phase Driver ATK +5%, HP +5%

2100:
Weakened Damage +70
Damage to Weakened targets +20%

2250:
+2000 ATK, +8000 HP

2400:
Phase Driver ATK +5%, HP +5%

2550:
Vulnerability +70
Lightning zone strike Vulnerability max stacks +4

2700:

+2500 ATK, +10000 HP

2850:

Phase Driver ATK +5%, HP +5%

3000:

Vulnerability +90

Lightning zone strike Vulnerability max stacks +4

3500:

+3000 ATK, +12000 HP

4000:

Phase Driver ATK +5%, HP +5%

4500:

Twinborn Magvolt Fang Damage +20%

5000:

+3500 ATK, +14000 HP

5500:

Phase Driver ATK +5%, HP +5%

6000:

Skill Damage +90

Skill Damage +20%

6500:

+4000 ATK, +16000 HP

7000:

Phase Driver ATK +5%, HP +5%

7500:

Weakened Damage +90

Damage to Weakened targets +20%

8000:

+6000 ATK, +24000 HP

8500:

Phase Driver ATK +5%, HP +5%

9000:

Vulnerability +110

Skill Damage +100

Weakened Damage +100

Lightning zone strike Vulnerability max stacks +4

Skill Damage +10%

Damage to Weakened targets +10%

9500:

+7500 ATK, +30000 HP

10000:

Phase Driver ATK +5%, HP +5%

10500:

Skill Damage +120

Max lightning zone strikes +3

Skill Damage +20%

11000:

+7500 ATK, +30000 HP

11500:

Phase Driver ATK +5%, HP +5%

12000:

Vulnerability +150

Lightning zone strike Vulnerability max stacks +8

12500:

+7500 ATK, +30000 HP

13000:

Phase Driver ATK +5%, HP +5%

13500:

Weakened Damage +120

Twinborn Magvolt Fang Damage +50%

Damage to Weakened targets +20%

14000:

+7500 ATK, +30000 HP

14500:

Phase Driver ATK +5%, HP +5%

15000:

Vulnerability +175

Skill Damage +150

Weakened Damage +150

Lightning zone strike Vulnerability max stacks +5

Skill Damage +30%

Damage to Weakened targets +30%

Database flags

TwinbornID = TB_BOOMERANG

Mode = Boomerang / Magvolt Fang

PrimaryStats = Vulnerability, SkillDamage, DamageToWeakened

Debuff = Vulnerability, Weaken

DPSRelevant = High

BestUse = Debuff stacking / Weaken synergy

IgnoreForDPS = HP

3. Twinborn Drill Resonance

Main association

Twinborn Drill

Twinborn Frostfire Dualist

Fire Whistleshot

Ice Whistleshot

Ice Focus

Fire Bomb

Shield Damage

Chilled Damage

Crit Damage

Lacerated Damage

Antimatter Maintainer

Stat columns

Shield Damage

Chill Damage
Crit Damage
Lacerate Damage

Optimizer identity

Twinborn Drill is a **Chill + Crit + Laceration + Shield Damage** tech. It has a big late-game identity because it stacks Ice Focus and boosts damage to Chilled/Lacerated enemies.

DPSRelevant = High / Very High

DebuffFamily = Chill, Laceration

ScalingType = Ice Focus stacks + Chilled target damage + Lacerated target damage

Extracted resonance rows

50:

+200 ATK, +800 HP

100:

Shield Damage +15

Fire Whistlehot Damage +18%

Shield Damage Boost +15%

150:

+400 ATK, +1600 HP

200:

Chill Damage +30

Ice Focus:

When Ice Whistlehot hits 10 times, gain 1 Ice Focus stack

+30% damage to Chilled targets for each stack

Max 1 stack

250:

+600 ATK, +2400 HP

300:

Chill Damage +40

Ice Whistlehot damage +18%

Damage to Chilled targets +10%

350:

+800 ATK, +3200 HP

400:

Antimatter Maintainer ATK +5%, HP +5%

450:

Chill Damage +70

Crit Damage +20

Required hits for Ice Focus -5

Max stacks +1

Crit Damage +20%

500:

+1000 ATK, +4000 HP

550:

Antimatter Maintainer ATK +5%, HP +5%

600:

Fire Bomb Damage +40%

700:

+1200 ATK, +4800 HP

800:

Antimatter Maintainer ATK +5%, HP +5%

900:

Chill Damage +100

Lacerate Damage +30

Max Ice Focus stacks +1

Extra damage to Lacerated targets +30%

1000:

+1400 ATK, +5600 HP

1100:

Antimatter Maintainer ATK +5%, HP +5%

1200:

After 10 Fire Whistleshot hits, Fire Whistleshot damage +18%

1350:

+1600 ATK, +6400 HP

1500:

Antimatter Maintainer ATK +5%, HP +5%

1650:

Crit Damage +40

Crit Damage +20%

1800:

+1800 ATK, +7200 HP

1950:

Antimatter Maintainer ATK +5%, HP +5%

2100:

After Ice Whistleshot hits a target, Fire Whistleshot damage to target +15%

2250:

+2000 ATK, +8000 HP

2400:

Antimatter Maintainer ATK +5%, HP +5%

2550:

Chill Damage +120

Damage to Chilled targets +20%

2700:

+2500 ATK, +10000 HP

2850:

Antimatter Maintainer ATK +5%, HP +5%

3000:

After Fire Whistleshot hits a target, Ice Whistleshot damage to target +60%

3500:

+3000 ATK, +12000 HP

4000:

Antimatter Maintainer ATK +5%, HP +5%

4500:

Shield Damage +35

Shield Damage Boost +20%

5000:

+3500 ATK, +14000 HP

5500:

Antimatter Maintainer ATK +5%, HP +5%

6000:

Fire Bomb Damage +30%

6500:

+4000 ATK, +16000 HP

7000:

Antimatter Maintainer ATK +5%, HP +5%

7500:

Lacerate Damage +60

Extra damage to Lacerated targets +30%

8000:

+6000 ATK, +24000 HP

8500:

Antimatter Maintainer ATK +5%, HP +5%

9000:

Chill Damage +150

Max Ice Focus stacks +1

9500:

+7500 ATK, +30000 HP

10000:

Antimatter Maintainer ATK +5%, HP +5%

10500:

Shield Damage +65

Crit Damage +70

Crit Damage +30%

Shield Damage Boost +30%

11000:

+7500 ATK, +30000 HP

11500:

Antimatter Maintainer ATK +5%, HP +5%

12000:

Chill Damage +160

Ice Whistlehot damage +18%

Damage to Chilled targets +10%

12500:

+7500 ATK, +30000 HP

13000:

Antimatter Maintainer ATK +5%, HP +5%

13500:

Chill Damage +190

Lacerate Damage +80

Required hits for Ice Focus -3

Max stacks +1

Extra damage to Lacerated targets +20%

14000:

+7500 ATK, +30000 HP

14500:

Antimatter Maintainer ATK +5%, HP +5%

15000:

Chill Damage +200

After Ice Whistlehot hits a target, Fire Whistlehot damage to target +45%

Damage to Chilled targets +10%

Database flags

TwinbornID = TB_DRILL

Mode = Drill / Frostfire Dualist

PrimaryStats = ChillDamage, CritDamage, ShieldDamage, LacerateDamage

Debuff = Chill, Laceration

DPSRelevant = HighVeryHigh

BestUse = Chill/Lacerate builds, late-game damage stacking

IgnoreForDPS = HP

4. Twinborn Rocket / RPG Resonance

Main association

Twinborn Rocket / RPG
Twinborn Blaster Sharkmaw Gun
Cluster Arrows
Mad Blight
Poisoned target damage
Shield Damage
Lacerated target damage
Skill Damage
Antimatter Maintainer

Stat columns

Skill Damage
Poison Damage
Shield Damage
Lacerate Damage

Optimizer identity

Twinborn Rocket/RPG is a **Poison + Shield + Skill Damage + Laceration** tech. The main special mechanic is **Mad Blight stacks**, which increase damage to Poisoned targets.

DPSRelevant = High later
DebuffFamily = Poison, Laceration, Chill
ScalingType = Mad Blight stacks + Sharkmaw Gun + Cluster Arrows

Extracted resonance rows

50:
+200 ATK, +800 HP

100:
Blaster Sharkmaw Gun Damage +50%

150:
+400 ATK, +1600 HP

200:

Skill Damage +25

Blaster Sharkmaw Gun Chills enemies on hit

Skill Damage +25%

250:

+600 ATK, +2400 HP

300:

Poison Damage +50

When Cluster Arrows hit, you gain 1 Mad Blight stack

Gain 10% damage to Poisoned targets for each stack

Max 5 stacks

350:

+800 ATK, +3200 HP

400:

Antimatter Maintainer ATK +5%, HP +5%

450:

Cluster Arrows +2

Cluster Arrows Damage +10%

500:

+1000 ATK, +4000 HP

550:

Antimatter Maintainer ATK +5%, HP +5%

600:

Poison Damage +100

Max Mad Blight stacks +5

700:

+1200 ATK, +4800 HP

800:

Antimatter Maintainer ATK +5%, HP +5%

900:

Shield Damage +30

Lacerate Damage +25

Restores 30% Shield after 5 Blaster Sharkmaw Gun shots

Shield Damage Boost +30%

Damage to Lacerated targets +25%

1000:

+1400 ATK, +5600 HP

1100:

Antimatter Maintainer ATK +5%, HP +5%

1200:

Blaster Sharkmaw Gun fires faster

Larger explosion radius

1350:

+1600 ATK, +6400 HP

1500:

Antimatter Maintainer ATK +5%, HP +5%

1650:

Skill Damage +45

Skill Damage +20%

1800:

+1800 ATK, +7200 HP

1950:

Antimatter Maintainer ATK +5%, HP +5%

2100:

Cluster Arrows +2

2250:

+2000 ATK, +8000 HP

2400:

Antimatter Maintainer ATK +5%, HP +5%

2550:

Poison Damage +120

Max Mad Blight stacks +2

2700:

+2500 ATK, +10000 HP

2850:

Antimatter Maintainer ATK +5%, HP +5%

3000:

Shield Damage +50

Shield Damage Boost +20%

3500:

+3000 ATK, +12000 HP

4000:

Antimatter Maintainer ATK +5%, HP +5%

4500:

Skill Damage +60

Shield Damage +65

Shield Damage Boost +15%

Skill Damage +15%

5000:

+3500 ATK, +14000 HP

5500:

Antimatter Maintainer ATK +5%, HP +5%

6000:

Blaster Sharkmaw Gun damage +60%

6500:

+4000 ATK, +16000 HP

7000:

Antimatter Maintainer ATK +5%, HP +5%

7500:

Poison Damage +150

Max Mad Blight stacks +3

8000:

+6000 ATK, +24000 HP

8500:

Antimatter Maintainer ATK +5%, HP +5%

9000:

Lacerate Damage +50

Damage to Lacerated targets +25%

9500:

+7500 ATK, +30000 HP

10000:

Antimatter Maintainer ATK +5%, HP +5%

10500:

Shield Damage +100

Shield Damage Boost +35%

11000:

+7500 ATK, +30000 HP

11500:

Antimatter Maintainer ATK +5%, HP +5%

12000:

Skill Damage +100

Skill Damage +40%

12500:

+7500 ATK, +30000 HP

13000:

Antimatter Maintainer ATK +5%, HP +5%

13500:

Poison Damage +200

Max Mad Blight stacks +5

14000:

+7500 ATK, +30000 HP

14500:

Antimatter Maintainer ATK +5%, HP +5%

15000:

Lacerate Damage +75

Blaster Sharkmaw Gun damage +60%

Cluster Arrows damage +25%

Damage to Lacerated targets +25%

Database flags

TwinbornID = TB_ROCKET_RPG

Mode = Rocket / Blaster Sharkmaw Gun

PrimaryStats = SkillDamage, PoisonDamage, ShieldDamage, LacerateDamage

Debuff = Poison, Laceration, Chill

SpecialMechanic = MadBlightStacks

DPSRelevant = High

BestUse = Poison/lacerate/shield damage builds

IgnoreForDPS = HP

Batch 2 optimizer takeaways

Best DPS mechanics from this batch

Lightning:

- Vulnerability
- Crit Damage
- Skill Damage
- Thunderburst field damage

Boomerang:

- Vulnerability stacks
- Damage to Weakened targets
- Skill Damage

Drill:

- Damage to Chilled targets
- Ice Focus stacks
- Damage to Lacerated targets
- Crit Damage

Rocket/RPG:

- Mad Blight stacks
- Damage to Poisoned targets
- Damage to Lacerated targets
- Shield Damage
- Skill Damage

Important debuff families added

Vulnerability
Weaken
Chill
Laceration
Poison

Best calculator fields

TwinbornID
ModeName
ResonanceAmount
TotalATK
TotalHP
SkillDamage
CritDamage
CritRate
Vulnerability
WeakenedDamage
ChillDamage
PoisonDamage
LacerateDamage
ShieldDamage
SpecialMechanic
BuffText
DPSRelevant
DebuffFamily
Condition
SourceImage
SourceDate

Batch 3 — Twinborn Durian, Soccer + Normal Drone/RPG/Molotov Resonance

Sources extracted from the images you gave:

Twinborn Durian Resonance
Twinborn Soccer Resonance
Normal Drone Resonance

Normal RPG Resonance

Normal Molotov Resonance

Game versions shown:

- Twinborn charts: v4.5, updated 2026-02-23
- Normal resonance chart: v3.3, updated 2024-10-30

HP is stored but marked **IgnoreForDPS**. ATK, crit, skill damage, shield damage, poison damage, chill damage, lacerate damage, vulnerability, and Deep Wound are **DPSRelevant**.

1. Twinborn Durian Resonance

Main association

Twinborn Durian

Twinborn Caltrops

Transgenic Caltrops

EM Explosions

Overload EM Explosion

Lightning Balls

Poison

Mutation Effect

Shield Damage

Skill Damage

Poison Damage

Quantum Nanobot

Stat columns

Shield Damage

Skill Damage

Poison Damage

Optimizer identity

Twinborn Durian is a **Poison + Skill Damage + Shield Damage** tech. It is not just Durian movement anymore. The important mechanics are:

Transgenic Caltrops trigger EM explosions

Overload EM explosions

Lightning balls apply Poison

Mutation effect permanently increases Skill Damage
Boost effect permanently increases damage to Poisoned targets

Extracted resonance rows

50:

+200 ATK, +800 HP

100:

Transgenic Caltrops trigger EM explosions every 2s.

150:

+400 ATK, +1600 HP

200:

After every 2 EM explosions, the third becomes an Overload explosion with +50% damage.

250:

+600 ATK, +2400 HP

300:

Overload EM explosion also fires 5 lightning balls in all directions, inflicting Poison on hit targets for 20s.

350:

+800 ATK, +3200 HP

400:

Quantum Nanobot ATK +5%, HP +5%

450:

Shield Damage +20

Transgenic Caltrops damage +35%.

Shield damage boost +20%.

Transgenic Caltrops upgrades no longer require passive.

500:

+1000 ATK, +4000 HP

550:

Quantum Nanobot ATK +5%, HP +5%

600:

Skill Damage +45

Transgenic Caltrops damage +40%.
Every 20 hits on targets by Transgenic Caltrops triggers mutation effect.
Mutation effect permanently increases Skill Damage by 45%.
Does not stack.

700:
+1200 ATK, +4800 HP

800:
Quantum Nanobot ATK +5%, HP +5%

900:
Poison Damage +50
Transgenic Caltrops damage +45%.
Every 10 lightning ball hits on targets triggers boost effect.
Boost effect permanently increases damage to Poisoned targets by 50%.
Does not stack.

1000:
+1400 ATK, +5600 HP

1100:
Quantum Nanobot ATK +5%, HP +5%

1200:
Shield Damage +30
EM explosion damage +50%.
Additional shield damage boost +10%.

1350:
+1600 ATK, +6400 HP

1500:
Quantum Nanobot ATK +5%, HP +5%

1650:
Transgenic Caltrops damage +10%.
Required lightning ball hits to trigger boost effect -5.

1800:
+1800 ATK, +7200 HP

1950:
Quantum Nanobot ATK +5%, HP +5%

2100:
Shield Damage +40
Additional shield damage boost +10%.

2250:
+2000 ATK, +8000 HP

2400:
Quantum Nanobot ATK +5%, HP +5%

2550:
Skill Damage +70
Required Transgenic Caltrops hits to trigger mutation effect -15.
Mutation effect permanently increases Skill Damage by an additional 25%.

2700:
+2500 ATK, +10000 HP

2850:
Quantum Nanobot ATK +5%, HP +5%

3000:
Poison Damage +65
Transgenic Caltrops damage +10%.
Lightning ball damage +30%.
Boost effect permanently increases damage to Poisoned targets by an additional 15%.

3500:
+3000 ATK, +12000 HP

4000:
Quantum Nanobot ATK +5%, HP +5%

4500:
Skill Damage +100
Mutation effect permanently increases Skill Damage by an additional 30%.

5000:
+3500 ATK, +14000 HP

5500:
Quantum Nanobot ATK +5%, HP +5%

6000:

Shield Damage +50

EM explosion damage +45%.

Overload EM explosion damage +45%.

Additional shield damage boost +10%.

6500:

+4000 ATK, +16000 HP

7000:

Quantum Nanobot ATK +5%, HP +5%

7500:

Poison Damage +80

Lightning ball damage +50%.

Boost effect permanently increases damage to Poisoned targets by an additional 15%.

8000:

+6000 ATK, +24000 HP

8500:

Quantum Nanobot ATK +5%, HP +5%

9000:

Poison Damage +100

Lightning ball damage +50%.

Boost effect permanently increases damage to Poisoned targets by an additional 20%.

9500:

+7500 ATK, +30000 HP

10000:

Quantum Nanobot ATK +5%, HP +5%

10500:

Poison Damage +120

Lightning ball damage +50%.

Boost effect permanently increases damage to Poisoned targets by an additional 20%.

11000:

+7500 ATK, +30000 HP

11500:

Quantum Nanobot ATK +5%, HP +5%

12000:

Shield Damage +60

EM explosion damage +75%.

Additional shield damage boost +10%.

12500:

+7500 ATK, +30000 HP

13000:

Quantum Nanobot ATK +5%, HP +5%

13500:

Skill Damage +165

Transgenic Caltrops damage +25%.

Mutation effect permanently increases Skill Damage by an additional 65%.

14000:

+7500 ATK, +30000 HP

14500:

Quantum Nanobot ATK +5%, HP +5%

15000:

Poison Damage +150

Overload EM explosion damage +75%.

Boost effect permanently increases damage to Poisoned targets by an additional 30%.

Database flags

TwinbornID = TB_DURIAN

Mode = Durian / Caltrops

PrimaryStats = ShieldDamage, SkillDamage, PoisonDamage

Debuff = Poison

SpecialMechanics = EMExplosion, OverloadExplosion, LightningBalls, MutationEffect,

PoisonBoostEffect

DPSRelevant = MediumHigh

IgnoreForDPS = HP

2. Twinborn Soccer Resonance

Main association

Twinborn Soccer
Twinborn Quantum Ball
Spiked Quantum Ball
Icy Quantum Ball
Super Soccer Ball
Hi-Energy
Protection
Hoarfrost
Chill
Crit Damage
Shield Damage
Quantum Nanobot

Stat columns

Shield Damage
Crit Damage
Chill Damage

Optimizer identity

Twinborn Soccer is a **Crit Damage + Chill Damage + Shield Damage** tech. It has several named internal effects:

Spiked Quantum Ball
Icy Quantum Ball
Hi-Energy
Protection
Hoarfrost
Super Soccer Ball

It is more useful when the build benefits from Chill damage or crit damage, but the guide still ranked Soccer/Durian lower than the other Twinborn pairs.

Extracted resonance rows

50:
+200 ATK, +800 HP

100:
Chill Damage +25

When Spiked Quantum Ball splits for the first time, it spawns 2 Icy Quantum Balls.

Icy Quantum Balls inflict Chill for 10s on targets hit.
Damage to Chilled targets +25%.

150:
+400 ATK, +1600 HP

200:
Shield Damage +10
Gain 0.5% Shield when firing Spiked Quantum Ball.
Max 15% in a single match.
Shield damage boost +10%.

250:
+600 ATK, +2400 HP

300:
After every 5s, kicks a Super Soccer Ball that explodes on impact.

350:
+800 ATK, +3200 HP

400:
Quantum Nanobot ATK +5%, HP +5%

450:
Crit Damage +30
Crit Damage +30%.
Twinborn Quantum Ball upgrade does not require passive skill.

500:
+1000 ATK, +4000 HP

550:
Quantum Nanobot ATK +5%, HP +5%

600:
Shield Damage +20
Chill Damage +50
Shield damage boost +10%.
Damage to Chilled targets +25%.

700:
+1200 ATK, +4800 HP

800:

Quantum Nanobot ATK +5%, HP +5%

900:

Crit Damage +60

Super Soccer Ball explosion damage +100%.

Crit Damage +30%.

1000:

+1400 ATK, +5600 HP

1100:

Quantum Nanobot ATK +5%, HP +5%

1200:

Crit Damage +70

Every 10 Spiked Quantum Ball hits triggers Hi-Energy.

Crit Damage +10%.

This effect does not stack.

1350:

+1600 ATK, +6400 HP

1500:

Quantum Nanobot ATK +5%, HP +5%

1650:

Shield Damage +30

Every 10 Spiked Quantum Ball hits triggers Protection.

Shield damage boost +10%.

This effect does not stack.

1800:

+1800 ATK, +7200 HP

1950:

Quantum Nanobot ATK +5%, HP +5%

2100:

Chill Damage +60

Every 10 Icy Quantum Ball hits triggers Hoarfrost.

Damage to Chilled targets +10%.

This effect does not stack.

2250:

+2000 ATK, +8000 HP

2400:

Quantum Nanobot ATK +5%, HP +5%

2550:

Crit Damage +85

Required hits to trigger Hi-Energy -5.

Additional Crit Damage +15%.

2700:

+2500 ATK, +10000 HP

2850:

Quantum Nanobot ATK +5%, HP +5%

3000:

Shield Damage +45

Required hits to trigger Protection -5.

Additional shield damage boost +15%.

3500:

+3000 ATK, +12000 HP

4000:

Quantum Nanobot ATK +5%, HP +5%

4500:

Chill Damage +75

Required hits to trigger Hoarfrost -5.

Additional damage to Chilled targets +15%.

5000:

+3500 ATK, +14000 HP

5500:

Quantum Nanobot ATK +5%, HP +5%

6000:

Shield Damage +60

Chill Damage +90

Protection gains additional +15% Shield damage boost.

Hoarfrost gains additional +15% damage to Chilled targets.

6500:

+4000 ATK, +16000 HP

7000:

Quantum Nanobot ATK +5%, HP +5%

7500:

Crit Damage +100

Super Soccer Ball explosion damage +100%.

Hi-Energy gains additional +15% Crit Damage.

8000:

+6000 ATK, +24000 HP

8500:

Quantum Nanobot ATK +5%, HP +5%

9000:

Crit Damage +120

Chill Damage +110

Hoarfrost gains additional +20% damage to Chilled targets.

Hi-Energy gains additional +20% Crit Damage.

9500:

+7500 ATK, +30000 HP

10000:

Quantum Nanobot ATK +5%, HP +5%

10500:

Chill Damage +145

Hoarfrost gains additional +35% damage to Chilled targets.

11000:

+7500 ATK, +30000 HP

11500:

Quantum Nanobot ATK +5%, HP +5%

12000:

Shield Damage +80

Protection gains additional +20% Shield damage boost.

12500:
+7500 ATK, +30000 HP

13000:
Quantum Nanobot ATK +5%, HP +5%

13500:
Crit Damage +150
Chill Damage +180
Hoarfrost gains additional +35% damage to Chilled targets.
Hi-Energy gains additional +30% Crit Damage.

14000:
+7500 ATK, +30000 HP

14500:
Quantum Nanobot ATK +5%, HP +5%

15000:
Shield Damage +100
Crit Damage +180
Protection gains additional +20% Shield damage boost.
Hi-Energy gains additional +30% Crit Damage.

Database flags

TwinbornID = TB_SOCCER
Mode = Soccer / QuantumBall
PrimaryStats = ShieldDamage, CritDamage, ChillDamage
Debuff = Chill
SpecialMechanics = SpikedQuantumBall, IcyQuantumBall, SuperSoccerBall, HiEnergy, Protection, Hoarfrost
DPSRelevant = MediumHigh
IgnoreForDPS = HP

3. Normal Drone Resonance

Main association

Drone
Destroyer

Attack Wingbit
Augment Wingbit
Precision Device
Wingbit Damage
Wingbit Interval
Wingbit Shots

Important note from chart

Assumes Eternal tech part, R5 relic, and same ATK throughout.
Normal Drone resonance max shown: 9000.

Optimizer identity

Normal Drone resonance is a **new-player / pre-Twinborn Destroyer resonance path**. The guide says once Twinborn Destroyer exists, normal Destroyer resonance becomes mostly for newer players.

Still, it is extremely important before Twinborn.

Extracted resonance rows

50:

Total ATK 200, Total HP 1200
+200 ATK, +1200 HP

100:

+1 Attack Wingbit
Increase from previous 4.21%
Increase from base 4.21%

150:

Total ATK 600, Total HP 4200
+400 ATK, +2400 HP

200:

Attack Wingbit Damage +25%
Increase from previous 1.06%
Increase from base 2.00%

250:

Total ATK 1200, Total HP 8000
+600 ATK, +3200 HP

300:

+1 Augment Wingbit

Increase from previous 8.24%

Increase from base 14.00%

350:

Total ATK 2000, Total HP 12000

+800 ATK, +3200 HP

400:

Precision Device ATK +5%

450:

Augment Wingbit Damage Enhancement +10%

Increase from previous 3.81%

Increase from base 18.34%

500:

Total ATK 3000, Total HP 17000

+1000 ATK, +4000 HP

550:

Precision Device ATK +5%

600:

Attack Wingbit Interval -2s

Increase from previous 2.33%

Increase from base 21.10%

700:

Total ATK 4200, Total HP 23000

+1200 ATK, +4800 HP

800:

Precision Device ATK +5%

900:

Augment Wingbit Damage +10%

Increase from previous 3.58%

Increase from base 25.44%

1000:

Total ATK 5600, Total HP 30000

+1400 ATK, +5600 HP

1100:

Precision Device ATK +5%

1200:

Attack Wingbit Damage +50%

Increase from previous 2.64%

Increase from base 28.75%

1350:

Total ATK 7200, Total HP 38000

+1600 ATK, +6400 HP

1500:

Precision Device ATK +5%

1650:

Augment Wingbit Interval -1s

Increase from previous 2.70%

Increase from base 32.22%

1800:

Total ATK 9000, Total HP 47000

+1800 ATK, +7200 HP

1950:

Precision Device ATK +5%

2100:

Attack Wingbit +5 shots, Interval -1s

Increase from previous 6.58%

Increase from base 40.93%

2250:

Total ATK 11000, Total HP 57000

+2000 ATK, +8000 HP

2400:

Precision Device ATK +5%

2550:

Augment Wingbit Damage Enhancement +30%

Increase from previous 11.08%

Increase from base 56.55%

2700:

Total ATK 13500, Total HP 69500
+2500 ATK, +10000 HP

2850:

Precision Device ATK +5%

3000:

+1 Attack Wingbit
Increase from previous 12.97%
Increase from base 76.86%

3500:

Total ATK 16500, Total HP 84500
+3000 ATK, +12000 HP

4000:

Precision Device ATK +5%

4500:

Attack Wingbit Damage +25%, +5 shots
Increase from previous 9.53%
Increase from base 93.71%

5000:

Total ATK 20000, Total HP 102000
+3500 ATK, +14000 HP

5500:

Precision Device ATK +5%

6000:

Augment Wingbit Interval -1s
Increase from previous 4.70%
Increase from base 102.82%

6500:

Total ATK 24000, Total HP 122000
+4000 ATK, +16000 HP

7000:

Precision Device ATK +5%

7500:

Attack Wingbit Damage +25%

Increase from previous 3.54%

Increase from base 110.01%

8000:

Total ATK 30000, Total HP 152000

+6000 ATK, +24000 HP

8500:

Precision Device ATK +5%

9000:

Augment Wingbit Damage Enhancement +20%

Increase from previous 6.20%

Increase from base 123.02%

Database flags

TechResonanceID = NORMAL_DRONE

MainTech = Drone / Destroyer

SpecialUnits = AttackWingbit, AugmentWingbit

SupportUnit = PrecisionDevice

MaxResonance = 9000

DPSRelevant = VeryHighBeforeTwinborn

LateGameNote = Lower priority after Twinborn Destroyer

IgnoreForDPS = HP

4. Normal RPG Resonance

Main association

RPG

Sharkmaw Gun

C4 Bomb

Hi-Maintainer

C4 Explosion Radius

C4 Launch Interval

C4 Explosion Delay

C4 Chill

Sharkmaw Gun Damage

Optimizer identity

Normal RPG resonance is a **C4 / Sharkmaw Gun explosion path**. The newer guide says RPG and Molotov normal resonance are mostly not worth focusing once Twinborn versions exist, but the data is still useful for earlier accounts or source completeness.

Extracted resonance rows

50:

Total ATK 200, Total HP 1200
+200 ATK, +1200 HP

100:

Sharkmaw Gun launches C4 bomb.
Sharkmaw Gun damage +10%.

150:

Total ATK 600, Total HP 4200
+400 ATK, +2400 HP

200:

C4 bomb explosion radius +20%.
Sharkmaw Gun damage +10%.

250:

Total ATK 1200, Total HP 8000
+600 ATK, +3200 HP

300:

C4 +1.
Sharkmaw Gun damage +20%.

350:

Total ATK 2000, Total HP 12000
+800 ATK, +3200 HP

400:

Hi-Maintainer HP +5%

450:

C4 explosion radius +10%.
C4 launch interval -0.2s.
Sharkmaw Gun damage +20%.

500:

Total ATK 3000, Total HP 17000
+1000 ATK, +4000 HP

550:

Hi-Maintainer HP +5%

600:

C4 explosion delay -0.5s.
Sharkmaw Gun damage +20%.

700:

Total ATK 4200, Total HP 23000
+1200 ATK, +4800 HP

800:

Hi-Maintainer HP +5%

900:

C4 Chills for 30s.
C4 attack radius +30%.
Launch radius +20%.
C4 launch interval -0.5s.
Sharkmaw Gun damage +20%.

1000:

Total ATK 5600, Total HP 30000
+1400 ATK, +5600 HP

1100:

Hi-Maintainer HP +5%

1200:

C4 explosion damage +35%.
Sharkmaw Gun damage +20%.

1350:

Total ATK 7200, Total HP 38000
+1600 ATK, +6400 HP

1500:

Hi-Maintainer HP +5%

1650:

C4 explosion delay -0.5s.
Sharkmaw Gun damage +20%.

1800:
Total ATK 9000, Total HP 47000
+1800 ATK, +7200 HP

1950:
Hi-Maintainer HP +5%

2100:
C4 attack radius +20%.
C4 explosion delay -0.5s.
Sharkmaw Gun damage +20%.

2250:
Total ATK 11000, Total HP 57000
+2000 ATK, +8000 HP

2400:
Hi-Maintainer HP +5%

2550:
C4 explosion radius +20%.
C4 explosion delay -0.5s.
Sharkmaw Gun damage +20%.

2700:
Total ATK 13500, Total HP 69500
+2500 ATK, +10000 HP

2850:
Hi-Maintainer HP +5%

3000:
C4 explosion damage +35%.
Sharkmaw Gun damage +20%.

3500:
Total ATK 16500, Total HP 84500
+3000 ATK, +12000 HP

4000:
Hi-Maintainer HP +5%

4500:

C4 explosion radius +20%.

Sharkmaw Gun damage +10%.

5000:

Total ATK 20000, Total HP 102000

+3500 ATK, +14000 HP

5500:

Hi-Maintainer HP +5%

6000:

C4 explosion delay -1s.

Sharkmaw Gun damage +10%.

6500:

Total ATK 24000, Total HP 122000

+4000 ATK, +16000 HP

7000:

Hi-Maintainer HP +5%

7500:

C4 explosion damage +30%.

Launch radius +20%.

Sharkmaw Gun damage +10%.

8000:

Total ATK 30000, Total HP 152000

+6000 ATK, +24000 HP

8500:

Hi-Maintainer HP +5%

9000:

C4 explosion radius +30%.

C4 launch interval -0.2s.

Sharkmaw Gun damage +10%.

Database flags

TechResonanceID = NORMAL_RPG

MainTech = RPG / SharkmawGun

SpecialMechanics = C4Bomb, ExplosionRadius, LaunchInterval, ExplosionDelay, Chill
SupportUnit = HiMaintainer
MaxResonance = 9000
DPSRelevant = MediumHigh
LateGameNote = Lower priority than Twinborn Drill/RPG
IgnoreForDPS = HP

5. Normal Molotov Resonance

Main association

Molotov
Fuel Barrel
SF / Soulfire
Energy Diffuser
Poison
Deep Wound
Fuel Barrel Fire Damage
SF Damage
SF Generation Interval

Optimizer identity

Normal Molotov resonance is a **fire + poison + Deep Wound** path. It adds extra SF effects, poison duration, and Deep Wound bonuses.

Guide/meta note: normal Molotov resonance is lower priority once Twinborn exists, but this data is still useful for early/mid-game or exact source modeling.

Extracted resonance rows

50:

Total ATK 200, Total HP 1200
+200 ATK, +1200 HP

100:

Every 4 rounds of Fuel Barrel attacks, the next Fuel Barrel attack spawns 3 additional Soulfires. Soulfires will lock on to the nearest target.
Lock-on only triggers once.

150:

Total ATK 600, Total HP 3600
+400 ATK, +2400 HP

200:
Passive skill not required to evolve.
Soulfire adds a 5s Poison effect to the target it hits.
Does not stack.

250:
Total ATK 1200, Total HP 6800
+600 ATK, +3200 HP

300:
Soulfire generated +1.
Fuel Barrel fire damage +5%.

350:
Total ATK 2000, Total HP 10000
+800 ATK, +3200 HP

400:
Energy Diffuser ATK +5%

450:
Soulfire damage +20%.
When 3 Soulfires hit the same target, it is inflicted with +5% Deep Wound for 10s.

500:
Total ATK 3000, Total HP 14000
+1000 ATK, +4000 HP

550:
Energy Diffuser ATK +5%

600:
Soulfire spawn round interval -1

700:
Total ATK 4200, Total HP 18800
+1200 ATK, +4800 HP

800:
Energy Diffuser ATK +5%

900:

Soulfire Damage +30%

1000:

Total ATK 5600, Total HP 24400

+1400 ATK, +5600 HP

1100:

Energy Diffuser ATK +5%

1200:

Soulfire generated +1.

Fuel Barrel fire damage +5%.

1350:

Total ATK 7200, Total HP 30800

+1600 ATK, +6400 HP

1500:

Energy Diffuser ATK +5%

1650:

Soulfire Poison effect damage +100%.

Poison duration +2s.

1800:

Total ATK 9000, Total HP 38000

+1800 ATK, +7200 HP

1950:

Energy Diffuser ATK +5%

2100:

Soulfire generation interval -1

2250:

Total ATK 11000, Total HP 46000

+2000 ATK, +8000 HP

2400:

Energy Diffuser ATK +5%

2550:

Soulfire Poison effect damage +100%.

Fuel Barrel fire damage +5%.

2700:

Total ATK 13500, Total HP 56000
+2500 ATK, +10000 HP

2850:

Energy Diffuser ATK +5%

3000:

Soulfire damage +30%.
Deep Wound effect +5%.

3500:

Total ATK 16500, Total HP 68000
+3000 ATK, +12000 HP

4000:

Energy Diffuser ATK +5%

4500:

Soulfire generated +1.
Fuel Barrel fire damage +3%.

5000:

Total ATK 20000, Total HP 82000
+3500 ATK, +14000 HP

5500:

Energy Diffuser ATK +5%

6000:

Soulfire Damage +15%

6500:

Total ATK 24000, Total HP 98000
+4000 ATK, +16000 HP

7000:

Energy Diffuser ATK +5%

7500:

Soulfire generated +1.
Fuel Barrel fire damage +5%.

8000:

Total ATK 30000, Total HP 122000

+6000 ATK, +24000 HP

8500:

Energy Diffuser ATK +5%

9000:

Soulfire Damage +10%.

Deep Wound effect +5%.

Database flags

TechResonanceID = NORMAL_MOLOTOV

MainTech = Molotov / FuelBarrel

SpecialMechanics = Soulfire, Poison, DeepWound, FuelBarrelFireDamage

SupportUnit = EnergyDiffuser

MaxResonance = 9000

DPSRelevant = MediumHigh

LateGameNote = Lower priority than Twinborn tech paths

IgnoreForDPS = HP

Batch 3 optimizer takeaways

Strongest mechanics in this batch

Twinborn Durian:

- Poison application
- Permanent Skill Damage mutation
- Permanent Poisoned-target damage boost
- Shield Damage

Twinborn Soccer:

- Crit Damage
- Chill Damage
- Shield Damage
- Hoarfrost / Hi-Energy / Protection effects

Normal Drone:

- Attack Wingbit

- Augment Wingbit
- Best normal resonance path before Twinborn Destroyer

Normal RPG:

- C4 explosion scaling
- C4 Chill
- Sharkmaw Gun damage

Normal Molotov:

- Soulfire generation
- Poison effect
- Deep Wound
- Fuel Barrel fire damage

Highest-value DPS fields added

PoisonDamage
ChillDamage
CritDamage
ShieldDamage
SkillDamage
DeepWound
DamageToPoisonedTargets
DamageToChilledTargets
AttackWingbitCount
AugmentWingbitCount
C4ExplosionRadius
C4ExplosionDelay
SoulfireGenerated

Recommendation logic update

If player has no Twinborn:

Normal Drone Resonance is very important.

If player has Twinborn Drone/Forcefield:

Shift priority away from normal Drone resonance into Twinborn resonance/overload.

If player is choosing between normal RPG/Molotov resonance and Twinborn:

Prefer Twinborn paths unless the player is too early to unlock them.

Twinborn Durian/Soccer:

Store fully, but rank lower than Drone/Forcefield, Drill/RPG, Lightning/Boomerang, Laser/Guardian, and Molotov/Bri

Batch 4 — Tech Parts: Twinborn full effects + Twinborn collectible sets

Sources extracted from your images:

Tech Parts - Twinborn

Tech Parts - Offensive / Defensive cleanup

Game version 4.5

Updated 2026-02-23

This batch is about **what each Twinborn tech does at rarity stages** and what collectible sets are associated with each Twinborn.

1. Twinborn Durian / Twinborn Caltrops

Pair association

TwinbornPairID = TB_DURIAN_SOCCER

Mode = Durian side

BaseTech = Durian

PairTech = Soccer Ball

CanEvolveInto = Twinborn Caltrops

Effects by rarity

Blue:

ATK +100

HP +400

Purple:

Durian turns into Twinborn Durian in a match and can evolve into Twinborn Caltrops.

Twinborn Durian:

Flight control.

Durian will not stray too far from you.

Soccer Ball:

Ball damage +30%.

Yellow:

ATK +400

HP +1600
Red:
Twinborn Caltrops:
Iron and Blood.
Spikes also inflict Bleed.
Adds 1 Transgenic Caltrops.
Skill damage for entire match +75%.

Quantum Ball:
Ball duration +1s.
Eternal / Rainbow:
Twinborn Durian:
Leaves behind pungent whiff during flight.
Follows bosses on hit.

Twinborn Caltrops:
Leaves behind fire trail during flight.
Leaves behind 1 fire bomb every 1s that explodes after 3s.
Transgenic Caltrops deal +20% damage.

Crit Damage for entire match +75%.

Soccer Ball:
+1 Ball.

Quantum Ball:
+1 Ball.
Quantum entanglement occurs after the balls eventually split up.

Collectible set

CollectibleSetName = Close to Creation
AssociatedTwinborn = Twinborn Durian / Twinborn Caltrops
Each 3 yellow stars:
Twinborn Caltrops spike damage +100%.

Each 3 red stars:
Twinborn Durian and Twinborn Caltrops damage +100%.
Shield Damage Boost +10%.

Total 25 stars:
Twinborn Durian and Twinborn Caltrops damage +100%.
Shield Damage Boost +15%.

Optimizer flags

Debuff = Bleed

Stats = SkillDamage, CritDamage, ShieldDamage

SpecialMechanics = TransgenicCaltrops, FireBombs, FireTrail, PungentWhiff

DPSRelevant = Medium/High

MetaPriority = Lower Twinborn priority, but real poison/bleed/skill scaling exists

2. Twinborn Soccer Ball / Twinborn Quantum Ball

Pair association

TwinbornPairID = TB_DURIAN_SOCCER

Mode = Soccer side

BaseTech = Soccer Ball

PairTech = Durian

CanEvolveInto = Twinborn Quantum Ball

Effects by rarity

Blue:

ATK +100

HP +400

Purple:

Soccer Ball skill becomes Twinborn Soccer Ball in a match and can evolve into Twinborn Quantum Ball.

Twinborn Soccer Ball:

Twinborn Soccer Ball damage +30%.

Durian:

Flight control.

Durian will not stray too far from you.

Yellow:

ATK +400

HP +1600

Red:

Twinborn Quantum Ball:

Quantum balls turn into Spiked Quantum Balls.

Inflicts Bleed on hit for 20s.
Spiked Quantum Ball duration +1s.
Crit Damage for entire match +80%.

Caltrops exclusive:
Iron and Blood.
Caltrop spikes inflict Bleed.
Eternal / Rainbow:
Twinborn Soccer Ball:
Kicks 1 more Soccer Ball.

Twinborn Quantum Ball:
Fires 1 more Spiked Quantum Ball.
Damage to Chilled targets +30%.
Shield Damage Boost +20%.

Durian:
Leaves a pungent trail during flight.
Follows bosses on hit.

Caltrops:
Leaves a fiery trail during flight.
Drops a fire bomb every 1s that explodes after 3s.

Collectible set

CollectibleSetName = Open Void Gate
AssociatedTwinborn = Twinborn Soccer Ball / Twinborn Quantum Ball
Each 3 yellow stars:
Twinborn Quantum Ball +1s.
Shield Damage Boost +5%.

Each 3 red stars:
Twinborn Soccer Ball damage +30%.
Skill Damage +10%.

Total 25 stars:
Twinborn Soccer Ball damage +45%.
Shield Damage Boost +10%.
Crit Damage +10%.

Optimizer flags

Debuff = Bleed, Chill synergy

Stats = CritDamage, SkillDamage, ShieldDamage, DamageToChilled
SpecialMechanics = SpikedQuantumBall, QuantumEntanglement
DPSRelevant = Medium/High
MetaPriority = Lower than Drone/Forcefield and Drill/RPG, but important for crit/chill builds

3. Twinborn Drone / Twinborn Destroyer

Pair association

TwinbornPairID = TB_DRONE_FORCEFIELD
Mode = Drone side
BaseTech = Drone
PairTech = Forcefield
CanEvolveInto = Twinborn Destroyer

Effects by rarity

Blue:

ATK +100

HP +400

Purple:

Drone skills become Twinborn Drone in a match and can evolve into Twinborn Destroyer.

Twinborn Drone:

Auto-Guidance.

Missiles home in on enemies in firing range, but with reduced fire rate.

Forcefield:

Energy Compression.

Range -25%, 2x damage.

Yellow:

ATK +400

HP +1600

Red:

Twinborn Destroyer exclusive:

Newtype Warhead.

Restores fire rate and missile explosion range is doubled.

Twinborn Destroyer generates a destruction forcefield.

EXP +25%.

Twinborn Destroyer missile damage +6%.

Skill Damage +75%.

Force Barrier exclusive:

Gain extra EXP when enemies are defeated in Forcefield range.

Eternal / Rainbow:

Twinborn Drone:

Starts at Lv.3.

Twinborn Destroyer:

Homing range up.

More missiles.

Twinborn Destroyer missile damage +10%.

+50% damage to Weakened targets.

Forcefield:

Restores Forcefield range.

Fires a large-area pulse every few seconds.

Force Barrier:

Fires a suppression pulse every few seconds.

Monsters hit by pulse are Snared, excluding bosses.

Collectible set

CollectibleSetName = Transgalactic Tentacle

AssociatedTwinborn = Twinborn Drone / Twinborn Destroyer

10x yellow stars:

Twinborn Drone missile count +5%.

Single shot damage +5%.

20x yellow stars:

Twinborn Drone missile count +10%.

Single shot damage +10%.

10x red stars:

Twinborn Drone missile count +10%.

Single shot damage +10%.

20x red stars:

Twinborn Drone missile count +10%.

Single shot damage +10%.

Optimizer flags

Debuff = Weakened target damage

Stats = SkillDamage, DamageToWeakened, MissileDamage
SpecialMechanics = HomingMissiles, DestructionForcefield, ForcefieldPulse, SuppressionPulse
DPSRelevant = Very High
MetaPriority = Highest Twinborn priority

This is still the king tech path for your optimizer.

4. Twinborn Forcefield / Twinborn Force Barrier

Pair association

TwinbornPairID = TB_DRONE_FORCEFIELD
Mode = Forcefield side
BaseTech = Forcefield
PairTech = Drone
CanEvolveInto = Twinborn Force Barrier

Effects by rarity

Blue:
ATK +100
HP +400
Purple:
In-match Forcefield skill turns into Twinborn Forcefield.
Can evolve into Twinborn Force Barrier.

Twinborn Forcefield:
Energy Compression.
Range -25%.
2x damage.

Drone:
Auto-homing.
Missiles auto-track enemies in range, but fire rate is reduced.

Yellow:

ATK +400
HP +1600

Red:

Twinborn Force Barrier exclusive:

Enemies defeated inside Twinborn Force Barrier yield bonus EXP.
Skill Damage +120%.

Destroyer exclusive:
Newtype Warhead.
Restores fire rate and missile explosion range is doubled.
Eternal / Rainbow:
Twinborn Forcefield:
Restores Twinborn Forcefield range.
Fires a large-area pulse every few seconds.
Crit Damage +160%.
Crit Rate +20%.

Twinborn Force Barrier:
Fires a suppression pulse every few seconds.
Monsters hit by pulse are Snared, excluding bosses.

Drone:
Starts at Lv.2.

Destroyer:
Homing range up.
More missiles.

Collectible set

CollectibleSetName = Impression Idols
AssociatedTwinborn = Twinborn Forcefield / Twinborn Force Barrier
10x yellow stars:
Twinborn Forcefield and Twinborn Force Barrier damage +21%.

20x yellow stars:
Twinborn Forcefield and Twinborn Force Barrier damage +21%.
Vulnerability effect +10%.
Crit Damage +15%.

10x red stars:
Twinborn Forcefield and Twinborn Force Barrier damage +21%.
Damage rate +25%.

20x red stars:
Twinborn Forcefield and Twinborn Force Barrier damage +21%.
Vulnerability effect +10%.
Crit Damage +15%.

Skill Damage +20%.

Optimizer flags

Debuff = Snare, Vulnerability from collectible

Stats = SkillDamage, CritDamage, CritRate, DamageRate

SpecialMechanics = ForcefieldPulse, SuppressionPulse, BonusEXP

DPSRelevant = High

MetaPriority = Part of best Twinborn pair because it unlocks Drone/Forcefield

5. Twinborn Drill Shot / Twinborn Frostfire Dualist

Pair association

TwinbornPairID = TB_DRILL_RPG

Mode = Drill side

BaseTech = Drill Shot

PairTech = RPG

CanEvolveInto = Twinborn Frostfire Dualist

Effects by rarity

Blue:

ATK +100

HP +400

Purple:

Drill Shot turns into Twinborn Drill Shot.

Can evolve into Twinborn Frostfire Dualist.

Fire Whistleshot + Ice Whistleshot.

Twinborn Drill Shot:

Drill Shot flight speed +30%.

RPG:

Firing cooldown -0.5s.

Yellow:

ATK +400

HP +1600

Red:

Twinborn Frostfire Dualist:
After Fire Whistleshot hits, a Fire Bomb is installed on the target.

Sharkmaw Gun exclusive:
Double shot.
Fires 2 shots each time.

Crit Damage +50%.
Eternal / Rainbow:
Twinborn Drill Shot:
Targets hit by Twinborn Drill Shot take extra 5% damage.

Twinborn Frostfire Dualist:
Fire Whistleshot damage doubled.
Targets hit by Fire Whistleshot take extra 900% damage from Ice Whistleshot for 1.5s.
Launches an extra fire missile at targets hit by fire bomb explosions.

RPG:
Explosion gains pull effect.

Sharkmaw Gun:
Triggers mini-explosion on enemy contact.

Shield Damage Boost +75%.

Collectible set

CollectibleSetName = Multiverse Perspective
AssociatedTwinborn = Twinborn Drill Shot / Twinborn Frostfire Dualist
10x yellow stars:
Twinborn Drill and Twinborn Frostfire Dualist damage +10%.
Fire Bomb Damage +15%.

20x yellow stars:
Twinborn Drill and Twinborn Frostfire Dualist damage +18%.
Fire Bomb Damage +25%.

10x red stars:
Twinborn Drill and Twinborn Frostfire Dualist damage +18%.
Fire Bomb Damage +25%.

20x red stars:
Twinborn Drill and Twinborn Frostfire Dualist damage +25%.
Fire Bomb Damage +35%.

Optimizer flags

Debuff = DamageTaken, Chill/Fire interaction

Stats = CritDamage, ShieldDamage, FireBombDamage

SpecialMechanics = FireWhistleshot, IceWhistleshot, FireBomb, PullEffect

DPSRelevant = Very High late

MetaPriority = Second Eternal Twinborn priority from guide

6. Twinborn RPG / Twinborn Blaster Sharkmaw Gun

Pair association

TwinbornPairID = TB_DRILL_RPG

Mode = RPG side

BaseTech = RPG

PairTech = Drill Shot

CanEvolveInto = Twinborn Blaster Sharkmaw Gun

Effects by rarity

Blue:

ATK +100

HP +400

Purple:

RPG skill turns into Twinborn RPG.

Can evolve into Twinborn Blaster Sharkmaw Gun.

Twinborn RPG:

Fire rate cooldown -0.5s.

Drill Shot:

Drill flight speed +30%.

Yellow:

ATK +400

HP +1600

Red:

Twinborn Blaster Sharkmaw Gun:

Blaster Sharkmaw Gun fires Cluster Arrows upon explosion, inflicting Poison on targets.

Whistling Arrow EX:
Light and shadow at once.
Gets 2 arrows.

Shield Damage Boost +50%.
Eternal / Rainbow:
Twinborn Blaster Sharkmaw Gun:
Explosions pull enemies in.
Cluster Arrows can split 1 time.

Drill Shot:
Targets hit by Drill Shot take +5% damage.

Whistling Arrow:
First Whistling Arrow damage doubled.
When target is hit by first Whistling Arrow, second Whistling Arrow deals +900% damage for 1.5s.

Skill Damage +75%.

Collectible set

CollectibleSetName = Hyperrift Tech
AssociatedTwinborn = Twinborn RPG / Twinborn Blaster Sharkmaw Gun
10x yellow stars:
Blaster Sharkmaw Gun damage +24%.

20x yellow stars:
Blaster Sharkmaw Gun damage +48%.
Damage to Lacerated targets +5%.

10x red stars:
Blaster Sharkmaw Gun damage +36%.

20x red stars:
Cluster Arrows damage +25%.
Damage to Lacerated targets +5%.

Optimizer flags

Debuff = Poison, Laceration synergy
Stats = SkillDamage, ShieldDamage, DamageToLacerated
SpecialMechanics = ClusterArrows, Poison, PullExplosion
DPSRelevant = High

MetaPriority = Strong late Twinborn path, especially with Poison/Laceration

7. Twinborn Lightning Emitter / Twinborn Maelstorm

Pair association

TwinbornPairID = TB_LIGHTNING_BOOMERANG

Mode = Lightning side

BaseTech = Lightning Emitter

PairTech = Boomerang

CanEvolveInto = Twinborn Maelstorm

Effects by rarity

Blue:

ATK +100

HP +400

Purple:

Lightning Emitter skill turns into Twinborn Lightning Emitter.

Can evolve into Twinborn Maelstorm.

Twinborn Lightning Emitter:

Bolt count at all levels +2.

Boomerang:

You can catch the boomerang and reset its cooldown.

Yellow:

ATK +400

HP +1600

Red:

Twinborn Maelstorm:

Falling thunder generates swirling Thunderburst fields.

Subsequent lightning strikes that hit the field trigger Thunderbursts.

Magnetic Rebounder EX:

EM induction.

Rebounder generates lightning arcs.

Crit Damage +80%.

Eternal / Rainbow:
Twinborn Maelstorm:
The last lightning strike is more powerful.

Boomerang:
Each hit deals additional damage.

Magnetic Rebounder:
Fires a chain lightning at nearby enemies every second.
Bounces to 3 targets.

Skill Damage +80%.

Collectible set

CollectibleSetName = Rewriting the Stars' Memories
AssociatedTwinborn = Twinborn Lightning / Twinborn Maelstorm
10x yellow stars:
Twinborn Maelstorm damage +9%.

20x yellow stars:
Twinborn Maelstorm damage +9%.

10x red stars:
Twinborn Maelstorm damage +9%.

20x red stars:
Twinborn Maelstorm damage +17%.

Optimizer flags

Debuff = Vulnerability from resonance rows
Stats = CritDamage, SkillDamage
SpecialMechanics = ThunderburstField, ChainLightning, FallingThunder
DPSRelevant = High/Very High
MetaPriority = Third Eternal Twinborn priority from guide

8. Twinborn Boomerang / Twinborn Magvolt Fang

Pair association

TwinbornPairID = TB_LIGHTNING_BOOMERANG

Mode = Boomerang side

BaseTech = Boomerang

PairTech = Lightning Emitter

CanEvolveInto = Twinborn Magvolt Fang

Effects by rarity

Blue:

ATK +100

HP +400

Purple:

Boomerang skill turns into Twinborn Boomerang.

Can evolve into Twinborn Magvolt Fang.

Twinborn Boomerang:

You can catch the boomerang and reset its cooldown.

Lightning Emitter:

Bolt count at all levels +2.

Yellow:

ATK +400

HP +1600

Red:

Twinborn Magvolt Fang:

Magvolt Fang generates a lightning zone within its projectiles, zapping enemies inside it.

Supercell EX:

Generated shockwave spread distance doubled.

Damage to Weakened targets +50%.

Eternal / Rainbow:

Twinborn Magvolt Fang:

Magvolt Fang hits trigger multiple lightning strikes.

Lightning Emitter:

Falling thunder adds another Thunderburst strike.

Supercell:

Falling thunder adds another Thunderburst damage.

Shockwave spread speed increased and returns.

Skill Damage +100%.

Collectible set

CollectibleSetName = Goldfinger

AssociatedTwinborn = Twinborn Boomerang / Twinborn Magvolt Fang

10x yellow stars:

Skill Damage +20%.

20x yellow stars:

Twinborn Magvolt Fang damage +30%.

Lightning Zone strike Vulnerability max stacks +2.

10x red stars:

Skill Damage +20%.

Damage to Weakened targets +10%.

20x red stars:

Lightning Zone strike Vulnerability max stacks +3.

Damage to Weakened targets +20%.

Optimizer flags

Debuff = Vulnerability, Weaken

Stats = SkillDamage, DamageToWeakened

SpecialMechanics = LightningZone, MagvoltFang, Thunderburst

DPSRelevant = High

MetaPriority = Better than normal Boomerang; becomes real DPS through Twinborn

9. Twinborn Guardian / Twinborn Defender

Pair association

TwinbornPairID = TB_LASER_GUARDIAN

Mode = Guardian side

BaseTech = Guardian

PairTech = Laser Launcher

CanEvolveInto = Twinborn Defender

Effects by rarity

Blue:

ATK +100

HP +400

Purple:

In-match Guardian skill becomes Twinborn Guardian.

Can evolve into Twinborn Defender.

Twinborn Guardian:

Knockback distance doubled.

Laser Launcher:

Lasers slow enemies on hit.

Yellow:

ATK +400

HP +1600

Red:

Twinborn Guardian:

Guardian becomes Twinborn Guardian.

Damage +200%.

Twinborn Defender EX:

Twinborn Defender damage +200%.

Chills targets for 5s.

Damage to Chilled targets +70%.

Death Ray EX:

Snares enemies.

Bosses and elites first.

Eternal / Rainbow:

Twinborn Guardian:

Enemies hit by Guardian deal -10% damage.

Additional Twinborn Guardian damage +300%.

Twinborn Defender EX:

Additional Twinborn Defender damage +300%.

Twinborn Defender inflicts Lacerate for 5s on targets hit.

Skill Damage +100%.

Laser Launcher:

Triggers delayed explosions along the laser path.

Death Ray:

Triggers small-range explosion when rays gather.

Collectible set

CollectibleSetName = Cyber Wonderland

AssociatedTwinborn = Twinborn Guardian / Twinborn Defender

10x yellow stars:

Twinborn Defender damage +17%.

20x yellow stars:

Twinborn Defender damage +17%.

Skill Damage +10%.

10x red stars:

Damage to Poisoned targets +10%.

20x red stars:

Twinborn Defender damage +33%.

Damage to Poisoned targets +10%.

Skill Damage +10%.

Optimizer flags

Debuff = Chill, Laceration, Poisoned target synergy

Stats = SkillDamage, DamageToChilled, DamageToPoisoned

SpecialMechanics = Defender, Freeze/Chill, Lacerate

DPSRelevant = Medium/High

MetaPriority = Good but behind Drone/Forcefield and Drill/RPG

10. Twinborn Laser Launcher / Twinborn Death Ray

Pair association

TwinbornPairID = TB_LASER_GUARDIAN

Mode = Laser side

BaseTech = Laser Launcher

PairTech = Guardian

CanEvolveInto = Twinborn Death Ray

Effects by rarity

Blue:

ATK +100

HP +400

Purple:

In-match Laser Launcher skill becomes Twinborn Laser Launcher.

Can evolve into Twinborn Death Ray.

Twinborn Laser Launcher:

Lasers slow enemies on hit.

Guardian:

Knockback distance doubled.

Yellow:

ATK +400

HP +1600

Red:

Twinborn Death Ray EX:

Continuously tracks and locks onto enemies.

Bosses and elites first.

Crit Rate +20%.

Crit Damage +40%.

Defender EX:

Defender spreads out and retreats.

Eternal / Rainbow:

Twinborn Laser Launcher:

Triggers delayed explosions along the laser path.

Twinborn Death Ray:

Triggers small-range explosion when rays gather.

Twinborn Death Ray damage +10%.

Skill Damage +80%.

Crit Damage +40%.

Guardian:

Enemies hit by Guardian deal -10% damage.

Defender:

Each Defender revolves around several smaller Guardians.

Collectible set

CollectibleSetName = Try All Possibilities

AssociatedTwinborn = Twinborn Laser Launcher / Twinborn Death Ray

10x yellow stars:
Crit Damage +20%.

20x yellow stars:
Shield Damage Boost +20%.
Damage to Poisoned targets +10%.

10x red stars:
Twinborn Death Ray damage +4%.
Skill Damage +20%.

20x red stars:
Crit Damage +20%.
Damage to Poisoned targets +10%.

Optimizer flags

Debuff = Slow/Snare, Poisoned target synergy from collectible
Stats = CritRate, CritDamage, SkillDamage, ShieldDamage
SpecialMechanics = LockOn, DeathRayExplosion, LaserDelayedExplosions
DPSRelevant = High
MetaPriority = Strong if crit/skill damage scaling is needed

11. Twinborn Brick / Twinborn 1-ton Iron

Pair association

TwinbornPairID = TB_MOLOTOV_BRICK
Mode = Brick side
BaseTech = Brick
PairTech = Molotov
CanEvolveInto = Twinborn 1-ton Iron

Effects by rarity

Blue:
ATK +100
HP +400
Purple:
In-match Brick skill turns into Twinborn Brick.
Can evolve into Twinborn 1-ton Iron.

Twinborn Brick:
Heavy like it.
Penetrates enemies.

Molotov:
Fires will spread.

Yellow:

ATK +400

HP +1600

Red:

Twinborn 1-ton Iron:

On hit, Twinborn 1-ton Iron shatters space, dealing small-area damage.

Shatter Damage +150%.

Crit Damage +80%.

Fuel Barrel EX:

Burnt enemies inflicted with Deep Wounds.

+10% damage taken.

Eternal / Rainbow:

Twinborn Brick:

Count +2 at all levels.

Twinborn 1-ton Iron:

Count +2.

1-ton Iron Weakens targets hit for 10s.

Twinborn 1-ton Iron damage +170%.

Damage to Weakened targets +70%.

Molotov:

Damage rate up.

Fuel Barrel:

Fire trail follows character.

Collectible set

CollectibleSetName = Brewing Recipe

AssociatedTwinborn = Twinborn Brick / Twinborn 1-ton Iron

10x yellow stars:

Twinborn Brick damage +5%.

Shield Damage Boost +10%.

20x yellow stars:

Twinborn Brick damage +5%.
Skill Damage +20%.

10x red stars:
Twinborn Brick damage +5%.
Crit Damage +20%.

20x red stars:
Twinborn Brick damage +5%.
Shield Damage Boost +20%.

Optimizer flags

Debuff = Deep Wound, Weaken
Stats = CritDamage, SkillDamage, ShieldDamage, DamageToWeakened
SpecialMechanics = ShatterSpace, 1TonIronCount, WeakeningHits
DPSRelevant = High
MetaPriority = Strong Legendary priority because Molotov/Brick is second Legendary Twinborn path

12. Twinborn Molotov / Twinborn Fuel Spray

Pair association

TwinbornPairID = TB_MOLOTOV_BRICK
Mode = Molotov side
BaseTech = Molotov
PairTech = Brick
CanEvolveInto = Twinborn Fuel Spray

Effects by rarity

Blue:
ATK +100
HP +400
Purple:
In-match Molotov skill becomes Twinborn Molotov.
Can evolve into Twinborn Fuel Spray.

Twinborn Molotov:
Fires will spread.

Brick:
Heavy like it.
Brick penetrates all enemies.

Yellow:

ATK +400

HP +1600

Red:

Twinborn Fuel Spray:

Shoots a Blazing Jet forward, inflicting Deep Wound for 10s.

Targets take additional 10% damage.

When the skill ends, throws Spray Canister that explodes on hit.

Blazing Jet and Spray Canister ignite elites and bosses on hit, burning them for 10s.

Skill Damage +50%.

1-ton Iron EX:

Light as a Feather.

1-ton Iron spins as they fly out.

Eternal / Rainbow:

Twinborn Molotov:

Damage rate up.

Twinborn Fuel Spray:

Spray Canister hits grant +2.5% Shield to main character.

Shield Damage Boost +50%.

Skill Damage +30%.

Brick:

Brick count at all levels +2.

1-ton Iron:

Count +2.

Shatters space on hit, dealing damage in small area.

Collectible set

CollectibleSetName = Wind Totem

AssociatedTwinborn = Twinborn Molotov / Twinborn Fuel Spray

10x yellow stars:

Twinborn Molotov Deep Wound +10%.

20x yellow stars:

Twinborn Molotov Deep Wound +15%.

10x red stars:

Twinborn Molotov Damage +10%.

20x red stars:

Twinborn Molotov Damage +20%.

Optimizer flags

Debuff = Deep Wound, Burn

Stats = SkillDamage, ShieldDamage

SpecialMechanics = BlazingJet, SprayCanister, BossBurn, DeepWound

DPSRelevant = High

MetaPriority = Good Legendary priority, lower Eternal priority

Batch 4 optimizer table structure

Use this table for the database:

TwinbornEffectID

TwinbornPairID

ModeName

BaseTech

PairedTech

Rarity

EffectText

AffectedStat

StatAmount

AssociatedDebuff

AssociatedSkill

Condition

DPSRelevant

IgnoreForDPS

SourceImage

SourceDate

Example rows:

TB_DRONE_RED_DESTROYER | TB_DRONE_FORCEFIELD | Twinborn Destroyer | Drone | Forcefield | Red | Newtype Warhead restores fire rate and doubles explosion range; creates

destruction forcefield; EXP +25%; missile damage +6%; skill damage +75% | SkillDamage | +75% | None | Twinborn Destroyer | Evolved | Yes | No | Tech Parts - Twinborn | 2026-02-23
TB_LASER_RED_DEATHRAY | TB_LASER_GUARDIAN | Twinborn Death Ray | Laser
Launcher | Guardian | Red | Tracks and locks onto enemies, bosses and elites first; Crit Rate +20%; Crit Damage +40% | CritRate/CritDamage | +20%/+40% | None | Twinborn Death Ray | Evolved | Yes | No | Tech Parts - Twinborn | 2026-02-23
TB_BRICK_ETERNAL_1TON | TB_MOLOTOV_BRICK | Twinborn 1-ton Iron | Brick | Molotov | Eternal | 1-ton Iron count +2; weakens targets for 10s; damage +170%; damage to Weakened targets +70% | DamageToWeakened | +70% | Weaken | Twinborn 1-ton Iron | Evolved | Yes | No | Tech Parts - Twinborn | 2026-02-23

Biggest takeaways from Batch 4

Best Twinborn pair overall:

Drone + Forcefield

Best pure late-game damage pair after that:

Drill + RPG

Best debuff/crit scaling pair:

Lightning + Boomerang

Best crit/laser scaling pair:

Laser + Guardian

Best burn/deep wound pair:

Molotov + Brick

Lowest priority pair:

Soccer + Durian

For the optimizer, the strongest Twinborn stats to prioritize are:

Skill Damage

Crit Rate

Crit Damage

Vulnerability

Damage to Weakened

Damage to Chilled

Damage to Lacerated

Deep Wound

Shield Damage Boost

Missile count/damage
Fire Bomb damage
Cluster Arrow damage

Batch 5 — Tech Resonance Cost, Rules, Chip Cost, Overload

Sources:
[Tech_Part_Resonance_Cost_2024-11-28](#), [Resonance_2024-10-30](#), normal resonance charts, and tech guide text.

A) Resonance energy by tech rarity

Tech rarity / level	Resonance energy
Yellow / Epic	50
Yellow +1	100
Yellow +2	150
Yellow +3	200
Red / Legendary	300
Red +1	400
Red +2	550
Red +3	700
Red +4	850
Eternal / Rainbow	1000

Optimizer fields

TechRarity | ResonanceEnergy | CanBeAssist | DPSRelevant
Yellow | 50 | Yes | Indirect

Red	300	Yes	Indirect
Eternal	1000	Yes	Indirect

B) Resonance multiplier chip cost

Multiplier upgrade	Chip cost
1.0 → 1.2	1
1.2 → 1.4	1
1.4 → 1.6	2
1.6 → 1.8	3
1.8 → 2.0	5
2.0 → 2.2	5
2.2 → 2.4	7
2.4 → 2.6	9
2.6 → 2.8	12
2.8 → 3.0	15
Total to 3.0	60 chips

Important optimizer rule

Resonance chip upgrades are slot-specific.

Example: chips spent on Drone resonance do not boost RPG, Molotov, or Twinborn slots.

C) Normal tech resonance rules

Rule	Data
Unlock requirement	Legendary tech
Max resonance	9000

Assist restriction	Offensive → Offensive, Defensive → Defensive
Same tech as assist	Allowed
Assist tech lost?	No, can unequip/re-equip
Max setup shown	3 Eternal assists + 3x multiplier
Chip cost to max multiplier	60 chips

Normal resonance table fields

NormalResonanceID
 MainTech
 MainTechType
 AssistTechTypeAllowed
 MaxResonance
 CurrentResonance
 Multiplier
 ChipInvested
 BuffUnlocked
 DPSRelevant

D) Twinborn resonance rules

Rule	Data
Unlock requirement	Legendary rarity of both paired techs
Eternal Twinborn requirement	2 Eternal techs
Max resonance	15000
Assist restriction	No type restriction
Offensive assist allowed?	Yes
Defensive assist allowed?	Yes
Same tech as assist	Allowed
Can equip both modes?	No

Max setup shown	3 Eternal techs + 10 Legendary techs + 90 chips
Main advantage	Bigger resonance table + Tech Overload access

Twinborn pair rule

One Twinborn pair has 2 modes, but only 1 mode can be equipped.

Example:

Drone + Forcefield pair can use Twinborn Destroyer OR Twinborn Force Barrier, not both.

E) Normal vs Twinborn resonance comparison

System	Unlock	Max resonance	Fodder restriction	Chip max noted	Priority
Normal tech	Legendary tech	9000	Same side only	60	Early/mid
Twinborn tech	Legendary pair	15000	Any tech type	90	Late/current meta

Optimizer rule

If Twinborn is unlocked:

prioritize Twinborn resonance over normal RPG/Molotov resonance.

If no Twinborn:

normal Drone resonance is still high value.

F) Tech Overload

Field	Data
Unlock requirement	3000 resonance on a Twinborn
Max levels	18

Full access requirement	15000 resonance
Upgrade material	Harmony/Resonance chips
Applies to	Twinborn tech overload system

Logic

```
if TwinbornResonance < 3000:
    TechOverloadUnlocked = False
    MaxOverloadLevel = 0
```

```
if TwinbornResonance >= 3000:
    TechOverloadUnlocked = True
```

```
if TwinbornResonance == 15000:
    MaxOverloadLevel = 18
```

Database fields

TechOverloadID
TwinbornID
RequiredResonance
OverloadLevel
ChipCost
EffectText
AffectedStat
DPSRelevant

G) Progression priority from tech guide

Excellent tech priority

Rank	Tech
1	Drone / Destroyer
2	Molotov
3	Guardian

Legendary Twinborn priority

Rank	Twinborn pair
1	Drone + Forcefield
2	Molotov + Brick
3	Laser + Guardian
4	Lightning + Boomerang
5	Drill + RPG
6	Soccer + Durian

Eternal Twinborn priority

Rank	Twinborn pair
1	Drone + Forcefield
2	Drill + RPG
3	Lightning + Boomerang
4	Laser + Guardian
5	Molotov + Brick
6	Soccer + Durian

H) Optimizer scoring rules

1. Do not recommend resonance before Legendary tech.
 2. Do not recommend Twinborn before both paired techs are Legendary.
 3. Do not recommend Tech Overload before 3000 Twinborn resonance.
 4. Value ATK, skill damage, crit, vulnerability, poison, chill, laceration, weaken, and shield damage.
 5. Store HP but mark IgnoreForDPS.
 6. Normal Drone resonance is good before Twinborn.
 7. Normal RPG/Molotov resonance becomes lower value once Twinborn options exist.
 8. Resonance chips should be spent by DPS gain per chip, not evenly across every tech.
-

I) Clean database tables needed

TechResonanceEnergyTable
TechMultiplierChipCostTable
NormalTechResonanceRules
TwinbornResonanceRules
TechOverloadRules
TechProgressionPriority

Batch 6 — Equipment Guide Extraction

Source: [Equipment_Meta_Guide_Griffith_2026](#) from your pasted Discord guide. This is **meta/priority data**, not exact stat-chart data. Use it for recommendations, warnings, and progression logic.

A) Gear priority by slot

Slot	Item	What it does / value	Optimizer note
Weapon	SS Weapon / Twin Lance	Best-in-slot for almost all modes. Best boss damage, decent crowd control.	Highest priority SS item.
Weapon	Kunai	Good early Ender's Echo and boss damage. Weaker late game.	Save one Excellent Kunai.
Weapon	Void Power	Best early/mid survival weapon. Best crowd control, weak boss damage. Purple enough until about Chapter 100.	Rush early. Do not overinvest forever.
Weapon	Lightchaser	Best for first 50 chapters. Falls off when mobs get high HP.	Early only.
Weapon	Sword of Disorder	"Dollar-store SS weapon." Good boss damage, weak crowd control.	Low priority.
Weapon	Other weapons	Trash / fodder.	Feed/salvage later.

Weapon TLDR

Get Purple Void Power + Excellent Kunai early.
Then focus on SS Weapon / Twin Lance.

B) Necklace guide

Necklace	What it does / value	Optimizer note
SS Necklace	Super end-game. Good when passive levels and crit are high. Needs about 40%+ crit to beat Void Necklace.	Do not rush too early.
Voidwaker Emblem	Best necklace with Yang and Metallia because both lack crit rate. Strong in Ender's Echo, especially with Lunar Mine and Armor of Quietus synergy.	Main necklace target early to late.
Eternal Necklace	Weak / out of meta.	Do not prioritize.
Metal Neckguard	Very good for chapters if passives are ordered well, especially Ronin Oyoroi + Fitness Guide. Huge effective HP.	Useful for hard chapter bosses. Mostly survival.
Chaos Necklace	Useless.	Avoid.

Necklace TLDR

Rush Red Void Necklace.
Make SS Necklace only after 40%+ crit rate and when you can AF it enough.
Keep Red Void Necklace.

C) Gloves guide

Gloves	What it does / value	Optimizer note
SS Gloves	Best-in-class. Gives crit rate and damage burst cycle. Helps transition to Metallia faster.	End-game target.

Voidwaker Handguards	Very good vs bosses and mobs, but SS Gloves beat them.	Build to red/AF3, later salvage/use for SS.
Eternal Gloves	Strong mid-game, pairs with Eternal Necklace, but Eternal Necklace is out of meta.	Usually skip switching back/forth.
Fingerless Gloves	Very good non-S early glove.	Make Epic early.
Shiny Wristguard	Best for Chapter 4 kill farming/events. AF3 improves AFK kill farming.	Keep one Excellent.
Chaos Gloves	Useless.	Avoid.
Other gloves	Not great.	Fodder.

Gloves TLDR

Make Epic Fingerless Gloves.
Keep Excellent Shiny Wristguard for kill farming.
Then build Red Void Gloves for future SS Gloves.

D) Armor / suit guide

Armor	What it does / value	Optimizer note
Armor of Quietus	Weak before red, but red is best-in-class for chapters/Zone Ops/AFK if you can kill during ghost mode.	Rush red Chaos Armor first.
Eternal Suit	Best all-rounder. Strong for Ender's Echo, clan bosses, Extreme Ops.	Build after Chaos Armor. Target AF3.
SS Armor	Weak at base. Needs lots of relic cores/AF to beat Eternal Suit.	Last SS priority. Do not make early.
Voidwaker Windbreaker	Strong for chapters/daily challenges but low utility elsewhere.	Skippable.
Full Metal Suit	Good fallback if no S suit. Similar to Eternal Suit but weaker.	Temporary armor.
Other suits	Garbage.	Fodder.

Armor TLDR

Rush Red Armor of Quietus first.
Then build Red/AF3 Eternal Suit.
SS Armor is last priority.

E) Belt guide

Belt	What it does / value	Optimizer note
SS Belt	OP at base and stronger with Astral Forge. Mix of Void Sash + Twisting Belt. All-rounder.	Main belt target.
Twisting Belt / Chaos Belt	Gives damage increase.	Get purple early.
Stylish Belt	Very good for waves. Better than Twisting if stats are low and pushing chapters.	Make Epic early.
Eternal Belt	Good only if bosses are hard and fights last 1+ minute. Outdated for DPS meta.	Low priority.
Voidwaker Sash	Good vs waves, useless vs bosses. Astral effects allow mob invincibility loops, but boss value low.	Not main DPS belt.

Belt TLDR

Make Epic Stylish Belt.
Get Purple Twisting/Chaos Belt.
Then work toward SS Belt.

F) Boots guide

Boots	What it does / value	Optimizer note
SS Boots	Best-in-class. Very strong at base, but requires constant movement.	Major SS priority after Twin Lance.
Voidwaker Treads	Best boots until SS Boots.	Build to red, later salvage/use path for SS Boots.

Light Runners	Very good at Epic for early chapters/waves.	Make Epic early.
Eternal Boots	Good until Epic, bad after.	Do not overinvest.
Chaos Shoes	Third best. Strong damage reduction in reverse state, but awkward before red.	Later/optional.

Boots TLDR

Get Epic Light Runners.
Focus on Red Void Boots.
Then move to SS Boots.

G) Full progression paths

Early game

Weapon: Kunai + Purple Void Power
Armor: Rush Red Armor of Quietus
Necklace: Metal Neckguard
Belt: Stylish Belt
Gloves: Fingerless Gloves
Boots: Light Runners

Mid game

Weapon: Void Power + Twin Lance
Armor: Red Armor of Quietus, then Red Eternal Suit
Necklace: Red Void Necklace
Belt: Purple Chaos/Twisting Belt
Gloves: Red Void Gloves
Boots: Red Void Boots

End game

Weapon: Twin Lance
Armor: Eternal Suit + Armor of Quietus + SS Armor only much later
Necklace: SS Necklace + keep Void Necklace
Belt: SS Belt
Gloves: SS Gloves
Boots: SS Boots

H) SS gear creation order

Rank	SS item	Reason
1	Twin Lance / SS Weapon	Best weapon, huge boss damage.
2	SS Boots	Strong at base, big DPS if moving.
3	SS Belt	Stronger at base than AF3 Chaos Belt; scales with hits and Destroyer.
4	SS Gloves	Strong burst cycle, crit/damage value.
5	SS Necklace	Crit-dependent; needs higher crit and AF.
6	SS Armor	Weak at base, needs many relic cores.

Optimizer rule

Do not recommend SS Armor early.

Do not recommend SS Necklace unless crit rate and forge level are good enough.

Twin Lance always comes first.

I) Wishlist rules

Rule	Meaning
Wishlist applies only to S-key chests	Does not apply to EDF keys or random S from events.
Wishlist blocks counterparts	If Void Power is wishlisted, Lightchaser and Sword of Disorder are blocked from that weapon slot.
One wishlist item is not enough	You can still pull from the other 5 slots.
Six wishlist items	You can only pull the 6 selected S items, blocking the other 12.
Do not wishlist Chaos just for Chaos cores early	Eternal/Void fragments are more needed early; Chaos is more late-game.

J) Wishlist priority by slot

Slot	First target	Later target
Weapon	Void Power until unlocked	Lightchaser / Void Power / Sword of Disorder for cores or SS Weapon AF
Necklace	Voidwaker Emblem until red	Eternal / Void / Chaos necklace for cores or SS Necklace AF
Gloves	Voidwaker Handguards until AF3 red	Eternal / Void / Chaos gloves for cores or SS Gloves AF
Armor	Armor of Quietus until red	Eternal Suit until AF3, then S armor for SS Armor AF
Belt	Unlock Twisting Belt / Chaos Belt	Eternal / Void / Chaos belts for cores or SS Belt AF
Boots	Voidwaker Treads until red	Eternal / Void / Chaos boots for cores or SS Boots AF

K) Selector priority

Eternal Selector

1. Eternal Suit until AF3
2. Eternal gear only if needed for Eternal Forge on SS gear

Void Selector

1. Void Power if not unlocked
2. Red Void Necklace
3. Red Void Boots
4. Red Void Gloves
5. Void gear for Void Forge on SS gear

Chaos Selector

1. Armor of Quietus until AF3
2. Chaos gear for Chaos Forge on SS gear

S Selector

1. Unlock Void Power
2. Red Armor of Quietus
3. Red Void Necklace
4. Red Void Boots
5. Red Void Gloves
6. SS forge materials later

Warning

Using selectors directly for cores is a last resort.
Prefer salvaging useless S gear, random S items, and wishlist pulls first.

L) Astral Forge rules

Rule	Data
Unlock	Have at least 1 Legendary gear
Location	Merge button → bottom right Astral Forge
Eligible gear	Legendary and SS only
Non-S Legendary requirement	Epic S item from same slot category
S Legendary requirement	Extra Epic copy of same S item
SS gear paths	Eternal 5, Void 5, Chaos 10
Can unforge?	Yes, returns materials

M) Chaos Forge rules

Rule	Data
------	------

Unlock Chaos Forge	Need total 3 Eternal/Void forge levels on that SS item
Example unlocks	E3, E1V2, E2V1
Chaos level cap	Chaos level cannot exceed total Eternal + Void levels
Main purpose	Chaos Fusion Power for Twin Lance
Chaos Fusion source	Total Chaos Forge levels across SS gear

N) Xeno Transmute rules

Rule	Data
Unlock	E4 + V4 on the gear
Levels	13
Current target	Twin Lance only, for now
F2P source	Universal exchange
Monthly F2P amount	6 Xeno Trans Cores
Cost	Relic Core + 5,000 gems per XTC
Events	Usually expensive / not worth

Xeno trigger priority

Rank	Trigger condition	Unlock
1	20 Energy Overload triggers	XT10
2	100 Drone hits	XT6
3	10 Forcefield hits	Base/available earlier

4	50 Void Power hits	Base/available earlier
5	Defeat 100 enemies	XT2

Xeno trigger effects

Effect	Unlock
Chilled	Default
Poison	XT3
Weakened	XT7

O) Optimizer warnings to hardcode

Never feed spare S gear randomly into other gear.

Save S gear for:

- Astral Forge
- SS crafting
- SS forge fodder
- Salvage cores
- Future selectors/core needs

Do not rush:

- SS Armor
- SS Necklace before enough crit
- Chaos wishlist early just for Chaos cores
- Normal RPG/Molotov resonance if Twinborn path is available

Combined Remaining Extraction — Survivor.io Optimizer Source Pack

This covers the rest of the main systems still needing extraction:

Boss Debuffs

Shield / Skill Damage Multipliers
Relic Core Usage
Astral Forge / Merge Rules
Normal Gear Effects
Clan Shop / Clan Rewards
Collectibles
Survivor Upgrade / Stat Boosts
Survivor Pass / Pass EXP
Energy Essence / Design Costs
Item Values
Wishlist / Merging
Gems / Oil Sources
Ender's Echo Rewards

1. Boss Debuffs / Damage-taken effects

Sources:

Boss_Debuffs_2024-12-27
Shield_and_Skill_Damage_Multipliers_2024-12-27
Crit_Rate_And_Damage_2025-03-09
SS_Gear_2025-02-06
Tech/Twinborn charts

Main debuff families

Debuff	What it does	Associated systems	DPS relevance
Vulnerability	Increases damage taken by enemy. Often boss-focused.	Yang, Joey, April, Laser, Boomerang, Lightning, Xeno, SS Boots, Twinborn charts	Very high
Weaken	Reduces enemy attack, but also enables “damage to Weakened targets.”	SS Gloves, Boomerang, Brick, Xeno, Donatello, boss debuffs	Very high if build has Weaken scaling
Chill	Slows/freezes enemies and enables damage to Chilled targets.	SS Boots, Drill, Guardian, Soccer, RPG, Xeno	High

Poison	DOT/debuff and enables damage to Poisoned targets.	Metallia, Croaky, Durian, RPG, Molotov, Xeno	High
Laceration	Enables damage to Lacerated targets.	Raphael, Leonardo, Michelangelo, Donatello, Laser, Drill, RPG, Guardian	High
Bleed	Damage/debuff; sometimes connects to laceration-like packages.	Raphael, Michelangelo, Durian/Caltrops, Soccer/Quantum Ball, Leather Gloves	Medium/high
Deep Wound	Increases damage taken; fire/Molotov linked.	Molotov, Fuel Barrel, Fuel Spray, Wind Totem	High
Snare / Slow	Control effect; not always direct DPS, but increases uptime/hit consistency.	Laser, Guardian, Forcefield, Fish, Splinter	Medium
Shield Damage Boost	Separate damage multiplier layer tied to energy shield systems.	SS Necklace, SS Armor, SS Boots, Twinborn charts	High
Skill Damage	Damage multiplier for skills.	SS Necklace, SS Gloves, Yang, tech, pets, collectibles	Very high
Crit Rate / Crit Damage	Crit multiplier layer.	Survivors, gear, pets, collectibles, combat harmony, SS Belt	Very high

Optimizer table

DebuffID
 DebuffName
 EffectType
 DamageTakenIncrease
 Duration
 StackLimit
 StackBehavior
 SourceSystem
 SourceItem
 SourceSkill
 Condition

DPSRelevant
Notes

Key rule

Do not count “damage to Chilled / Weakened / Poisoned / Lacerated targets” unless the build has a source of that debuff and realistic uptime.

Example:

SS Boots give damage to Chilled enemies.

That only scores high if Chill uptime is high through SS Boots trail, Drill, Soccer, Guardian, or Xeno.

2. Shield Damage / Skill Damage Multipliers

Shield Damage

Source type	What it means	DPS scoring
Shield Damage Boost	Boosts damage while shield mechanics are active or shield-related attacks trigger.	High if player has shield uptime
Energy Shield	Shield HP layer used by SS Necklace/Armor/Boots.	Indirect DPS
Shield restore	Keeps shield active longer.	Medium/high if shield damage exists
Shield threshold effects	Some gear gives more damage when shield is high.	High but conditional

Associated systems:

SS Necklace
SS Armor
SS Boots
SS Gloves
Twinborn Guardian

Twinborn Rocket/RPG
Twinborn Durian
Twinborn Soccer
Twinborn Molotov/Brick
Collectibles

Skill Damage

Skill Damage is one of the best generic DPS stats.

Main sources:

SS Necklace Clarity
SS Gloves Berserk
SS Armor Super Enhance
Yang skills
Pet Encouragement
Twinborn resonance
Combat Harmony
Collectibles
Tech parts

Optimizer logic:

TotalSkillDamage =
GlobalSkillDamage
+ GearSkillDamage
+ SurvivorSkillDamage
+ PetSkillDamage
+ TechSkillDamage
+ ConditionalSkillDamage

Do not mix Skill Damage with Crit Damage. They are separate multiplier groups unless the calculator later confirms exact game formula.

3. Relic Core Usage

Sources:

Relic_Core_Usage_-_Enders_Echo_2024-11-29
Relic_Core_Usage_-_Enders_Echo_mini_2024-11-29

Relic_Core_Usage_-_Clan_Expedition_2024-11-29
Relic_Core_Usage_-_Clan_Expedition_mini_2024-11-29
SS_Gear_2025-02-06
Equipment guide text

What relic cores do

Relic cores are used for SS gear forge upgrades.

They affect:

Eternal Forge levels
Void Forge levels
Chaos Forge unlock path
Relic core allocation by game mode
SS gear power spikes

Mode-specific usage

Mode	What matters most
Ender's Echo	Boss DPS, crit breakpoints, single-target damage, SS Belt/Gloves/Weapon value
Clan Expedition	Boss damage plus mode-specific thresholds; may differ from EE path
Chapters / Zone Ops	Survival, movement, crowd control, Armor of Quietus, SS Boots, Void Power/SS Weapon

Key relic core rule

Spend relic cores by DPS gain per relic core, not by “highest rarity” or “newest item.”

Important notes

SS Weapon and SS Belt are usually high-value relic core targets.
SS Gloves depend heavily on crit/passive scaling.
SS Necklace becomes better once crit rate and passive level systems are strong.
SS Armor is weak at base and should not be early priority.

Relic core recommender fields

Mode
RelicCoreCount

CurrentSSWeapon_E/V/C
CurrentSSBelt_E/V/C
CurrentSSGloves_E/V/C
CurrentSSBoots_E/V/C
CurrentSSNecklace_E/V/C
CurrentSSArmor_E/V/C
CritRate
CritDamage
SkillDamage
RecommendedNextUpgrade
Reason
DPSGainEstimate

4. Astral Forge / Merge / Salvage

Sources:

Astral_Forge_2024-10-06
Merging_-_Equip_2024-11-28
Merging_-_Pets_2024-11-28
Merging_-_Tech_Parts_2024-11-28
Equipment guide text

Merge system

System	What merging does
Equipment merge	Raises gear rarity: normal → excellent → epic → legendary → higher red variants
Pet merge	Raises pet rarity/star path
Tech merge	Raises tech part rarity, required for resonance and Twinborn
S gear saving	S gear should be saved for AF, SS gear, cores, or exact future needs

Hard warning

Do not feed spare S gear randomly into other gear.

Save S gear for:

- Astral Forge
- SS crafting

- SS forge materials
- Salvage cores
- Wishlist/selector strategy

This warning is directly supported by the equipment guide text.

Astral Forge unlock

Requires at least 1 Legendary gear.
Only Legendary and SS gear can be Astral Forged.

Astral Forge requirement rules

Gear type	Requirement
Non-S Legendary	Epic S gear from same slot category + matching fragments
S Legendary	Extra Epic copy of same S item + matching fragments
SS Gear	Relic cores + faction cores/fragments + gear/fodder requirements

Chaos Forge

Unlock Chaos Forge when SS gear has 3 total Eternal/Void forge levels.
Chaos level cannot exceed total Eternal + Void level.
Chaos Fusion Power is based on total Chaos Forge levels across SS gear.

5. Normal Gear Effects — Equipment slot extraction

Sources:

Weapons_V2_2_2024-11-28
Necklaces_V2_2024-11-28
Eternal_Necklace_2024-11-28
Gloves_V2_2024-11-28
Armor_V2_2024-11-28
Belts_V2_2024-11-28
Boots_V2_2024-11-28
Equipment guide text

Weapons

Weapon	Role	DPS / optimizer note
Twin Lance / SS Weapon	End-game best weapon	Highest weapon priority
Void Power	Early/mid survival and crowd control	Purple enough for a long time; weak boss damage
Kunai	Early boss/Ender's Echo weapon	Save one Excellent
Lightchaser	Early chapters/crowd control	Falls off later
Sword of Disorder	Boss damage but weak crowd control	Low value once SS exists
Bat / Katana / Shotgun / Revolver	Normal weapons	Mostly fodder unless early temporary

Necklaces

Necklace	Role	Optimizer note
Voidwaker Emblem	Best practical necklace for Yang/Metallia because of crit issue	Rush red
SS Necklace	End-game, crit/passive dependent	Only after enough crit and forge
Metal Neckguard	Chapter survival/effective HP	Good for hard bosses, not DPS-first
Eternal Necklace	Out of meta/weak	Low priority
Chaos Necklace	Bad	Avoid

Gloves

Gloves	Role	Optimizer note
SS Gloves	End-game burst/crit/damage	High priority after Belt/Boots

Voidwaker Handguards	Strong pre-SS gloves	Build to red/AF3
Eternal Gloves	Strong mid-game crit	Less priority now
Fingerless Gloves	Best early non-S glove	Make Epic
Shiny Wristguard	Kill farming	Keep Excellent for Chapter 4/events
Chaos Gloves	Bad	Avoid

Armor

Armor	Role	Optimizer note
Armor of Quietus	Chapter/Zone Ops/AFK ghost mode	Rush red first
Eternal Suit	All-rounder, EE/clan/extreme ops	Build to AF3
SS Armor	Late relic-heavy item	Last SS priority
Voidwaker Windbreaker	Strong chapters but low cross-mode value	Optional/skip
Full Metal Suit	Temporary fallback	Use if no S armor

Belts

Belt	Role	Optimizer note
SS Belt	Best belt; crit/boss scaling	High SS priority
Twisting Belt / Chaos Belt	Damage increase	Get purple early
Stylish Belt	Early waves/chapter push	Make Epic
Eternal Belt	Bosses only, outdated	Low priority
Voidwaker Sash	Waves/survival	Weak boss value

Boots

Boots	Role	Optimizer note
SS Boots	Best boots, movement-based DPS	Very high priority
Voidwaker Treads	Best pre-SS boots	Build red
Light Runners	Early chapters/waves	Make Epic
Eternal Boots	Good only early	Don't overinvest
Chaos Shoes	Later/optional damage reduction	Not main DPS path

6. Clan System

Sources:

[Clan_Shop_2025-02-17](#)
[Clan_Level_Rewards_2024-11-28](#)
[Clan_Expedition_Rewards_2024-11-28](#)
[Chapter_Unlocks-2024-09-17](#)

Unlock

Clan unlocks after Chapter 3 win.

Clan features

Feature	What it does
Clan shop	Spend clan currency on rare resources
Clan expedition	Clan boss/content rewards
Clan level rewards	Unlocks perks, shop stock, member cap/boosts
Clan member boosts	Clan-wide ATK/HP/quick earnings type bonuses

Clan shop item categories

The clan shop chart includes items in these groups:

S selectors / S chests
Survivor shards
Tech keys
Collectible keys
Pet-related materials
Equipment-related materials
Relic/core-style materials
Awakening / resonance resources
General upgrade currencies

Optimizer valuation rule

ClanShopValue =
DPSValue
+ BottleneckValue
+ GemEquivalentValue
+ AccountStageValue
- ClanCurrencyCost

High priority usually:

Relic cores / rare cores
S selectors or S chests
Tech keys if chasing tech/Twinborn
Collectible keys later
Survivor shards if they unlock crit/ATK breakpoint
Pet awakening materials if pet path is bottlenecked

Low priority:

HP-only value
Common resources that are not bottlenecked
Items with poor gem-equivalent value

7. Clan Expedition Rewards

Reward types

Clan currency
Clan EXP
Chest rewards
Ranking rewards

Expedition tier rewards
Tech/equipment/material rewards

Optimizer role

Clan Expedition should be treated as:

Recurring reward source
Boss DPS mode
Relic core / shop currency pathway
Clan progression pathway

Mode scoring

For Clan Expedition builds, prioritize:

Boss damage
Crit rate breakpoints
Skill damage
Vulnerability
Weaken/Chill/Poison/Laceration uptime
Pet direct damage
SS Belt/Gloves/Weapon synergy
Relic core allocation from Clan-specific guide

8. Collectibles

Sources:

Collectible_Chest_Odds_2024-11-28
Collectible_Costs_2024-10-11
Collectibles_Edition_1_2024-11-28
Collectibles_Edition_2_2024-11-28
Collectibles_Edition_3_2024-11-28
SS_Gear_2025-02-06
Tech Parts - Twinborn collectible sets
Crit_Rate_And_Damage_2025-03-09

What collectibles do

Collectibles give:

Global ATK
Crit Rate
Crit Damage
Skill damage
Gear-specific bonuses
SS gear-specific bonuses
Twinborn tech-specific bonuses
Damage to debuffed targets
Shield damage boosts

Collectible cost system

Collectibles require shards/stars and scale by rarity.

Important fields:

CollectibleID
Edition
Rarity
ShardCost
StarLevel
OwnedStars
SetBonus
AffectedSystem
DPSRelevant

Collectible chest odds

Need store as probability table:

ChestType
Rarity
DropRate
ShardRate
FullItemRate
ExpectedValue

Collectible editions

Edition 1/2/3 charts define collectible pools. The optimizer needs:

CollectibleName
Edition
Rarity
AssociatedGear
AssociatedSkill
AssociatedTech
AssociatedSurvivor
AffectedStat
DropPool

If exact names are icon-only/unclear, store:

Collectible_IconID
ExactNamePending

Do not guess names from icons.

9. SS Gear Collectible Sets

From SS gear chart bottom.

Collectible set	Associated system	Effects
Erudite Heirloom	SS Weapon / Starforged Havoc	Higher thrust damage, Energy Overload stack damage, Disordered Death/Void Fissure damage
Otherworld Treasure	SS Necklace / shield	Omnicombat shield requirement reduction, Clarity skill damage, Reverse Scale skill damage
Meaning of Life	SS Gloves / Fury / Weaken	Skill damage during Fury, damage to Weakened, Stealth Attack crit damage
Interdimension Movement	SS Armor / energy level	Cabalistic shield boost, Suppression skill damage, Eternal Recharge energy damage multiplier

Optimizer rule

Collectibles are not generic only.
Some must attach directly to gear effect IDs.

Example:

Interdimension Movement 25 stars modifies SS Armor Eternal Recharge.

Meaning of Life modifies SS Gloves Stealth Attack / Weaken scaling.

10. Twinborn Collectible Sets

From Tech Parts - Twinborn charts.

Set	Twinborn pair/mode	Main bonuses
Close to Creation	Durian/Caltrops	Caltrops spike damage, Durian/Caltrops damage, shield damage
Open Void Gate	Soccer/Quantum Ball	Quantum Ball duration, shield damage, Soccer damage, skill damage, crit damage
Transgalactic Tentacle	Drone/Destroyer	Missile count, single-shot damage
Impression Idols	Forcefield/Force Barrier	Forcefield damage, vulnerability, crit damage, damage rate, skill damage
Multiverse Perspective	Drill/Frostfire	Drill/Frostfire damage, Fire Bomb damage
Hyperrift Tech	RPG/Sharkmaw	Sharkmaw damage, Cluster Arrows, lacerated damage
Rewriting the Stars' Memories	Lightning/Maelstorm	Maelstorm damage
Goldfinger	Boomerang/Magvolt	Skill damage, vulnerability max stacks, weakened damage
Cyber Wonderland	Guardian/Defender	Defender damage, skill damage, poisoned target damage
Try All Possibilities	Laser/Death Ray	Crit damage, shield damage, poisoned target damage, Death Ray damage, skill damage
Brewing Recipe	Brick/1-ton Iron	Brick damage, shield damage, skill damage, crit damage

Wind Totem

Molotov/Fuel Spray

Deep Wound, Molotov damage

11. Survivor Stat Boosts

Sources:

Survivor_Stat_Boosts_2024-12-12

Survivor_Upgrade_Costs_2024-12-19

Survivors_2025-01-26

Survivors_S_2025-01-26

Survivors_SP_2025-01-26

Crit_Rate_And_Damage_2025-03-09

Main milestone types

Milestone	What it gives
Level 40	Often ATK, crit damage, HP, or healing
Level 80	Often crit rate or healing
Level 120	Often crit rate, crit damage, ATK
5 yellow stars	Often global ATK/HP/crit depending survivor
Red awakening	Active/passive skill upgrades and Teamwork Passive slots

Important global DPS sources

Yang level 120: Crit Rate +10%

Metallia level 120: Crit Damage +10%

Joey level 120: Crit Damage +10%

King/Tsukuyomi/Worm/Wesson level 80/120 crit sources

TMNT 5-star: each gives All Survivors Crit Rate +8%

Combat Harmony: up to 30% global Crit Rate

Survivor Synergy: up to 30% global Crit Damage

Upgrade costs

Need store costs by survivor type:

Non-S survivor essence costs
S survivor essence costs
SP survivor pizza/shard/book costs
Awakening core costs
Blue material costs
Shard costs

Optimizer rule

Survivor investment should be scored by permanent DPS:

- Global ATK
- Global Crit Rate
- Global Crit Damage
- Active survivor DPS gain
- Teamwork passive unlock
- Debuff source unlock

HP/healing mostly ignored for damage-only optimizer.

12. Survivor Pass / Survivor Pass EXP

Sources:

Survivor_Pass_2024-12-19
Survivor_Pass_Experience_2024-09-16

Unlock

Survivor Pass unlocks after Chapter 8 win.

Pass categories

Pass type	Reward focus
Survivor pass	Shards, essence, awakening core, reset vial key, collectible key
Tech parts pass	Epic tech chest, tech keys, tech parts, resonance chip
Equipment pass	Designs, epic equipment, S keys, S chest, relic core

Notable rewards

Joey shards
Energy essence
Survivor shard boxes
Reset vial key
Awakening core
Collectible key
Epic tech part
Tech part key
Resonance chip
Designs
Epic equipment
S keys
S chest
Relic core

Pass EXP sources

Mission	Daily max	Weekly max	Season max
Dailies	130	910	3640
Defeat monsters 180,000	90	630	2520
Clear chapters/challenges	100	700	2800
Regular Challenge	0–150	300–450	1350–1500
Path of Trials	50	350	1400
Ender's Echo	60	420	1680
Clan Expedition	—	200	800
Special Ops	50	350	1400
Total	480–630	3860–4010	15590–15740

Optimizer use

PassValue =
Sum(ItemValue * Quantity)
- Cost

High-value pass items:

Relic core
Awakening core
Resonance chip
S chest / S keys
Tech keys
Collectible keys
Survivor shards

13. Energy Essence / Survivor Upgrade Costs

Sources:

Energy_Essence_Cost_2024-11-28
Energy_Essence_Cost_Full_2024-11-28
Survivor_Upgrade_Costs_2024-12-19

What it tracks

Essence cost to level survivors
Level 40/80/120 breakpoints
Total essence required
Different cost curves for normal, S, and SP survivors

Optimizer scoring

$\text{EssenceSpendValue} = \text{DPSGainAtNextBreakpoint} / \text{EssenceCost}$

Best upgrades are usually:

Level 40 if gives crit damage or ATK
Level 80 if gives crit rate
Level 120 if gives crit rate/crit damage/ATK

Bad upgrades for DPS:

HP-only
Healing-only

14. Equip Design Costs

Source:

Equip_Design_Costs_2024-11-14

What it tracks

Designs are used to level equipment.

Fields:

GearSlot
GearLevel
DesignCost
GoldCost
ATKGain
HPGain
DPSRelevant

Optimizer rule

Weapon/glove/necklace/belt/boots ATK-related leveling gets DPS value.
Armor/HP-only gains are lower for damage-only scoring.

Design bottleneck should be part of upgrade recommendation.

15. Item Values

Source:

Item_Values_2024-10-02

What it does

Assigns approximate gem-equivalent value to items.

Used for:

Event shop buying
Clan shop priority
Pass value
Gem spending
Black Friday saving
Selector value
Chest value

Value table fields

ItemID
ItemName
Category
GemEquivalent
DPSRelevant
BottleneckTier
Source
Notes

High bottleneck items:

Relic Core
Awakening Core
Resonance Chip
Xeno Trans Core
S selector / S chest
Collectible keys
Tech keys
Survivor shards
Pet awakening crystals

16. Sources of Gems

Sources:

Sources_Of_Gems_2024-10-07
Sources_of_Gems_2024-12-18

Gem source categories

Dailies
Weeklies
Events
Special Ops
Ender's Echo
Clan systems
Chapter clears
Zone Ops
Path of Trials
Privilege card
Monthly card
Ads/free shop
Achievements/missions
Compensation/mail

Optimizer use

MonthlyGemIncomeEstimate
EventBudget
BlackFridaySavingsProjection
GemSpendWarning

Rule:

Spend gems only when expected DPS/item value beats future event value.

17. Sources of Oil

Sources:

Sources_Of_Oil_2024-11-03
Sources_of_Oil_2024-12-18

Oil use

Oil is tied to talent/account progression.

DPS-relevant if it unlocks:

ATK
Crit Rate
Crit Damage

Damage multipliers
Key talent breakpoints

Store:

OilSource
OilAmount
Frequency
UnlockRequirement
DPSRelevant

18. Ender's Echo Rewards

Source:

EE_Rewards_2024-09-15

What it does

Ender's Echo is a recurring boss damage mode.

Rewards are based on:

Participation
Damage/ranking
Season/round rewards
Mode progression

Optimizer use

EE build should prioritize:

Single-target boss DPS
Crit breakpoints
Yang/Metallia/Joey active build
Pet damage
SS Weapon
SS Belt
SS Gloves
Vulnerability
Weaken/Lacerate/Chill/Poison uptime

Relic core recommendation path for EE

Do not overvalue:

- Wave-only effects
 - Mob-only kill farming
 - Survival HP unless dying
-

19. Wishlist Probabilities

Source:

Wishlist_Probabilities_2024-11-28
Equipment guide text

Rules

Rule	Meaning
Applies to S-key chests only	Not EDF keys or random S from events
Wishlist blocks counterparts	Pick one item per slot to block other S items in same slot
6 wishlisted items	Pulls are limited to those 6 S items instead of all 18
One selected item	Does not guarantee that item; other slots can still appear

Optimizer fields

- ChestType
- WishlistSlot
- SelectedItem
- BlockedItems
- PullProbability
- ExpectedKeysToTarget
- ExpectedGemCost

Wishlist priority

Weapon: Void Power first, then S weapon cores/fodder

Necklace: Voidwaker Emblem first
Gloves: Voidwaker Handguards first
Armor: Armor of Quietus first, then Eternal Suit
Belt: Twisting/Chaos Belt first
Boots: Voidwaker Treads first

20. Timed Muster Medal Conversion

Source:

Timed_Muster_Medal_Conversion_2024-12-12

What it does

Shows conversion/value for Timed Muster medals.

Use for:

- Event shop efficiency
- Medal conversion value
- Should spend vs save decision

Fields:

- MedallItem
 - ConversionCost
 - RewardItem
 - GemEquivalent
 - DPSRelevant
 - Priority
-

21. Chapter Unlocks — final rule table

Source:

Chapter_Unlocks-2024-09-17

	System	Unlock
Equipment		Chapter 1 after losing

Shop	Chapter 1 after losing
Talent Tree	Chapter 1 after losing
Growth Fund / Daily Shop / Friends	Chapter 2 after losing
Missions / Main Challenges / Quick Earnings / Events	Chapter 2 after winning
Clans	Chapter 3 after winning
Durian / Laser skill	Chapter 4 after winning
Gold Mine	Chapter 5 after winning
Regular Challenge	Chapter 6 after losing
Tech Parts	Chapter 6 through Regular Challenge
Ender's Echo	Chapter 8 after winning
Survivor Pass	Chapter 8 after winning
Special Ops	Chapter 8 after winning
Pets	Chapter 9 after winning
Survivor leveling	Chapter 9 after winning
Mine/Bomb skills	Chapter 10 after losing
Permanent Privilege Card	Chapter 11 after losing
Zone Ops / Extreme Ops	Chapter 30 after winning

Optimizer rule:

Do not recommend a system before the player has unlocked it.

22. Final database schema list

To turn the whole dataset into your calculator, use these tables:

FeatureUnlocks
ItemValues

GemSources
OilSources
EquipmentBaseEffects
EquipmentMetaPriority
SSGearEffects
SSForgeCosts
RelicCoreRecommendations
AstralForgeRules
ChaosForgeRules
XenoTransmuteRules
WishlistRules
SelectorPriority
MergeRules

SkillBaseEffects
BossDebuffs
DamageMultiplierRules
CritSources

TechBaseEffects
NormalTechResonance
TechResonanceEnergy
ResonanceChipCosts
TwinbornPairs
TwinbornBaseEffects
TwinbornResonanceRows
TwinbornCollectibleSets
TechOverloadRules

PetBaseEffects
PetSkills
PetAwakeningCosts
PetToyBonuses
PetRecommendations

SurvivorBaseEffects
SurvivorStatBoosts
SurvivorUpgradeCosts
SurvivorAwakening
CombatHarmony
SurvivorSynergy

CollectibleChestOdds
CollectibleCosts

CollectiblePools
CollectibleSetBonuses

ClanShop
ClanLevelRewards
ClanExpeditionRewards
SurvivorPassRewards
SurvivorPassEXP
EndersEchoRewards

23. Final optimizer priority logic

Damage-first scoring

Score high:

- ATK
- Crit Rate
- Crit Damage
- Skill Damage
- Vulnerability
- Weaken uptime + damage to Weakened
- Chill uptime + damage to Chilled
- Poison uptime + damage to Poisoned
- Laceration uptime + damage to Lacerated
- Deep Wound
- Shield Damage Boost when shield uptime exists
- Boss damage / single-target damage
- Pet direct DPS
- Drone/Twinborn missile scaling

Ignore or down-rank

HP
Healing
Damage reduction
Revival
Movement speed unless it enables SS Boots / hit uptime
EXP gain unless it affects build ramp
Crowd control unless it improves DPS uptime

Spend recommendation logic

BestUpgrade =
Highest DPS gain
divided by scarce cost
adjusted for unlock state
adjusted for mode
adjusted for player's current bottleneck

Where scarce costs include:

Gems
Relic cores
Awakening cores
Resonance chips
Xeno cores
S selectors
S gear fodder
Tech fodder
Pet awakening crystals
Collectible shards
Energy essence
Designs

24. Final source-type warning

Use each source for the right job:

Source type	Use for
Charts/images	Exact stats, breakpoints, costs
Discord/meta guide	Priority, warnings, progression logic
Item value chart	Gem-equivalent value
Relic core charts	SS upgrade order by mode
Crit/debuff charts	Damage formula inputs
Chapter chart	Unlock gating

Do not use meta opinion as exact stat data unless a chart confirms it.

Final Mount Batch — Survivor.io Optimizer Data

1. Mount system basics

SystemName = Mounts / Vehicles

DPSRelevant = Yes

MainPurpose = Extra damage source + component stat board + synced stats from inactive mounts

UnknownMounts = Ignore

ConfirmedMounts = Electric Scooter, Tech Hoverboard, Doomsteed

Mounts matter in **two different ways**:

1. Equipped mount skill damage/effects
2. Component board stats from equipped + inactive synced mounts

Important rule:

Components not installed in any mount = 0 value.

Components in equipped mount = 100% value.

Components in inactive mounts = partial value based on that mount's Sync Rate.

2. Confirmed mount list

Mount	Unlock cost	Upgrade material	Starting sync rate	Use style
Electric Scooter	30 Scooter shards	Scooter shards only	0.20	Auto damage, easiest
Tech Hoverboard	50 Hoverboard shards	Hoverboard shards only	0.30	Shield stack / avoid getting hit
Doomsteed	80 Doomsteed shards	Doomsteed shards + Mount Cores	0.40	Boss hugging / movement scaling

Database row format:

MountID

MountName

UnlockShardCost
UsesMountCoreForUpgrade
BaseSyncRate
MainDamageStyle
DPSRelevant
Source

Rows:

MOUNT_SCOOTER | Electric Scooter | 30 | No | 0.20 | Auto discs / lightning / laceration | Yes | Mount Discord + image
MOUNT_HOVERBOARD | Tech Hoverboard | 50 | No | 0.30 | Thorn Shield / Shockwave Blast / Chilled damage | Yes | Mount Discord + image
MOUNT_DOOMSTEED | Doomsteed | 80 | Yes | 0.40 | Blazing Charge / collision / poison + boss scaling | Yes | Mount Discord + image

3. Mount selection rule

This is not a “priority chart.” This is a system rule from the guide:

Use the highest unlocked mount.
Doomsteed > Tech Hoverboard > Electric Scooter.

Reason:

Once the better mount is unlocked, use it even if the older mount has more stars.
Older mounts still matter because their installed components give partial stats through Sync Rate.

Optimizer logic:

if DoomsteedUnlocked:
 EquippedMount = Doomsteed
elif TechHoverboardUnlocked:
 EquippedMount = TechHoverboard
else:
 EquippedMount = ElectricScooter

4. Mount upgrade cost table

Electric Scooter

Unlock cost:

Open = 30 Scooter shards

Upgrade costs:

Stage	Cost
Yellow 1	20 shards
Yellow 2	25 shards
Yellow 3	40 shards
Yellow 4	50 shards
Red 1	75 shards
Red 2	100 shards
Red 3	120 shards
Red 4	160 shards
Total upgrade cost	590 shards
Total including unlock	620 shards

MountID = MOUNT_SCOOTER

OpenCost = 30

UpgradeCostOnly = 590

TotalCostWithUnlock = 620

CoreCost = 0

Tech Hoverboard

Unlock cost:

Open = 50 Hoverboard shards

Upgrade costs:

Stage	Cost
Yellow 1	10 shards
Yellow 2	15 shards
Yellow 3	20 shards
Yellow 4	35 shards
Red 1	45 shards
Red 2	60 shards
Red 3	75 shards
Red 4	100 shards
Total upgrade cost	360 shards
Total including unlock	410 shards

MountID = MOUNT_HOVERBOARD

OpenCost = 50

UpgradeCostOnly = 360

TotalCostWithUnlock = 410

CoreCost = 0

Doomsteed

Unlock cost:

Open = 80 Doomsteed shards

Upgrade costs:

Stage	Shards	Mount Cores
Yellow 1	20	1
Yellow 2	30	1

Yellow 3	45	2
Yellow 4	65	2
Red 1	100	4
Red 2	140	4
Red 3	180	4
Red 4	220	6
Total upgrade cost	800	24
Total including unlock	880	24

MountID = MOUNT_DOOMSTEED

OpenCost = 80

UpgradeCostOnly = 800

TotalCostWithUnlock = 880

CoreCost = 24

Important:

Only Doomsteed needs Mount Cores.

Scooter and Hoverboard only need their shards.

5. Sync Rate table

Sync Rate decides how much value you get from components installed in inactive mounts.

Equipped mount component value = 100%

Inactive mount component value = ComponentStat * MountSyncRate

Sync rates by mount/star

Stage	Scooter	Hoverboard	Doomsteed
Base / 0 star	0.20	0.30	0.40
Yellow 1	0.22	0.33	0.44

Yellow 2	0.24	0.36	0.48
Yellow 3	0.28	0.40	0.55
Yellow 4	0.32	0.45	0.62
Red 1	0.38	0.50	0.71
Red 2	0.44	0.55	0.80
Red 3	0.52	0.60	0.90
Red 4	0.60	0.75	1.00

Optimizer formula:

ComponentEffectiveValue =
ComponentStatValue * SyncRate

For equipped mount:

SyncRate = 1.00

Example from guide:

Using Doomsteed as main mount.

Gold Skill Damage component = 27%.

If installed in Doomsteed:

$27\% * 1.00 = 27\%$

If installed in 0-star Hoverboard:

$27\% * 0.30 = 8.1\%$ / roughly 9% in guide example

If installed in 0-star Scooter:

$27\% * 0.20 = 5.4\%$

Total if installed across all 3 mounts:

$27 + 8.1 + 5.4 = 40.5\%$

This is why:

Do not merge too early.

Fill all unlocked boards first.

Then merge extra components.

6. Electric Scooter full extraction

Mount identity

MountID = MOUNT_SCOOTER

MountName = Electric Scooter

UnlockCost = 30 Scooter shards

DamageStyle = Auto discs + shock lightning

Debuffs = Laceration, Shock

BestUseLogic = Easiest mount, no input needed

DPSRelevant = Yes

Special skill

SkillName = Volta Disc

When used, fires 2 hi-voltage discs.

Hitting the same target twice inflicts Shock.

Shock triggers a lightning strike at the target's location.

Discs inflict Laceration for 2 seconds on hit.

Fields:

AppliesLaceration = Yes

LacerationDuration = 2s

AppliesShock = Yes

ShockTrigger = Same target hit twice

LightningStrike = Yes

Skill/star effects

Stage	Sync rate	Effect
Yellow 1	0.22	Shock triggers 2 lightning strikes
Yellow 2	0.24	Every 3 lightning strikes grants 1 Laceration Mastery stack. Each stack gives +2% damage to Lacerated targets, up to 5 stacks
Yellow 3	0.28	Disc Damage +33%
Yellow 4	0.32	Damage to Weakened targets +20%; Crit Damage +55%

Red 1	0.38	Shock triggers 3 lightning strikes
Red 2	0.44	Damage to Weakened targets +35%; Crit Damage +90%
Red 3	0.52	Shock Damage +33%
Red 4	0.60	Each Laceration Mastery stack gives +4% damage to Lacerated targets

Line effects

Line	Unlock	Effect
Line 1	Default	Damage to Weakened targets +10%
Line 2	Default	Triggering skill fires 1 additional disc
Line 3	Default	Damage to Weakened targets +15%; Crit Damage +20%
Line 4	Default	On disc hit, Crit Damage +1%, max 35%
Line 5	Yellow 2	Every 3 lightning strikes grants 1 Laceration Mastery stack. Each stack gives +2% damage to Lacerated targets, up to 5 stacks
Line 6	Yellow 4	Damage to Weakened targets +20%; Crit Damage +55%
Line 7	Red 2	Damage to Weakened targets +35%; Crit Damage +90%
Line 8	Red 4	Each Laceration Mastery stack gives +4% damage to Lacerated targets

Optimizer notes

Main buckets:

- Crit Damage
- Damage to Weakened
- Damage to Lacerated
- Disc Damage
- Shock Damage

Debuffs:

- Laceration
- Shock

Condition:

- Needs disc hits / shock triggers / line completion

DPSRelevant = Yes

UseStyle = Passive auto damage

7. Tech Hoverboard full extraction

Mount identity

MountID = MOUNT_HOVERBOARD

MountName = Tech Hoverboard

UnlockCost = 50 Hoverboard shards

DamageStyle = Thorn Shield + Shockwave Blast

Debuffs = Chilled damage scaling

BestUseLogic = Avoid getting hit, stack shield

DPSRelevant = Yes

Special skill

SkillName = Thorn Shield

When used, gain 1 Thorn Shield stack.

Thorn Shield can stack up to 5 times.

Each shield stack blocks 1 hit when taking damage and triggers Shockwave Blast.

At full stacks, gaining Thorn Shield triggers an explosion.

Fields:

ThornShieldStackMax = 5

BlocksHit = Yes

OnHitTrigger = Shockwave Blast

FullStackTrigger = Explosion

How to use

Try to avoid getting hit.

Stack Thorn Shield.

Shield stack is visible on the hoverboard icon in battle.

Higher stacks = more damage boost from the board.
When hit, Shockwave Blast triggers.

Skill/star effects

Stage	Sync rate	Effect
Yellow 1	0.33	Thorn Shield quality upgrade: Additional Shockwave Blast damage +16.7%
Yellow 2	0.36	For each Shockwave Blast triggered, damage to Chilled targets +6%, max 60%
Yellow 3	0.40	Thorn Shield quality upgrade: Additional Shockwave Blast damage +16.7%
Yellow 4	0.45	Damage to Chilled targets +40%; Skill Damage +30%
Red 1	0.50	Accelerated Charging: Energy gain rate +25%
Red 2	0.55	Damage to Chilled targets +60%; Skill Damage +45%
Red 3	0.60	Accelerated Charging: Energy gain rate +25%
Red 4	0.75	Max Shield Resonance stacks +6

Line effects

Line	Unlock	Effect
Line 1	Default	Damage to Chilled targets +15%; Skill Damage +10%
Line 2	Default	When Thorn Shield fades, 20% chance to not consume stacks
Line 3	Default	Damage to Chilled targets +25%; Skill Damage +15%
Line 4	Default	When gaining Thorn Shield, gain 1 Shield Resonance stack. Shield Damage Boost +10% for each stack, up to 4 stacks
Line 5	Yellow 2	For each Shockwave Blast triggered, +6% damage to Chilled targets, max 60%

Line Yellow 4 Damage to Chilled targets +40%; Skill Damage +30%
6

Line Red 2 Damage to Chilled targets +60%; Skill Damage +45%
7

Line Red 4 Max Shield Resonance stacks +6
8

Optimizer notes

Main buckets:

- Skill Damage
- Damage to Chilled targets
- Shield Damage Boost
- Shockwave Blast Damage

Debuffs:

- Needs Chilled source to fully use DamageToChilled
- Hoverboard itself is more of a Chilled damage scaler than a Chill source unless another source confirms it applies Chill

Conditions:

- Thorn Shield stacks
- Shockwave Blast triggers
- Avoiding hits helps stack/maintain value
- Line completion matters

DPSRelevant = Yes

UseStyle = Stack shield, avoid hits, convert shield events into damage

8. Doomsteed full extraction

Mount identity

MountID = MOUNT_DOOMSTEED

MountName = Doomsteed

UnlockCost = 80 Doomsteed shards

UpgradeMaterial = Doomsteed shards + Mount Cores

DamageStyle = Blazing Charge collision damage

Debuffs = Poisoned damage, Lacerated damage, Boss damage, Root

BestUseLogic = Touch/hug/circle boss
DPSRelevant = Yes

Special skill

SkillName = Blazing Charge
SecondaryName = Iron Resolve

Effects:

Iron Resolve:
When charging energy, damage taken -10%.

Blazing Charge:
When skill is used, Doomsteed turns into Flamesteed.
Consumes 10 energy every second.
During Blazing Charge, gain damage immunity.
During Blazing Charge, gain +3 max bonus Movement Speed.
Collisions instakill minions.
Collisions deal damage to bosses and elites.
Collision damage can happen once every 0.5 seconds.

Fields:

EnergyDrain = 10 energy per second
DamageImmunity = Yes
MaxBonusMovementSpeed = +3
MinionCollision = Instakill
BossEliteCollisionDamage = Yes
CollisionIntervalBase = 0.5s

How to use

Keep touching/hugging the boss as much as possible.
Circle around the boss.
This stacks Doomsteed value and also works with SS Boots movement/Glacial stacks.

Skill/star effects

Stage	Sync rate	Effect
-------	--------------	--------

Yellow 1	0.44	Tempest Rush: Blazing Charge collision damage to bosses/elites scales with Movement Speed. When Movement Speed exceeds 12, +5% damage for each additional point, up to 4 stacks
Yellow 2	0.48	Each energy stack provides +1 Movement Speed
Yellow 3	0.55	Blazing Charge collision damage interval becomes 0.4s
Yellow 4	0.62	Damage to Poisoned targets +60%; Damage to Lacerated targets +30%
Red 1	0.71	Blazing Charge: after triggering collision damage, roots bosses/elites for 3s. Root reduces Movement Speed to 0. Rooted targets cannot move. Triggers once per unit during each Blazing Charge
Red 2	0.80	Damage to Poisoned targets +80%; Damage to Bosses +5%
Red 3	0.90	Max additional Tempest Rush stacks +8
Red 4	1.00	For every 160% damage to Poisoned targets, +5% damage to bosses, up to 25%

Line effects

Line	Unlock	Effect
Line 1	Default	Damage to Poisoned targets +20%
Line 2	Default	Movement Speed +2
Line 3	Default	Damage to Poisoned targets +40%; Skill Damage +50%
Line 4	Default	For every 100 collision kills on minions or 5 collisions on bosses, gain 1 Charge stack. +30% Skill Damage for each stack, max 5 stacks
Line 5	Yellow 2	Each energy stack provides +1 Max Movement Speed
Line 6	Yellow 4	Damage to Poisoned targets +60%; Damage to Lacerated targets +30%

Line Red 2 Damage to Poisoned targets +80%; Damage to Bosses +5%
7

Line Red 4 For every 160% damage to Poisoned targets, +5% damage to bosses, up
8 to 25%

Optimizer notes

Main buckets:

- Skill Damage
- Damage to Poisoned targets
- Damage to Lacerated targets
- Damage to Bosses
- Movement Speed scaling
- Collision damage
- Charge stacks

Debuffs:

- Poisoned target damage
- Lacerated target damage
- Root utility

Conditions:

- Needs movement/contact uptime
- Best when hugging/circling boss
- Better when Movement Speed > 12
- Better with SS Boots-style movement stacking
- Better if Poison/Laceration uptime exists

DPSRelevant = Very High

UseStyle = Active boss contact / movement scaling

9. Component stat values

Components roll a stat randomly. The **shape does not decide the stat**. The percentage is fixed by stat type and rarity.

Component stat table

Component stat	Green	Blue	Purple	Purple +1	Gold	Gold +1	Red
----------------	-------	------	--------	-----------	------	---------	-----

Shield Damage	5%	7%	9%	11%	14%	18%	21%
Boss Damage	1%	2%	2.5%	3%	3.5%	4%	5%
Lacerated Damage	2%	3%	4%	5%	5%	7%	8%
Skill Damage	10%	15%	19%	23%	27%	35%	42%
Crit Damage	8%	12%	15%	18%	22%	28%	33%
Weakened Damage	10%	15%	19%	23%	27%	35%	42%
Chilled Damage	10%	15%	19%	23%	27%	35%	42%
Poisoned Damage	10%	15%	19%	23%	27%	35%	42%

Database row format:

ComponentStatID
StatName
Rarity
ValuePercent
DamageBucket
DPSRelevant

Example rows:

COMP_SKILL_GOLD | Skill Damage | Gold | 27% | SkillDamage | Yes
COMP_SHIELD_RED | Shield Damage | Red | 21% | ShieldDamage | Yes
COMP_BOSS_RED | Boss Damage | Red | 5% | BossDamage | Yes
COMP_POISON_GOLD1 | Poisoned Damage | Gold+1 | 35% | DamageToPoisoned | Yes

10. Component merging

Merging rerolls/randomizes the boost unless a component is locked away and not merged.

Merge input	Output
2 Green	1 Blue
2 Blue	1 Purple
2 Purple	1 Purple +1

2 Purple +1 1 Gold
3 Gold 1 Gold +1
3 Gold +1 1 Red

Important notes:

+1 components have dots on the component corners.

Boosts are randomized after merging.

Locked components keep their boost but are not merged.

Eventually, to reach Red, locked components may need to be unlocked and merged.

Optimizer rule:

Do not auto-merge components unless:

1. All unlocked mount boards are filled, OR
 2. The merged component gives higher effective total value than keeping components spread across synced boards.
-

11. Component refining

SystemName = Component Refining

Material = Mount Vehicle Refine Essence

Source = Component chests + events

Effect = Changes component boost into random new boost

Rules:

Refining changes the component stat randomly.

Refining cost increases every time you refine the same component.

Better to refine Yellow or higher components.

If you refine below Yellow, then later merge it, the refined stat can be lost/rerolled.

Optimizer rule:

if ComponentRarity < Gold:

 RecommendRefine = False

else:

 RecommendRefine = True if player is targeting specific stat

For damage-only optimizer, target stats are:

Skill Damage

Shield Damage
Boss Damage
Lacerated Damage
Crit Damage
Chilled / Weakened / Poisoned only if that pool is already useful/positive

But store this as a fallback assumption, not a hard rule.

12. Component placement logic

Core rules

1. Put best components in equipped/main mount first.
2. Fill all unlocked mount boards before merging extras.
3. Components in inactive mounts still give partial value through Sync Rate.
4. Components not installed give no value.
5. Do not move components every day; set and forget unless doing a serious recalculation.

Why filling boards beats merging early

Example:

You have 3 Gold Skill Damage components.

Each Gold Skill Damage = 27%.

Option A:

Merge 3 Gold into 1 Gold+1.

Gold+1 Skill Damage = 35%.

Option B:

Install all 3:

Main mount = 27%

Inactive Hoverboard 0-star = $27\% * 0.30 = 8.1\%$

Inactive Scooter 0-star = $27\% * 0.20 = 5.4\%$

Total = 40.5%

$40.5\% > 35\%$

Optimizer should calculate this automatically:

SpreadValue = Sum(ComponentValue * MountSyncRateOr1)
MergeValue = NewComponentValue * BestAvailableSlotMultiplier

if SpreadValue > MergeValue:

 KeepSeparate

else:

 Merge

13. Mount line effects

Each mount has its own line effects. A line effect activates when a horizontal row is complete.

Completed row = no black/empty cell in that row.

Completed rows are highlighted gold.

Only line effects from the equipped mount count.

Unlock rules:

Line	Unlock requirement
------	--------------------

Line 1	Default
--------	---------

Line 2	Default
--------	---------

Line 3	Default
--------	---------

Line 4	Default
--------	---------

Line 5	Mount at 2 Yellow stars
--------	-------------------------

Line 6	Mount at 4 Yellow stars
--------	-------------------------

Line 7	Mount at 2 Red stars
--------	----------------------

Line 8	Mount at 4 Red stars
--------	----------------------

Database fields:

MountLineEffectID

MountID

LineNumber

UnlockRequirement

EffectText

AffectedStat

StatAmount
Condition
DPSRelevant

14. Random component stats display

The in-game “Random Component Stats” section shows stats from the components installed in the mount you are using.

For optimizer, do **not** rely only on displayed stats.

Use two separate totals:

DisplayedMainMountStats =
Sum(components installed in equipped mount)

SyncedInactiveMountStats =
Sum(components installed in inactive mounts * that mount's sync rate)

TotalMountComponentStats =
DisplayedMainMountStats + SyncedInactiveMountStats

This matters because inactive mount stats may not show the same way, but they still matter through sync.

15. Mount material sources

From Discord text:

Mount Vehicle Core comes from Core Choice Chest.

Mount Vehicle Component Key comes from Events, Operation Retreat, and Clan Shop.

Mount Vehicle Refine Essence comes from component chests and events.

Database format:

MountMaterialID
MaterialName
UsedFor
Source

DPSRelevant
Bottleneck

Rows:

MOUNT_SHARD_SCOOTER | Scooter Shards | Unlock/upgrade Electric Scooter |
Events/shops/chests | Yes | Medium
MOUNT_SHARD_HOVERBOARD | Hoverboard Shards | Unlock/upgrade Tech Hoverboard |
Events/shops/chests | Yes | Medium
MOUNT_SHARD_DOOMSTEED | Doomsteed Shards | Unlock/upgrade Doomsteed |
Events/shops/chests | Yes | High
MOUNT_CORE | Mount Core | Doomsteed upgrades | Core Choice Chest | Yes | High
MOUNT_COMPONENT_KEY | Vehicle Component Key | Open component chests |
Events/Operation Retreat/Clan Shop | Yes | High
MOUNT_REFINE_ESSENCE | Vehicle Refine Essence | Reroll component stat | Component
chests/events | Yes | Medium/High

16. Mount damage buckets added to optimizer

Mounts add these buckets:

MountSkillDamage
MountCollisionDamage
MountDiscDamage
MountShockDamage
MountShockwaveBlastDamage
MountDamageToWeakened
MountDamageToLacerated
MountDamageToChilled
MountDamageToPoisoned
MountDamageToBoss
MountCritDamage
MountShieldDamage
MountComponentSkillDamage
MountComponentShieldDamage
MountComponentBossDamage
MountComponentElementalDamage
MountMovementSpeed
MountSyncRate

Add these to global bucket map:

SkillDamage += MountSkillDamage + MountComponentSkillDamage
CritDamage += MountCritDamage + MountComponentCritDamage
ShieldDamage += MountShieldDamage + MountComponentShieldDamage
DamageToBoss += MountBossDamage + MountComponentBossDamage
DamageToWeakened += MountDamageToWeakened + MountComponentWeakened
DamageToLacerated += MountDamageToLacerated + MountComponentLacerated
DamageToChilled += MountDamageToChilled + MountComponentChilled
DamageToPoisoned += MountDamageToPoisoned + MountComponentPoisoned

Do not mix mount skill damage with pet skill damage or tech skill damage too early. Store source separately first.

17. Mount/debuff interaction

Mount	Debuff/damage family
Electric Scooter	Laceration, Weakened damage, Crit Damage, Shock
Tech Hoverboard	Chilled damage, Shield Damage, Skill Damage
Doomsteed	Poisoned damage, Lacerated damage, Boss damage, Movement Speed, Root

Optimizer logic:

DamageToPoisoned only scores high if the build has Poison uptime.
DamageToLacerated only scores high if the build has Laceration uptime.
DamageToChilled only scores high if the build has Chill uptime.
DamageToWeakened only scores high if the build has Weaken uptime.

But for components:

Still store all component stats exactly.
Do not delete “bad” component types.

18. Mount + SS gear synergy notes

Doomsteed + SS Boots

Doomsteed wants movement and boss contact.
SS Boots also reward movement and stacking.
So Doomsteed and SS Boots should have a synergy flag.

Fields:

SynergyID = DOOMSTEED_SSBOOTS
Condition = Player can keep moving around/touching boss
Stats = MovementSpeed, GlacialBloodline, BossCollisionDamage
DPSRelevant = High

Hoverboard + shield systems

Hoverboard uses Thorn Shield and Shield Resonance.
This links to Shield Damage Boost and shield-based damage systems.

Fields:

SynergyID = HOVERBOARD_SHIELD_DAMAGE
Stats = ShieldDamageBoost, SkillDamage, DamageToChilled
DPSRelevant = High

Scooter + Laceration / Weaken

Scooter applies Laceration through discs.
It also gives damage to Weakened targets and Crit Damage.

Fields:

SynergyID = SCOOTER_LACERATE_WEAKEN
Stats = DamageToLacerated, DamageToWeakened, CritDamage
DPSRelevant = Medium/High

19. Mount component stat recommendation logic

This is not a priority chart. This is how the optimizer should calculate value.

Component value should depend on player damage pool:

ComponentValue =
ComponentStatPercent
* SyncMultiplier
* DamageBucketNeed
* DebuffUptime
* ModeWeight

Where:

SyncMultiplier = 1.0 if installed in equipped mount
SyncMultiplier = mount sync rate if installed in inactive mount

DamageBucketNeed examples:

If SkillDamage is negative or low:
SkillDamage components gain high value.

If ShieldDamage is negative or low and player uses shield systems:
ShieldDamage components gain high value.

If Poisoned damage is already strong and Poison uptime is high:
Poisoned components can be valuable.

If Chilled damage is negative but Chill uptime is low:
Chilled component value stays low.

General guide text says:

Skill Damage is usually the best general component target.
Shield Damage can be second if negative/needed.
Boss Damage is small per component but direct.
Lacerated/Crit/Chill/Weaken/Poison depend on player pools and debuff uptime.

Store this as:

AssumptionRule = ComponentFallbackPreference
NotHardcodedTruth = True

20. Clean database schema for mounts

MountBase

MountID

MountName
UnlockShardType
UnlockShardCost
UsesMountCore
BaseSyncRate
MainMechanic
SpecialSkillName
SpecialSkillText
UseStyle
DPSRelevant
SourceImage
SourceDate

MountUpgradeCost

MountID
Stage
ShardCost
CoreCost
TotalShardCostToStage
TotalCoreCostToStage
SyncRateAtStage
DPSRelevant

MountStarEffect

MountID
Stage
SyncRate
EffectText
AffectedStat
StatAmount
AssociatedDebuff
AssociatedMechanic
Condition
DPSRelevant

MountLineEffect

MountID
LineNumber
UnlockRequirement
EffectText
AffectedStat

StatAmount
AssociatedDebuff
Condition
DPSRelevant

MountComponentStatValue

ComponentStatID
StatName
Rarity
ValuePercent
DamageBucket
DPSRelevant

MountComponentInventory

ComponentID
ShapeID
Rarity
StatName
ValuePercent
InstalledMountID
InstalledSlot
Locked
RefineCount
DPSRelevant

MountComponentMergeRule

InputRarity
InputCount
OutputRarity
BoostRerolls
Notes

MountSyncCalculation

EquippedMountID
ComponentID
InstalledMountID
BaseValue
SyncRateApplied
EffectiveValue

21. Final mount optimizer rules

Use these exact rules:

1. Ignore unknown/unused mount from old leak image.
 2. Only use Electric Scooter, Tech Hoverboard, Doomstead.
 3. Use highest unlocked mount as equipped mount.
 4. Still upgrade unused mounts because their sync rate improves inactive component value.
 5. Installed components in equipped mount give 100% value.
 6. Installed components in inactive mounts give value $\times$ that mount's sync rate.
 7. Uninstalled components give 0 value.
 8. Fill all unlocked boards before merging extras.
 9. Merging rerolls component stat unless locked/not merged.
 10. Refine mostly Yellow or higher components.
 11. Component stat value depends on rarity and stat type, not shape.
 12. Only completed horizontal rows activate line effects.
 13. Lines 1-4 default, Line 5 requires 2Y, Line 6 requires 4Y, Line 7 requires 2R, Line 8 requires 4R.
 14. Only equipped mount line effects count.
 15. Component stat display may show only equipped mount stats, but optimizer should calculate synced inactive stats too.
-

22. Final “is source pack complete?” answer

Status = Complete enough for v1 DPS optimizer
MissingMajorSystems = None
IgnoredSystems = Unknown leaked mount
MountDataStatus = Added

Only thing still not perfect:

Exact in-game formula testing is still needed for how every damage bucket multiplies.
But source data coverage is now good enough to build the optimizer.